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Emoji and inferential processes: the neurophysiology of bridging in multimodal settings

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For my family.

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*"The truest respect which you can pay to the reader's understanding, is to halve this matter amicably, and leave him something to imagine, in his turn, as well as yourself."*¹

¹ STERNE, Lawrence. **The life and opinions of Tristram Shandy, gentleman.** [S. l.]: Coradella Collegiate Bookshelf Editions, 2004. p. 155.

ABSTRACT

PIMENTEL DE OLIVEIRA, G. H. A. **Emoji and inferential processes: the neurophysiology of bridging in multimodal settings** . 2026. 142p. Dissertation (Master in Linguistics) – Faculty of Letters, Federal University of Rio de Janeiro, Rio de Janeiro, 2026.

This study investigates the neural correlates of the complex cognitive phenomenon known as inferential “bridging,” particularly when it is triggered across different modalities of human communication (text–emoji). To this end, we compare three different conditions in which we manipulate the informational dependence between a sentence and an emoji acting as an independent discourse unit (Grosz; Kaiser; Pierini, 2021, 2023): in the *given* condition, the emoji adds little to no information to the sentence; in the *bridged* condition, it triggers a bridging inference; and in the *new* condition, it has no clear relation to the context. The conditions mirror Burkhardt’s (2006) investigation of definite-DP-mediated bridging. Two experiments were conducted: a web-based norming study, using the PCIBex platform, and a main experiment, using electroencephalography under the Event-Related Potentials methodology. The 120 stimuli most consistently rated as expected in the norming procedure were used in the main experiment, alongside 120 fillers consisting of sentences where one or more elements had been replaced by an emoji. Participants then saw the stimuli in Rapid Serial Visual Presentation (RSVP) while having their neural activity measured. A Spatio-Temporal Cluster-Based Permutation analysis found an N400-like sustained negativity for conditions *bridged* and *new* relative to *given*, and no difference between *bridged* and *new*, indicating that the *given* condition is perceived as categorically different from the others, but that *bridged* and *new* are equally taxing, which might point to the similarly unconstrained and unpredictable nature of both. In line with other research on emoji (Tang; Chen, *et al.*, 2020), no P600 effects of integration were found, indicating that obtaining a P600 might depend on how structurally bound the emoji is to the sentence (Tang; Chen, *et al.*, 2024). The norming study’s results were subsequently used to fit a rERP (regression-based Event-Related Potentials) analysis, in an attempt to further refine the cluster test results, but no effects were found for any of the predictors.

Keywords: Bridging inferences. Multimodality. Emoji. ERPs. Neuroscience of language.

RESUMO

PIMENTEL DE OLIVEIRA, G. H. A. **Emojis e processos inferenciais: a neurofisiologia das anáforas indiretas em contexto multimodal**. 2026. 142p. Dissertação (Mestrado) – Faculdade de Letras, Universidade Federal do Rio de Janeiro, Rio de Janeiro, 2026.

Este estudo investiga os correlatos neurais do fenômeno cognitivo complexo conhecido como *bridging* inferencial ou anáfora indireta, particularmente quando ele é desencadeado entre diferentes modalidades da comunicação humana (texto–emoji). Para tanto, comparamos três condições distintas nas quais manipulamos a dependência informacional entre uma frase e um emoji que atua como unidade independente do discurso (Grosz; Kaiser; Pierini, 2021, 2023): na condição *given*, o emoji acrescenta pouca ou nenhuma informação à frase; na *bridged*, ele desencadeia uma inferência de *bridging*; e na *new*, ele não possui relação clara com o contexto. As condições espelham a investigação de Burkhardt (2006) sobre o *bridging* mediado por DPs definidos. Foram conduzidos dois experimentos: um estudo normativo, por meio da plataforma online PCibex, e um experimento principal, utilizando eletroencefalografia à luz da metodologia dos Potenciais Relacionados a Eventos (ERPs). Os 120 estímulos mais consistentemente avaliados conforme o esperado no estudo normativo foram utilizados no experimento principal, ao lado de 120 distrações (*fillers*) que consistiam em frases nas quais um ou mais elementos haviam sido substituídos por um emoji. Os participantes viram, então, os estímulos em Apresentação Visual Serial Rápida (RSVP) enquanto tinham sua atividade neural registrada. Uma análise de permutação baseada em clusters espaço-temporais revelou uma negatividade sustentada, semelhante ao N400, para as condições *bridged* e *new* em relação à condição *given*, e nenhuma diferença entre *bridged* e *new*, indicando que a condição *given* é percebida como categoricamente diferente das demais, mas que *bridged* e *new* são igualmente custosas, o que pode apontar para a natureza igualmente pouco restringida e imprevisível de ambas. Em consonância com outras pesquisas sobre emojis (Tang; Chen, *et al.*, 2020), não foram encontrados efeitos de P600 relacionados à integração, indicando que a obtenção de um P600 pode depender de quão estruturalmente integrado o emoji está à frase (Tang; Chen, *et al.*, 2024). Os resultados do estudo normativo foram, em seguida, usados em uma análise de *rERP* (*regression-based Event-Related Potentials*), na tentativa de refinar ainda mais os resultados do teste de clusters, mas não se encontrou nenhum efeito para qualquer dos preditores.

Palavras-chaves: Inferências (anáforas indiretas). Multimodalidade. Emojis. ERPs. Neurociência da linguagem.

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LIST OF ABBREVIATIONS AND ACRONYMS

CAAE — Certificado de Apresentação para Apreciação Ética (Certificate of Submission for Ethical Review)

CEP — Comitê de Ética em Pesquisa (Research Ethics Committee)

CI / CIs — Confidence Interval(s)

DP / DPs — Determiner Phrase(s)

EEG — Electroencephalogram / Electroencephalography

EOG — Electrooculogram / Electrooculography

ERP / ERPs — Event-Related Potential(s)

FDR — False Discovery Rate

fMRI — functional Magnetic Resonance Imaging

GIF / GIFs — Graphics Interchange Format(s)

IC / ICs — Independent Component(s)

ICA — Independent Component Analysis

ITI — Inter-Trial Interval

MEG — Magnetoencephalography

MMN — Mismatch Negativity

MNE — Minimum Norm Estimate (analysis software)

OLS — Ordinary Least Squares

PCibex — Penn Controller for Ibex (online experiment platform)

rERP — regression-based ERP

ROI / ROIs — Region(s) of Interest

RSVP — Rapid Serial Visual Presentation

UFRJ — Universidade Federal do Rio de Janeiro

VIF — Variance Inflation Factor

M — Mean

SD — Standard Deviation

Mentioned ERP components

N300 — Frontal negativity peaking ~300 ms after stimulus onset; correlated with the visual properties of stimuli

N400 — Centro-parietal negativity peaking ~400 ms after stimulus onset; correlated with semantic access / prediction / lexical access

Nref — Referential negativity; correlated to ambiguous-anaphor resolution processes

P200 — Positivity peaking ~200 ms after stimulus onset, correlated with irony effects

P600 — Centro-parietal (sometimes decomposed in centro-parietal and frontal) positivity peaking ~600 ms after stimulus onset; correlated with discourse updating / integration

LIST OF SYMBOLS

α — A discourse event (first event in a discourse relation, e.g., event α causes event β)

β — A discourse event (second event); also the vector of regression coefficients (e.g., β_0 = intercept; $\beta_1 - \beta_3$)

ε — Residual (error term) in the regression model

Ω — Ohm (unit of electrode impedance)

y — EEG amplitude (dependent variable in the regression model)

ms — Millisecond

Hz — Hertz

$\approx$ — Approximately equal to (e.g., given < bridged $\approx$ new)

< — Less than (amplitude gradations, e.g., given < bridged < new)

$\times$ — Multiplied by

— — Minus / negative (e.g., -0.2 s epoch start)

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1 INTRODUCTION

The digital environment, to which we are today permanently connected, has brought with it numerous innovations, among them a particular integration of the technology of writing with other resources. For years now we have been able to interact in real time with other individuals through text messages, comment on the photos they post, respond to their public thoughts and express our own; and we frequently do so within the rules and limitations of a highly multisemiotic environment.

GIFs, stickers, and emoji are used every day as a complement to traditional text, serving the most varied functions and compensating for the gestures and other extralinguistic cues that the virtual world lacks. They can, for instance, expand the meaning of messages, modulate their impact, or signal the presence of figures of speech. Various fields have turned their attention to the subject, and although written language has already been extensively investigated, this multimodal reality in which it is now embedded offers new and interesting perspectives for study and analysis.

The present work, situated within Neurolinguistics and research on discourse and online comprehension, aims to investigate precisely one of these intersections of language. More clearly, it aims to investigate the processing of inferences triggered by emoji, such as in the following example, where we infer why “Sue impressed Ann” from the basketball emoji — “Sue impressed Ann, by playing basketball” (Grosz; Kaiser; Pierini, 2021, p. 342):

1. “Kate said Sue impressed Ann 🏀”

We draw primarily on Grosz, Kaiser, and Pierini (2021, 2023), which demonstrate that activity emoji can, in certain contexts, behave like independent discourse units with anaphoric effect. Face emoji, on the other hand, would behave like expressives such as swear words, indexing first person.

Inspired by their work, we observed that this anaphoric relation could also be investigated, within Clark and Haviland’s (1977) given–new paradigm, as a bridging inference, and that by controlling the level of informational dependence between an emoji and its host clause, we could construct a neuroscience experiment to investigate the neural correlates of multimodal inferencing, and of information dynamics more broadly.

Our research questions are as follows: how are the given–new dynamics computed when the two discourse elements belong to distinct modalities — image and text? And what are the neural correlates of multimodal bridging?

Emoji research, though growing, has mostly concentrated on how emoji operate *within* the linguistic structure — replacing words — and, in neuroscience especially, has relied almost exclusively on face emoji, leaving the discourse level and non-face emoji largely unexplored. The available results are also mixed, Event-Related Potentials (ERPs) such as the N400 or the P600 surfacing depending on position, congruence, structural integration of the visual element, or discourse relation (Tang; Chen, *et al.*, 2020, 2024; Cohn; Kutas, 2015; Weissman; Tanner, 2018). Emoji sometimes mirror linguistic processes, and sometimes depart from them. By investigating how emoji influence the processing and integration of information, this work intends to contribute to research on inferential processes, on multimodal processing, and on ERPs in general.

The neural signature of bridging has already been traced in single-modality settings. Burkhardt (2006), investigating it in a purely linguistic setting, linked its processing to two ERPs: the N400 and the P600. Cohn and Kutas (2015), on the other hand, working in a purely visual setting, found a P600 that grew the more the stimuli prompted mental-model revision, but no classic N400 in their inference conditions.

It is Burkhardt’s given–bridged–new paradigm that we adapt and carry into the multimodal domain.

We operate on the degree of informational dependence between the emoji and the sentence that precedes it, holding the sentence constant and varying only the final image, in decreasing relation of dependence:

- a. *Given*: the emoji simply retrieves information previously mentioned — e.g., “He walked through the city 🚶”¹;
- b. *Bridged*: the emoji introduces new information that is dependent on the prior context through an inferential bridge — e.g., “He walked through the city 🌉”;
- c. *New*: the emoji refers to completely new information — e.g., “He walked through the city 🍕”.

Mirroring Burkhardt’s (2006) model, we expect a graded response: 1) attenuated N400 amplitudes for *given*, pronounced amplitudes for *new*, and intermediate values for *bridged* (our target, which is mixed in nature), indexing dependency formation (*given* < *bridged* < *new*); 2)

¹ “Ele caminhou pela cidade”, in Portuguese.

a P600 that patterns *given* < *bridged* $\approx$ *new*, indexing the integration of a new referent. These hypotheses are stated in full in Section 3.7.

To test them, we build a sentence–emoji corpus and conduct two experiments: a norming procedure, to ensure each stimulus is interpreted as intended, and a main experiment, using electroencephalography and the ERP methodology.

The dissertation is organized as follows: Chapter 2 situates emoji within discourse and develops the parallel between emoji anaphoricity, bridging, and the situation model. Chapter 3 introduces the ERP paradigm, reports previous results for inferential bridging in purely linguistic (Burkhardt, 2006) and purely visual settings (Cohn; Kutas, 2015), and reviews other relevant literature. Chapter 4 describes the creation of the sentence–emoji corpus and the norming study. Chapter 5 reports the EEG experiment. Chapter 6 draws the findings together and points to limitations and further research. Chapter 7 synthesizes the contribution and concludes the work.

1.1 OPEN SCIENCE AND DATA AVAILABILITY

All resources mentioned in this dissertation — except for raw, individualized data — are made available in the appendix and in a GitHub repository: https://github.com/ghapoliveira/multimodal_inferences. Large tables and scripts are available only in the repository.

1.2 A NOTE ON TRANSLATION

All essential resources here and in the referenced repository were translated to English, including the stimuli. Idiomaticity was prioritized over literalism, and we tried to avoid dwelling on translation problems such as the one regarding verbal tenses that are expressed differently between the languages. Looking only at an isolated sentence in Portuguese, it is not always clear whether it corresponds to the Present Perfect or the Simple Past in English. When the doubt arose, we used what seemed to make more sense in the context of receiving a message. The same applied to the distinction among “will”, “going to” and the Present Continuous, as well as to preposition usage.

1.3 ETHICAL COMPLIANCE

The following described procedures were duly approved by the Committee of Ethics in Research of the Federal University of Rio de Janeiro (CEP/UFRJ) and are registered under the

Certificate of Presentation for Ethical Evaluation (CAAE) number 93482425.1.0000.5286. In compliance with the Committee's ethical recommendations, no participant received monetary compensation, but only complementary hours certificates (upon request).

2 EMOJI AND THE GIVEN–NEW CONTRACT

2.1 EMOJI AS INDEPENDENT DISCOURSE UNITS

Emoji have recently been drawing attention for the ways they appear to integrate seamlessly into human communication and convey meaning. This attention has spread across a variety of domains, such as Philosophy, Communication, Computer Science¹, and Linguistics, where it has rekindled interest in multimodal language more broadly and sparked debate².

Within the growing but still relatively narrow field of emoji research, however, the focus has been mostly on how emoji operate within the linguistic structure. Cohn, Engelen, and Schilperoord (2019), for instance, asks whether emoji have a grammar of their own and are subject to sequential constraints; Scheffler *et al.* (2022), working in psycholinguistics, examines the processing effects of substituting emoji for words inside a sentence; Tang, Chen, *et al.* (2020) and Tang, Chen, *et al.* (2024), following an Event-Related Potentials (ERPs) paradigm — as we will comment more in Section 3.4, when approaching previous ERP research on emoji — also investigate emoji replacing words. Comparatively, little research has addressed emoji at the level of discourse³.

There is also a second limitation concerning the array of emoji used, especially in neuroscience research — it mostly uses face emoji. Activity, object, and place emoji, for instance, are still vastly unexplored.

In this context, Grosz, Kaiser, and Pierini (2021, 2023) offer a compelling framework. They propose that activity emoji — such as 🏀, 🏄, or 🚴, evoking the activities of playing basketball, surfing, and cycling, respectively — can act as independent discourse units: rather than having a direct compositionality with the sentence, they interact with it much as a free adjunct does, being sensitive to discourse relations in ways that are similar to pronoun resolution processes (Grosz; Kaiser; Pierini, 2021, p. 342). E.g. (p. 344):

- “My job is pretty fun 🎸⁴”

¹ See Bai *et al.* (2019) for an interesting review through many different fields of knowledge.

² They have, e.g., been compared to iconographic writing and used to argue against universalist claims about writing systems (Downey, 2014, although he refers to emoticons, an earlier version of emoji), as well as regarded as evidence for the multimodality of language (Cohn, 2024).

³ An exception would be Weissman and Tanner (2018), who investigate the ironic use of emoji, comparing it to irony through words.

⁴ The emoji used while citing Grosz, Kaiser and Pierini’s (2021), and later Pierini’s (2021) examples are slightly different from the ones we used in our corpus — they use the Apple renditions.

Here, the emoji sits outside the sentence proper, yet we resolve the relation almost immediately as something like “My job is pretty fun; playing the violin/being a violinist” (p. 344). The two discourse units are linked through a salient discourse relation, and the emoji interpretation will resolve depending on those, as is made clear through one of their central examples (p. 342):

- a. “Kate said Sue impressed Ann 🏀”
- b. “Kate said Sue admired Ann 🏀”

Although they differ only in the embedded verb, we read the sentences differently. In the first one, it is Sue that impresses Ann because of her basketball skills; in the second, it is Ann. The authors compare this with the phenomenon of implicit causality, which occurs with pronouns: “an ambiguous pronoun in a causal adjunct [...] is preferably resolved towards the stimulus argument of a main clause psych-predicate” (p. 346). If we replace the 🏀 with the neutral expression “because she was *hix* [nonce word to avoid world knowledge interference]” in both “a” and “b”, as they proceed to explain, the pronoun “she” will resolve towards Sue in the first case, with the stimulus-experiencer verb “to impress”, and towards Ann in the second, with the “experiencer-stimulus” verb “to admire”; exactly as with the emoji 🏀 (p. 346):

- a. “stimulus-experiencer verb: kate said **sue** impressed ann because **she** was hix”
- b. “experiencer-stimulus verb: kate said **sue**⁵ admired ann because **she** was hix”

They report that this pattern of resolution seems to happen systematically when the emoji is in an Explanation relation with the previous text (p. 347), and they offer multiple other examples, also demonstrating other discourse relations between emoji and text, such as (p. 345) Elaboration (event β is part of event α); Narration (event β temporally follows event α), and Result (event α causes event β)⁶.

Additionally, the paper mentions that face emoji behave differently from activity emoji, preferentially resolving towards the author of the message, and not any of the individuals referenced in the discourse — a first-person index, much like expressives “such as damn or f*cking” (p. 348). “kate said that sue sent the report to ann 😡” could be read as “kate said that sue sent the **f*cking** report to ann” (p. 348), and the Ann–Sue impress-admire situation is read completely differently (p. 350):

- a. “Kate said Sue impressed Ann 😊”

⁵ In each cited example, we preserved the same capitalization and highlighting style used by the authors.

⁶ See Section 4.4 for a more detailed exploration of these categories.

- b. “Kate said Sue admired Ann 😡”

Rather than indicating that Sue or Ann is 😡 (angry), it indicates that this is the state of the speaker.

There are many other interesting points made, such as the fact that the emoji is not easily paraphrasable — in “My job is pretty fun 🎻”, the 🎻 can either be read as “playing the violin” or “being a violinist” — which could point to its semantics being minimalistic, just indicating “a violin-involving eventuality/property” or “(there is a) violin” (p. 344). More central to our purposes, however, is the nature of the link between the emoji and its accompanying text (its ‘host clause’ (p. 344)), which is inferential; in fact, they observe that the parallels made between emoji and anaphora are in line with Hobbs and Kehler’s proposals that linguistic pronoun resolution is a part of more general inferencing processes of maintaining coherence (p. 342).

Pierini (2021), within the same framework, analyzes this link by tracing parallels with gesture semantics, using Schlenker’s typology. He proposes that, when placed at sentence-final position (functioning as independent adjuncts), emoji generate cosuppositions. The content of the emoji would be entailed by the content of the expression it modifies (p. 725). For instance (p. 725):

- “John trained today 🏋️”

The emoji “🏋️” would trigger the inference “John trained today, which involved weightlifting” (p. 720), and this inference would project much like gestures through negations — “John didn’t train today 🏋️” (but if he did, it would involve weightlifting); modals — “John might train today 🏋️” (if he does, weightlifting will be involved); question operators and quantifiers such as “each” — “Did each of these ten guys train today? 🏋️” (for each of them, if they trained today, weightlifting would be involved); etc. (p. 725).

However, insofar as the relation between emoji and text involves an inference used to maintain coherence, it is also parallel to a bridging inference — sometimes also called indirect anaphora.

2.2 BRIDGING INFERENCES AND THE SITUATION MODEL

Clark and Haviland (1977) define bridging as one of the procedures — alongside addition and restructuring, which are not in the scope of this work — by which the listener ⁷ can restore

⁷ Clark and Haviland (1977) used “speaker”/“listener”, as they were focused on spoken language. We retain their terminology while reporting their work, but the given–new contract can be generalized to other media.

discourse coherence upon facing an element with no direct antecedent. It is a resource to preserve the given–new contract, a concept they develop and that, deriving from the Gricean principle of cooperation⁸, establishes that listener and speaker are obliged to make a syntactic distinction between given and new information in the discourse — the speaker cannot mention elements about which the listener has no knowledge, and the listener presupposes that their interlocutor is giving them all information necessary for comprehension. The paradigm can be summarized through the “Maxim of Antecedence”: “Try to construct your utterance such that the listener has one and only one direct antecedent for any given information and that it is the intended antecedent” (p. 4). Take the following example (p. 6):

- “John is a democrat. Bill is honest too.”

The two sentences are in apparent disconnection. John is mentioned only in the first sentence; Bill is only mentioned in the second. The predicates related to them are also not repeated. Communication could break down. However, presupposing the speaker is being cooperative and has given us all the needed information to interpret (Maxim of Antecedence), and also that they are being relevant (Maxim of Relation), we are led to infer a relation, bridging the two discourse units. “In this case the listener assumes he was supposed to infer that the speaker believes all Democrats are honest”⁹.

The same process appears to happen when we are confronted with emoji as described in Grosz, Kaiser, and Pierini (2021, 2023). In “Kate said Sue impressed Ann 🏀”, the basketball is in apparent rupture with discourse — both in form and in semantic content. Nothing is repeated, and so we have to infer the relation to save discourse, retrieving the referent of 🏀. Because the situation “Kate said Sue impressed Ann” is open and “impressing someone in basketball” is a plausible linkage, we resolve the antecedent problem by the interpretation “Kate said Sue impressed Ann, by playing basketball”¹⁰.

⁸ Grice, in his classic work “Logic and conversation”, establishes what he believes are the essentials for successful communication. One such essential is the Cooperative Principle, directed towards the speaker: “Be cooperative”, i.e., help your interlocutor understand your message. From it, four maxims derive: the Maxim of Quantity — “Make your contribution no more and no less informative than is necessary”; of Quality — “Say only that which you both believe and have adequate evidence for”; of Relation — “Be relevant”; and of Manner — “Make your contribution easy to understand; avoid ambiguity, obscurity, and prolixity.” (Grice, H. P., as cited in Clark; Haviland, 1977).

⁹ The authors actually present this example by adding that a stress on the second “too” would indicate the information is given, as the paper is also making the point of word stress as a givenness marker. Because the sentence is here presented written, and we are making the point of bridging in text, we modified the context to refer to the reader, and not a hypothetical listener.

¹⁰ It could also be paraphrased as “by being a good basketball player”, “being good in basketball” or “in basketball”. Again, as Grosz, Kaiser, and Pierini (2021, 2023) mention, emoji semantics might be minimalistic, such as ‘(there is a) basketball’.

If emoji can behave like independent discourse units and if they can have anaphoric effect and be sensitive to discourse relations, then we can explore the nature of this link through the lens of information dynamics (given–new paradigm). We can, for instance, operate on the degree of informational dependence between the two elements. Let us analyze the following example, foreshadowing our own experimental corpus:

Item 96:

- a. I was taken by surprise 😮
- b. I was taken by surprise 🌀
- c. I was taken by surprise 🛡️¹¹

In the first sentence, there is not much information being added by the 😮. It resolves towards the speaker¹², but if the first sentence states the person “was taken by surprise”, the following emoji only depicts a person with a surprised face — the information is mostly already given. In the second sentence, on the other hand, the emoji resolves not towards the first person, but rather “surprise”, and we do not know what surprise is being entailed until we reach the emoji — as with the basketball in the Ann–Sue example, it gives us complementary information on the event (bridged). In the final case, the information is too far-fetched to resolve easily: we attempt to attribute an antecedent to 🛡️ (a shield¹³), but we cannot find a readily available plausible connection — it is completely new information.

One could propose the explanation of the speaker being a book character that was taken by surprise by seeing a mythological shield. With time, we can always construct a model to connect the emoji. The third sentence, however, seems to recruit an additional effort of *construction* that is not present in the other two examples.

This connects to another concept central to the work we intend to develop: the situation model, developed by Dijk and Kintsch (1983). While reading, the reader does not only activate their knowledge of words and their semantic content, or the relation between different clauses and propositions, but they also activate a situation representation of the text, its persons, objects, events (Dijk; Kintsch, 1983, p. 11–12). The situation model is constantly updated by new bottom-up

¹¹ Item 96 of our experimental corpus. “Eu fui pega de surpresa” (feminine gender), in Portuguese.

¹² Both because of the explicit use of first person and because, as mentioned, face emoji preferentially resolve towards the speaker (Grosz; Kaiser; Pierini, 2021).

¹³ This can potentially seem like an emoji that is not only far-fetched semantically, but which is also difficult to decode visually. However, we conducted a norming experiment, as described in Section 4.6, and we probed, through a 7-point Likert scale, “To what extent is the emoji visually identifiable?” (From “Impossible to identify” to “Perfectly identifiable”). 🛡️ obtained the surprisingly high average of 5.38 (using the post hoc analysis data).

information, while also exerting top-down influence on other levels of comprehension. Crucially, Dijk and Kintsch (1983) establish that, as processes related to local coherence, bridging and other inferences would likely pertain to the situation level of comprehension (p. 62). The model would explain the phenomenon of coreference, allowing multiple expressions to refer to the same entity (p. 338):

[...] expressions in discourse do not refer to other expressions or their underlying concepts in a text, but to individuals in the situation model. The expression my brother and the lawyer may have different conceptual meanings, but may both refer to the same individual, say John. The situation model provides the representation of this same individual.

Taking into account the situation model of comprehension, the difference among the three sentences in question can also be expressed in terms of how distant each emoji is from the situation prompted by the context “I was taken by surprise” — abstractly, a surprising event. 😲, visually depicting a surprised person (likely the speaker), easily slides into this representation. 🌀 can also easily be accepted as the cause of the surprise — the reader might even be looking for it before seeing the emoji¹⁴. 🌀, on the other hand, demands a complete revision of the situation model already initiated, which could potentially generate higher processing costs.

Burkhardt (2006), working from a mental-representation model distinct from, though compatible with, van Dijk and Kintsch’s situation model, and drawing on Avrutin (1999) and on her own work (2005), proposes a distinction between dependent and independent discourse entities. Discourse entities would be dependent on prior context either by identity, i.e., repetition of reference (“I found a cat on the street. The cat was injured.”) or through a licensed inference constructed from world or situation knowledge (Burkhardt, 2006, p. 160) — a bridging inference (“I found a cat on the street. The fur was all matted”). The formation of this dependency would rely on the existence of a salient element in prior discourse that would serve as an anchor (p. 160). Independent elements, on the other hand, would have no such anchor and would have to be integrated as new discourse referents, resulting in increased processing costs (p. 160). Dependency formation would emerge for given information (in her case, definite DPs, as described in Section 3.2), while the establishment of independent reference would be present for new information, with bridging patterning with both, as partially dependent and partially independent entities.

¹⁴ There is much debate about how much inferencing really takes place in comprehension — whether individuals only draw necessary inferences for coherence (McKoon; Ratcliff, 1992), or whether they draw more elaborate, predictive inferences (Magliano; Dijkstra; Zwaan, 1996). For the present purposes, however, we work on inferences for local coherence, for which there is much less debate.

Assembling the framework described, therefore, we would have that the emoji in the aforementioned triad would be: 😊 — given information, highly dependent on its prior context and coherent with the ongoing mental/situation model; 🔄 — bridged information, partially dependent and coherent with the ongoing mental/situation model; 🌊 — new information, completely independent and demanding a revision of the mental/situation model.

What we propose in the present research is to investigate the neural correlates of the differences within this triad. More precisely, we intend to investigate whether manipulating the level of informational dependence — from given to new information — between emoji and their host clause is reflected in neural processing, as well as to identify the neural correlates of multimodal bridging.

It is worth noting that we did not use activity emoji alone. As we discuss in Chapter 4, our decision not to repeat any emoji within the corpus led us to draw on many categories, including face emoji (as in the example above) — provided they could form a given, bridged, or new relation with the prior discourse. Grosz, Kaiser, and Pierini (2021, 2023), though they focus mostly on activity emoji, do not restrict the phenomenon to them; still, this broad use is an exploratory extension of our theoretical foundation, and we make it explicit here.

In the next chapter, we briefly explain the Event-Related Potentials (ERPs) methodology we will utilize and present some caveats of ERP research. Then, we move to an exposition of ERP research more closely related to inferences, first in purely linguistic settings (Burkhardt, 2006), and next in purely visual settings (Cohn; Kutas, 2015). We finish by referring to previous work on multimodality, and by offering a review of a key debate regarding our ERPs of interest — the N400 and the P600.

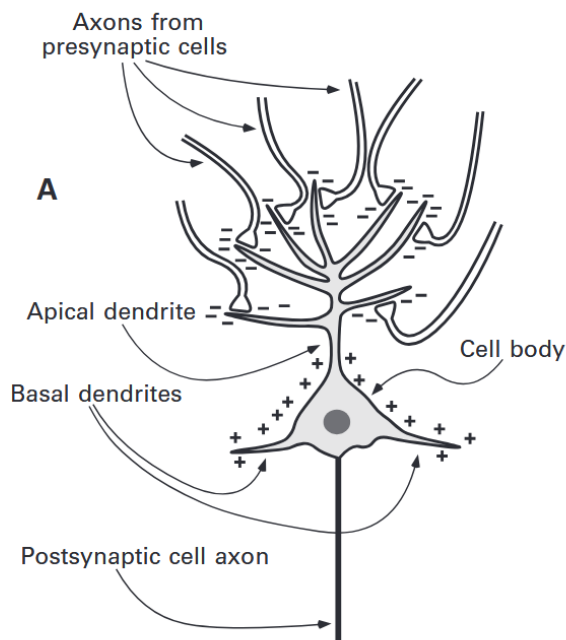
3 EVENT-RELATED POTENTIALS AND DISCOURSE COMPREHENSION

3.1 THE ERP PARADIGM

The Event-Related Potentials (ERPs) technique is an electroencephalography method in neuroscience research which allows us to examine the brain's electrical responses to experimental stimuli with high temporal precision.

The electroencephalogram (EEG) measures postsynaptic electric potentials that result from the coordinated activity of a group of neurons — when neurotransmitters bind to receptors on the membrane of the neuronal cell, they cause an electrical potential to propagate throughout the neuron. Figure 1 shows a neuronal cell at the moment an excitatory neurotransmitter is released on its apical dendrite, causing positive ions to enter the cell membrane, which makes its outside negative (-) and its inside positive (+). This positivity will then propagate throughout the cell body, forming a small dipole (one side positive, one negative):

Figure 1 – Synapse in a neuron cell. From Luck (2014).



When many neurons fire at the same time, this difference in electric potential forms a bigger dipole, measurable from outside the head. Because of the skull, membranes and scalp resistance, it diffuses, and hence the source of the effect cannot be unambiguously estimated. On the other

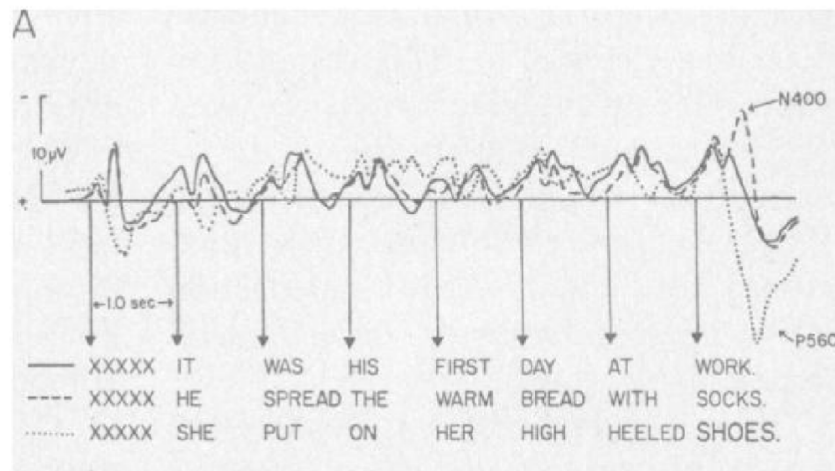
hand, the dipole forms almost instantaneously after the synapses, meaning that measuring its voltage fluctuations can provide high temporal resolution of underlying postsynaptic processes.

The ERP methodology takes advantage of this temporal resolution. Stimuli are presented to participants while the EEG records the neuronal activity. When there is an event of interest — such as the presentation of a word or, in the case of this study, an emoji — a trigger, placed at event onset time, is sent by the software that operates the stimulus presentation and merged with the recording: the neural activity is hence time-locked with the stimulus and can be inspected and compared depending on experimental hypotheses and conditions.

Say we want to compare two experimental conditions to investigate, for example, how the brain reacts to semantic anomalies. One way to approach this would be to work with a series of control type sentences (A), with no semantic anomaly, and another set of anomalous type sentences (B). First, we make sure the sentences are comparable, i.e., that each sentence of A shares the same target word or the same context with another sentence of B — for this hypothetical case, let us imagine that each sentence of A has an identical pair in B, except for the substitution of one word, which causes the anomaly. We would place a trigger on the onset of the differing words and time-lock the neuronal activity. Then, processing the obtained waveforms, we would select only the time window around the trigger events (epoching); average them per participant, so we can extract the mean responses for type A and type B sentences within a single person; and, finally, average together all individual averaged sets of A and B (grand averages) and subtract the averaged waveforms (difference waves). That difference wave would show how, in terms of amplitude and timing, the difference between A (control) and B (target) is perceived in the brain.

Through time and experience, different ERP “signatures” were discovered, i.e., groups of similar brain responses to certain types of stimuli, with similar head distribution (topography) and time of occurrence, allowing researchers to compare results. The explanation above roughly echoes a series of experiments developed by Kutas and Hillyard in the eighties (1980, 1984), and which are at the foundation of one such signature: the N400, a negative, centro-parietal deflection peaking at around 400 ms after stimulus onset, first discovered to pattern with semantic anomalies. Figure 2, taken from Kutas and Hillyard (1984), illustrates the effect — differently from the explanation, it is comparing different sentences, but the N400 is shown to appear when there is a semantic anomaly:

Figure 2 – The N400 effect. From Kutas and Hillyard (1984).



Since then, the understanding of the N400 has evolved, and semantic anomalies were found to be neither necessary nor sufficient for the effect to arise (Kutas; Federmeier, 2011, p. 4). In fact, every content word in discourse elicits an N400 (Van Berkum *et al.*, 2007), modulated by how well it fits the context, being thus influenced by priming (Franklin *et al.*, 2007), discourse constraints (Nieuwland; Van Berkum, 2006), and expectations (see Wang; Schumacher, 2013, on topicality). It has been proposed to reflect semantic access more broadly (Kutas; Federmeier, 2011), lexical retrieval (Aurnhammer *et al.*, 2023), or prediction (Kuperberg, 2016) (its amplitude being smaller the more constrained and predictable the word).

There are many other ERPs, such as the MMN (Mismatch Negativity), an early negativity that is elicited when a sensory pattern is broken, be it auditory, visual, or haptic (for instance, a sequence of identical standard tones suddenly interrupted by a tone of a different pitch¹, or a series of visual “X”s interrupted by an “O”); the N300, a frontal negativity related to the visual properties of objects or scenes (Kumar; Federmeier; Beck, 2021), peaking at around 300 ms after stimulus onset; or the P600, a centro-parietal positivity which peaks at around 600 ms after stimulus onset and was first attributed to syntactic anomalies, but has since expanded — it has been proposed to reflect the establishment of a new referent (Burkhardt, 2006), and discourse integration or updating processes (Burkhardt, 2006, 2007; Brouwer *et al.*, 2017; Aurnhammer *et al.*, 2023).

¹ See, e.g., Santos (2026) for an exploration of the MMN as an index of dialectal discrimination in Brazilian Portuguese.

It is not always easy to distinguish different ERPs or to point at their exact underlying function. They might share topography, polarity, and time of occurrence. The Nref effect, for example, related to the resolution of ambiguous anaphors (Van Berkum, 2009; Nieuwland; Coopmans; Sommers, 2019), has a similar onset to the N400 and is also a negativity, making it difficult to separate the two effects, depending on experimental design. The Nref is more frontally distributed, but ERPs can also behave differently depending on multiple factors, and the N400 has been reported to be somewhat more frontally distributed for experiments with pictures (Kutas; Federmeier, 2011, p. 8). A strong negativity might also “mask” or attenuate a subsequent positivity, as Delogu, Brouwer, and Crocker (2021) explain. Centrally, however, each ERP has a functional sensitivity (Kutas; Federmeier, 2011), established through prior research, and, depending on the phenomenon of interest, there might be no compelling reason to consider overlap.

This inherent ERP complexity has led researchers to sometimes refer to ERPs in the plural, such as “N400s” (Kutas; Federmeier, 2011), frontal negativities (Cohn; Kutas, 2015, p. 15), or positivities, with attempts being made to untangle them (Kuperberg; Brothers; Wlotko, 2020; Brothers *et al.*, 2020).

ERP research is fundamentally correlational — we know that an electric signature occurs at the time a given stimulus is presented, but we still do not fully grasp the internal mechanisms that give rise to it. A lot of research has been directed at understanding which exact neural processes are at the origin of ERPs, as well as at differentiating them. fMRI (functional Magnetic Resonance Imaging) research, which has good spatial but poor temporal resolution, has sometimes been employed alongside electroencephalography in order to find the exact brain regions that produce these electrical potentials. Promising studies have, for example, been linking the N400 to the left lateralized temporal cortex, a region known to mediate the retrieval of the semantic features of incoming words (Lin; Gina, 2021).

Despite its intricacies, however, the ERP methodology is still one of the best known methods to investigate brain processes, as well as one of the most employed. It is not as resource-intensive as fMRI and MEG (magnetoencephalography), its temporal resolution allows for a great variety of experimental explorations, and it is also attached to an extensive literature, allowing for fruitful comparisons. Building on this foundation, we now turn to the specific brain potentials directly linked to inferences, multimodal processing, and online comprehension.

3.2 THE NEURAL CORRELATES OF LINGUISTIC BRIDGING

In an already briefly mentioned seminal work from 2006 (Section 2.2), Burkhardt, also working under the given–new paradigm, set out to investigate bridging inferences in a similar context to ours, but in a purely linguistic medium, through definite DPs. In languages where those are available, the definite markers are frequently used to indicate givenness, as in the following example (Burkhardt, 2006, p. 160):

- “Sean met a girl yesterday. The girl talked about a recent trip to Boston.”

First, the girl is introduced into discourse by the indefinite determiner “a”, then, when she is mentioned again in the second sentence, she is referred to by the definite determiner “the” to indicate to the reader that the girl is a known entity which they should already know about and search in memory. When used without direct repetition, however, the definite DP can trigger a bridging inference, as the reader is prompted to search for a relation that is not explicit (p. 160):

- “Natalie went to a wedding in Italy. **The groom** talked about a recent trip to Boston.”

As we can observe, “the groom” was not mentioned in the previous context; nevertheless, the ensemble is easily interpreted and there is no need for all the relations to be spelled out — “Natalie went to a wedding in Italy. The wedding had a groom. The groom talked about a recent trip to Boston”. The presence of the definite article “the” indicates to the reader that “the groom” is a given entity, and to save discourse from apparent disconnection, the reader is led to construct an inferential bridge between “the groom” and the “wedding”. This bridge is constructed through world knowledge and in this case depends on salient roles in the mental model created for the context: a wedding can have a groom, a bride, a church, etc. Other discourse relations, such as causality or part-whole, could also serve as anchors (p. 160).

Burkhardt proposes that the process of DP integration encompasses two different stages: dependency formation and establishment of independent reference. The former would occur with given DPs, the latter with new DPs (p. 169). Bridged DPs would function as hybrids: while they have a dependency relation with former discourse — which takes the shape of a licensed inference made possible by an anchor — they also add new information. Based on prior findings that linked attenuated N400 amplitudes to semantic priming and repetition, as well as previous research with pronominal resolution and inferences, she proposes that the process of dependency formation would be correlated with the N400. The integration of a new referent, on the other

hand, would be linked to the P600, as it had been shown to co-occur with increased anaphoric complexity.

Given DPs, in this context, would show attenuated amplitudes for both N400 and P600, reflecting, respectively, effective dependency formation and low integration effort. New DPs, on the other extreme, would show greater N400 and P600 deflections, reflecting first the difficulty in establishing dependency, and then the effort of integrating a new referent. Bridged DPs would lie in the middle, showing first a greater N400 than given, as they are only partially dependent, and a P600 comparable to new. Essentially, a given < bridged ≤ new gradation was to be expected, according to her general hypotheses on which processes are indexed by each ERP.

To test her model, she built three conditions²: one “given”, in perfect identity of referent (“Tobias visited a conductor in Berlin. He said that the conductor was very impressive”); one “bridged”, with an indirect reference to prior context (“Tobias visited a concert in Berlin. He said that the conductor was very impressive.”); and one “new”, where the rupture in discourse was not easily recoverable (“Tobias talked to Nina. He said that the conductor was very impressive”).

Her results were in accordance with the hypotheses: an attenuated N400 (broadly distributed) for the “given” and “bridged” conditions — less attenuated for “bridged” — which she interpreted as dependency formation; and a P600 (left-posterior) attenuated for “given” but similar for “new” and “bridged”, which she interpreted as integration.

In a later study, Burkhardt (2007) further explored this P600-as-integration hypothesis by manipulating the degree of contextual constraint on the inference, from necessary to probable and inducible (Burkhardt, 2007, p. 1852):

- Necessary: “Yesterday a Ph.D. student was shot downtown. The press reported that ***the pistol*** was probably from army stocks.”
- Probable “Yesterday a Ph.D. student was killed downtown. The press reported that ***the pistol*** was probably from army stocks.”
- Inducible: “Yesterday a Ph.D. student was found dead downtown. The press reported that ***the pistol*** was probably from army stocks.”

She found no N400 and only a graded P600 effect — smallest for inducible, followed by probable and necessary, which did not differ between themselves — attributing the absence of

² Throughout this dissertation, we use quotation marks for this terminology as used by other authors (e.g., Burkhardt’s “given,” “bridged,” and “new” conditions) and italics for our own experimental conditions (*given*, *bridged*, *new*), reserving unmarked use for the general given–bridged–new distinction independent of any single study’s operationalization.

an N400 to differences in lexical-semantic and discourse-level processing: “semantic memory supports lexical-semantic and discourse level processes differently” (2007, p. 1853). For the graded P600, she proposes that it reflects increased processing costs for new information in discourse memory.

It is Burkhardt’s (2006) paradigm that our design carries into the multimodal domain. Where Burkhardt (2006) manipulated the referential dependency of a definite DP, we manipulate that of a sentence-final emoji, mirroring her given–bridged–new conditions explicitly:

- a. I was taken by surprise 😮 — *given*;
- b. I was taken by surprise 🕸 — *bridged*;
- c. I was taken by surprise 🛡 — *new*.

In accordance with her model and results, we would also expect an N400 gradation (*given* < *bridged* < *new*), reflecting dependency formation for *given* and *bridged* (with *bridged* being less dependent), but not for *new*; and a P600 duality or gradation (*given* < *bridged* $\cong$ *new*), reflecting establishment of a new referent in *bridged* and *new*, but not in *given* — as well as increased processing costs for *bridged* and *new*.

One caveat, however, distinguishes our *given* condition from Burkhardt’s (2006)³. Because she used words, her “given” condition was straightforward — the antecedent was simply repeated. An image, by contrast, can refer back in essentially two ways: by iconicity (i.e., via direct visual resemblance to what they refer to) or by semantic association⁴. Take these two sentences as examples (not present in the corpus):

- “I took my car to the mechanic 🧑🔧” — Iconicity.
- “I took my car to the mechanic 🗜” — Semantic association.

In the first sentence, the emoji portrays a mechanic, which is mentioned in the previous sentence. In the second, on the other hand, we see a wrench, which by semantic association can be connected to “mechanic” or to the situation of taking the car to the mechanic.

This is particularly relevant because the bridging inference, our main object of study, can also be viewed as an associative process. The difference between our conditions, nevertheless,

³ Unlike her, we also manipulated target, and not context — a point which we discuss in more detail in Section 4.3.

⁴ A fuller treatment could situate what we call “semantic association” within a Peircean framework, where iconicity, indexicality, and symbolicality interact — an emoji may resemble its referent, be contiguously associated with it, or stand for it by convention, often at once. We set this aside here, using “iconicity” and “semantic association” as a coarser working distinction sufficient for our experimental purposes; the finer semiotic decomposition is a matter for future work.

lies most centrally in the amount of information added in each case. In “I took my car to the mechanic ✨”, for example, there is substantially more information being added if we interpret it as indicating a crash⁵ — this inference is not present with the wrench or the mechanic, which overlap semantically with the described situation. We return to this point in Section 4.4 and Chapter 6.

3.3 THE NEURAL CORRELATES OF VISUAL BRIDGING

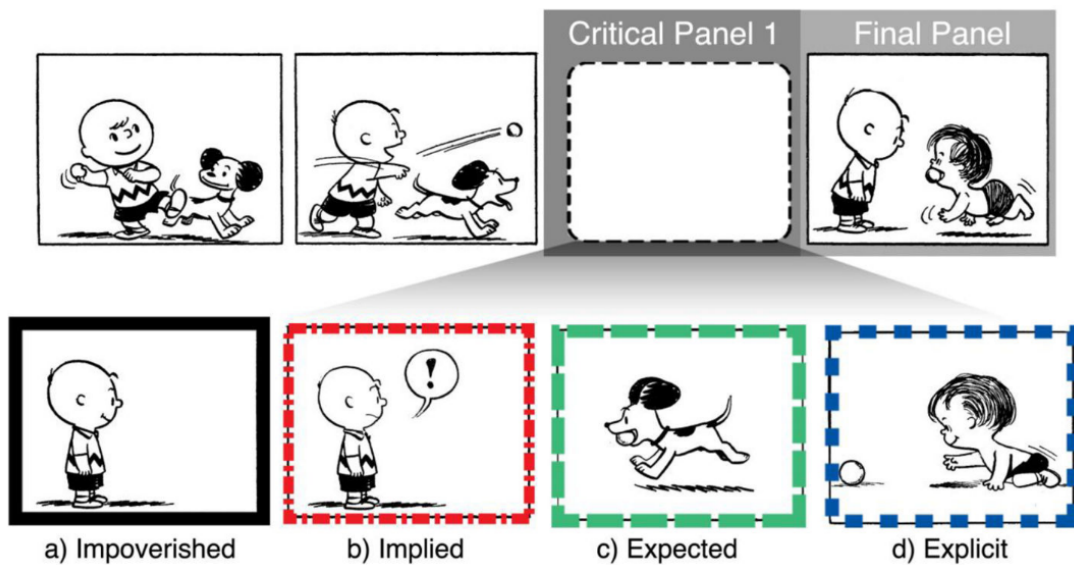
Cohn and Kutas (2015), working in a purely visual context, also approached bridging inferences. Their context differs from ours, as they investigated visual scenes (Peanuts strips), but their findings are important to us to the extent that they might indicate the general signature of the core phenomenon.

They raised two experimental questions: Is there evidence of a difference in neural processing between visual narratives that unfold explicitly and implicitly? And if there is, does the processing of implicit narratives differ, depending on the nature of the implications?

In order to answer these questions, they compared four experimental conditions controlling two panels of interest inside a Peanuts comic strip: Impoverished, Implied, Expected, and Explicit. In “Impoverished” sequences, Charlie Brown is shown throwing a ball for Snoopy to catch, then, at the critical panel, Charlie Brown passively waits for something that happens offscreen, after which Linus retrieves the ball instead of Snoopy. “Implied” adds a dialogue balloon with “!” in the panel where Charlie is waiting, indicating that something important is happening offscreen. “Expected” shows Snoopy retrieving the ball, instead of Linus — which causes a clash with the final panel, with Linus retrieving the ball. Finally, “Explicit” depicts Linus taking the ball and then retrieving it. Figure 3, taken from Cohn and Kutas (2015), illustrates the different conditions:

⁵ This is not a normed example and is not part of our experimental corpus.

Figure 3 – Cohn and Kutas' (2015) experimental conditions.



For the critical panel, they obtained a posterior late positivity that increased the more the scene changed, “in line with the proposal that late positivities (P600s) reflect the updating or revision of the mental model” (p. 12). Impoverished and Implied panels, which just show Charlie Brown passively looking/seeing something surprising offscreen, showed smaller positivities than Expected and Explicit, which depicted greater change in narrative. Images which showed more contiguity (Expected) also showed smaller positivities than those that broke it (Explicit), and Implied, which implied an event offscreen and contained more information than Impoverished conditions, had greater positivity than Impoverished. Overall, the positivity was graded as: Explicit > Expected > Implied > Impoverished.

For the final panel, Impoverished sequences showed the largest positivities, also in line with the P600 as an index of situation model updating or revision: “no prior events had either depicted or implied what was to happen, and thus [it] called for the greatest updating of structure” (Cohn; Kutas, 2015, p. 13). Impoverished was followed by Implied, which, by the usage of “!”, had indicated that something was happening offscreen, and had similar amplitudes to Explicit. Expected had the smallest amplitudes.

They also found a frontal negativity in response to the final panel of Impoverished and Implied sequences, between 200 and 600 ms, which they do not interpret as an N400. Interpreting it as an N400, they say, is neither compatible with its distribution nor with it showing only for Impoverished and Implied: the semantic incongruity in Expected final panels is arguably stronger

than the one in Impoverished and Implied sequences, and they did find a more characteristic, N400-looking peak for Expected final panels, preceded by an N300, which “typically precedes the N400s to visual images” (p. 14). The frontal negativity for Impoverished and Implied sequences patterned differently and contained no N300 (p. 14).

The frontal negativity, they posit, might rather be related to working memory demands created by omitting information (p. 15). They say frontal negativities have been hypothesized to reflect working memory demands on discourse processing — the effort of resolving ambiguous references (Nref), or of computing or maintaining inferred event information between scenes (p. 15).

Processing scenes is arguably different from processing still images with text — and the brain has different subsystems for images and scenes (Epstein; Kanwisher, 1998). They did not find the same N400 as Burkhardt (2006), for example. However, as the underlying brain signature of resolving implicit referents and bridging omitted events might transpose from scenes to text-image multimodality, this work serves as a useful comparison.

3.4 PREVIOUS ERP RESEARCH ON EMOJI

As commented above, ERP research on emoji is still a growing field, and it has mostly focused on the substitution of sentence elements with emoji — and usually face emoji. However, there are some findings that relate to our ERPs of interest (the N400 and the P600) that warrant mention.

Tang, Chen, *et al.* (2020), investigating how emoji are processed when compared to words, compared four different conditions: 1) An emoji incongruent with its context; 2) A word (parallel to the emoji) incongruent with its context; 3) An emoji congruent with its context; 4) A word (parallel to the emoji) congruent with its context (Tang; Chen, *et al.*, 2020, p. 2):

- Congruent:

- a. “Our project eventually succeeded, and I felt very 😊⁶”
- b. “Our project eventually succeeded, and I felt very happy.”

- Incongruent:

- a. “Our project eventually succeeded, and I felt very 😞”
- b. “Our project eventually succeeded, and I felt very sad.”

⁶ Apple Emoji Font.

They found that, while incongruous words elicited both an N400 and a P600 (the P600 can also be triggered by completely impossible interpretations, such as an inconsistency between the result and the context, as they explain (p. 4)), emoji only elicited a sustained N400. They observe that this is in line with other research that only obtained an N400 component for final words replaced by pictorial symbols, and mention that this could further direct the interpretation of the N400 as meaning-oriented and not language-oriented (p. 4)⁷.

Paradoxically, however, in another study, this one replacing sentence-intermediate words with face emoji⁸, they found the opposite result — the P600, but no N400. They explained these findings by saying that, perhaps, when the emoji is integrated into the sentence and occupying a structural role such as a verb, its semantic and syntactic properties might be just too interwoven, with a failure to integrate it semantically resulting in a complete collapse of syntactic processing (Tang; Chen, *et al.*, 2024, p. 1749), indexed by the P600.

They also argue that in Tang, Chen, *et al.* (2020) the emoji were more loosely connected to the sentence and hence their syntactic role was weakened — while their semantic role was strengthened (and so there was no P600, but only an N400).

Weissman and Tanner (2018), on the other hand, conducting three experiments that compared sentences ending with either congruent, incongruent, or ironic face emoji (with the emoji acting outside the sentence), found late positivities for incongruent and ironic emoji — to the extent that participants interpreted the emoji as ironic — attributing the effect to monitoring or reanalysis processes. They also found a P200 response to the ironic emoji, which parallels results for irony using words.

This raises the issue that, in multimodal emoji research, ERP components have proven somewhat elusive: emoji have been shown to elicit an N400 or a P600 depending on their position, their congruence, and how tightly they are integrated into sentence structure. They sometimes mirror linguistic processing, as in Weissman and Tanner (2018) irony effects, and sometimes diverge from it.

3.5 PREVIOUS ERP RESEARCH ON MULTIMODALITY

Willems, Özyürek, and Hagoort (2008), working not with emoji, but with a different multimodal pairing, have a design close to ours, but with auditory stimuli. They set out to

⁷ They also mention a potential masking of the P600 effect, due to the large and sustained N400 (p. 5).

⁸ Following the same congruent/incongruent pattern, but bringing the emoji into the sentence, as in (using WeChat emoji): “Seeing his funny face, I 🤪 for a while” (Tang; Chen, *et al.*, 2024, p. 1746).

investigate word versus picture integration into sentence context by using audio followed by images in four conditions (Willems; Özyürek; Hagoort, 2008, p. 1238):

- The target word and the image matched:

The man gave his wife a nice flower that evening



- The target word mismatched its context, but the image was correct:

The man gave his wife a nice cherry that evening



- The picture mismatched the correct word:

The man gave his wife a nice flower that evening



- And a “double mismatch”:

The man gave his wife a nice cherry that evening



They found an N400 effect with similar latency and amplitude in every mismatch condition (including double mismatch). The effect was strong, not preceded by an N300⁹, and had carryover effects (p. 1244).

⁹ They observe that an N300 was also not found in other studies with pictures in sentences (Willems; Özyürek; Hagoort, 2008, p. 1243). Research on emoji also does not report it.

To the extent that their stimuli have the image functioning outside discourse, as a complement, and that “The man gave his wife a nice flower that evening 🌹” and “The man gave his wife a nice flower that evening 🍒” are similar, respectively, to our *given* and *new* conditions, this is an essential reference. Their stimuli, however, did not have a counterpart to our *bridged* conditions.

Work on gestures also warrants a brief mention, since it also touches multimodal integration. Interestingly, in semantic violation or disambiguation paradigms — which are closer to the anaphoric relations with which we are concerned — it systematically finds only N400 effects, rarely reporting a P600 (Gunter; Weinbrenner, 2017, p. 5).

One exception would be Gunter and Weinbrenner (2017), which, through two different experiments, investigated the phenomenon of abstract pointing, where a referent absent from the physical scene is cued through pointing to empty gesture space: e.g., participants watched a person talking about Goethe or Shakespeare while pointing left or right abstractly while referring to either one. Following the Gricean principle, people usually perceive gestures as being meaningful; when they realize they are not, they only monitor them for their cue validity — a process indexed by a P600. They obtained an isolated P600 effect when unreliable pointing was being used and a biphasic N400–P600 pattern when pointing was reliable, with an attenuated N400–P600 for referent-matching pointing relative to baseline (no pointing). Citing Burkhardt (2006), they attribute the P600 to mental model updating (p. 8).

They propose that studies with gestures that only found the N400 mostly deal with iconicity, which is highly intertwined with context — both influence each other at a semantic level, “which increases the lexical and semantic retrieval demands from long-term memory leading in the end to an N400 [for mismatch or ambiguous conditions]” (p. 9). In those studies, they suggest, there was no need for mental model revision or updating. Abstract pointing, on the other hand, would recruit more resources (p. 9):

Because the vast majority of ERP studies on gestures reported an isolated N400, without a subsequent P600, it seems that there was no need for any update or revision processes in those studies and that gesture-related information only needed to be retrieved from memory. Thus, the basis of processing and integrating iconic gestures is directly related to the semantic content of the context. In contrast, referencing via pointing gestures (i.e., concrete deixis) has a different, more interactive, and social basis.

This seems to suggest that whether a P600 accompanies the N400 may hinge on how much or on what type of information the visual element adds, which is the dimension we manipulate in our experimental corpus.

3.6 THE N400, THE P600, AND DISCOURSE COMPREHENSION

Insofar as our discussion deals with the P600 and the N400, which are the two most debated potentials in the online comprehension literature, it also touches on a broader and controversial dispute, which we attempt to summarize below.

The P600, as we mentioned, was initially thought to reflect syntactic mismatches, with semantic anomalies being rather attributed to the N400. However, multiple examples of a “semantic P600” were increasingly reported:

- “The hearty meal was devouring the kids” (Kim; Osterhout, 2005).

This anomalous sentence, contrary to expectations, did not give rise to an N400 when compared to “The hearty meal was devoured by the kids”¹⁰, but only to a P600 (Kim; Osterhout, 2005). Demanding explanation, the phenomenon divided the literature roughly into two groups (as defined in Aurnhammer *et al.* (2023)): Multi-Stream models and Retrieval Integration.

Multi-Stream models posit two “streams” in online comprehension: one semantic, which would be indexed by the N400; and one syntactic, linked (indirectly) to the P600. The N400, being modulated by word expectation and plausibility, would index, at least partially, semantic integration, while the P600 would occur only when there is a mismatch between semantic and syntactic streams. For “the hearty meal was devouring”, they propose that, although there is a semantic anomaly, there is a locally available and semantically attractive option (devoured). Therefore, the reader/listener would be temporarily blind to the semantic violation (a kind of semantic illusion), which would be rather picked by the syntactic stream and reveal a conflict between semantics and syntax — indexed by the P600.

Retrieval-Integration, on the other hand, proposes delimited roles for each of these potentials: the N400 would index lexical retrieval, while the P600 would reflect discourse integration processes, varying continuously with integration effort. They argue that the absence of an N400 effect for “devouring/devoured” is justified by the level of associative priming both words receive from context (Aurnhammer *et al.*, 2023, p. 3), while the presence of a P600 is due to the increased difficulty in interpreting “meal” as an agent for “devouring” (p. 3).

There are other theoretical proposals that do not fit either side, such as the predictive model described in Kuperberg (2016) and Kuperberg, Brothers, and Wlotko (2020). Its explanation of

¹⁰ They actually had two controls, one passive (“The hearty meal was devoured by the kids”) and one active (“The hungry boy was devouring the cookies”).

why the semantic illusion happens is similar to the one given on Multi-Stream grounds, but the predictive model also seems to acknowledge that some integration might be indexed by later positivities (Kuperberg; Brothers; Wlotko, 2020, p. 19), although most of the integration would potentially already be done through the prediction process (p. 19). By this account, the N400 would rather reflect probabilistic inference and the access to semantic features¹¹. The reader/listener would actively try to infer the message, constructing statistical models and structures that depend on the strength of the available cues. This inference process would be done in multiple levels of representation, with the beliefs and hypotheses regarding the event/sentence/structure to be comprehended guiding low level predictions. The reader/listener would not, however, always use all the cues that are available to understand — this would depend on the strength of these cues, and the prediction prompted by them could sometimes be so strong that the interpreter might become “temporarily blind” to the incongruence (reading/listening “devoured” instead of “devouring”).

Nieuwland and Van Berkum (2005), exploring the limits of the semantic P600 phenomenon, set out to replicate it within discourse and with a stronger semantic rupture. In an auditory experiment, they compared two situations: one with a coherent continuation, and another with an anomalous one (Nieuwland; Van Berkum, 2005, p. 692):

Table 1 – Example story used by Nieuwland and Van Berkum (2005) (approximate translation from Dutch offered by the authors).

Introduction

A tourist wanted to bring his huge suitcase onto the airplane. However, because the suitcase was so heavy, the woman behind the check-in counter decided to charge the tourist extra. In response, the tourist opened his suitcase and threw some stuff out. So now, the suitcase of the resourceful tourist weighed less than the maximum twenty kilos.

Coherent continuation

Next, the woman told the *tourist* that she thought he looked really trendy. The tourist grabbed the woman’s hand and eagerly asked her for a date. But the woman reprimanded the tourist for being pushy and told him to just get on the plane right away.

Anomalous continuation

Next, the woman told the *suitcase* that she thought he looked really trendy. The suitcase grabbed the woman’s hand and eagerly asked her for a date. But the woman reprimanded the suitcase for being pushy and told him to just get on the plane right away.

¹¹ Supported by the mentioned theorized origin of the N400 in the left lateralized temporal cortex (Lin; Gina, 2021).

They also did not obtain N400 effects for the anomalous situation, but only a P600. As they themselves phrase, it was a quite surprising finding, given the literature (2005, p. 695):

In the light of what we know about the N400 in language comprehension, the absence of a regular N400 effect in the current study is a very surprising finding. In spoken as well as written language, differences in the degree to which a particular word fits the wider semantic context reliably show up in the amplitude of the N400 elicited by that word.

As Multi-Stream advocates observed, however, although a semantically attractive alternative is not available locally in Nieuwland and Van Berkum (2005), it is available *globally* — “tourist”, mentioned multiple times in the context, is a better alternative.

Aurnhammer *et al.* (2023), therefore, adapting the model in Nieuwland and Van Berkum (2005) and seeking to test the competing hypotheses, constructed an experiment where the availability of the semantically attractive alternative is controlled. Unlike Nieuwland and Van Berkum (2005), the stimuli were written, and the manipulation resided in the context, not the target. The test involved a context paragraph designed to prime equally a target word and a distractor, which was then followed by a verb that rendered the next word: both plausible and expected — Condition A (control); neither plausible nor expected, but available in the context as a distractor — Condition B; neither plausible nor expected, and the use is not licensed by the context — Condition C. E.g. (p. 7):

Item 2

A teacher saw an old world map in the showcase of an antique shop. Such an authentic artifact appeared suitable for his classroom and he approached the **saleswoman** ...

A: Then bought the teacher the map ...

B: Then kissed the teacher the map ...

C: Then filled the teacher the map ...

Multi-Stream models, proposing that the semantic P600 is only present when there is a mismatch between semantic and syntactic streams, would predict an N400, but no P600, when there is no plausible nor expected semantically attractive alternative. Retrieval-Integration, on the other hand, predicts no N400 in any of the conditions (because of the lexical priming both words receive), expecting only a graded P600 indicating integration effort.

Aurnhammer *et al.* (2023) found, in accordance with their predictions: 1) no N400 effect in the different conditions — conflicting with Multi-Stream views of the N400 as integration; 2) a P600 that pairs with plausibility; 3) that the P600 they obtained could potentially be further dissected into one with a parietal peak and one with a frontal peak — in line with other research

(Brothers *et al.*, 2020; Kuperberg; Brothers; Wlotko, 2020); 4) an early negativity that they argue does not match the classical N400 — which also patterns with Cohn and Kutas’ findings (Cohn; Kutas, 2015).

The functional interpretation of the N400 and the P600, however, is not yet completely settled, as the mixed results in multimodal settings, with or without emoji, illustrate.

3.7 OUR EXPERIMENT AND HYPOTHESES

Taking together the literature, Burkhardt’s study, working with linguistic bridging, (2006) points to an N400 indexing dependency formation followed by a P600 reflecting the establishment of an independent referent; Cohn and Kutas’ experiment for visual scenes points to a graded positivity indexing discourse updating and a frontal negativity in place of a classic N400 (Cohn; Kutas, 2015), potentially reflecting working memory demands related to anaphoric processing.

Emoji studies, although not working with inferences, also found N400 and P600 effects. Tang, Chen, *et al.* (2020) and Tang, Chen, *et al.* (2024) observed that obtaining one or the other might depend on the emoji’s structural integration with the context (Tang; Chen, *et al.*, 2020, 2024), with a weaker structural relation reflecting only an N400, but not a P600 (Tang; Chen, *et al.*, 2020). Weissman and Tanner (2018), however, working with emoji as independent units — hence further removed from the sentence structure — found P600 effects to irony with emoji.

Previous multimodal research using auditory and visual stimuli, where the image was also independent, found only N400 effects, possibly reinforcing this idea that structural integration might be necessary for the P600 effect. Finally, prior research on gestures is consistent with the possibility that information type or information addition might play a role in whether the P600 is found (Gunter; Weinbrenner, 2017).

Although recent literature has been pinning the N400 down to lexical access, the access of semantic features, or to prediction processes, and the P600 to integration effort, the exact role of these Event-Related Potentials is, as demonstrated, highly disputed.

Our research, far from offering a definitive solution, proposes to explore the underlying components involved in discourse comprehension within a still relatively unexplored path.

Purely linguistic or scenic bridging cannot help but insert the bridged element within a structure: “the groom” in “Natalie went to a wedding in Italy. **The groom** talked about a recent trip to Boston.” is inserted into a structural continuum with the preceding and following context — the informational relation here is cued by syntax, through the definite DP. Charlie Brown

throwing the ball and Linus retrieving it instead of Snoopy also follows a structure — in this case scenic rather than linguistic. In our stimuli, sentence and emoji function as two units of meaning that interact with each other, but which are not structurally knitted together. In that aspect, our experiment is more similar to Weissman and Tanner’s (2018), to the one in Willems, Özyürek, and Hagoort (2008), or to research on iconic gestures.

Weissman and Tanner’s design, however, involved reanalysis. Upon seeing the emoji at the end of the sentence, the participant, provided they interpreted the emoji as being ironic, had to reinterpret the sentence as its opposite. Our stimuli, on the other hand, do not prompt such deep reanalysis, functioning, as argued, more like free adjuncts.

The design we propose, therefore, can shed new light on isolating information effects and their neural correlates.

Since all mentioned studies found an early negativity — except for Tang, Chen, *et al.* (2024), which involved using emoji integrated in syntax, occupying verb position, and Aurnhammer *et al.* (2023), which had a very different design from ours, maximally priming target words and distractors — we can also expect it for our model. Then, because new and partially new information should be more difficult to integrate — and aligning with a view of the P600 as integration — we expect a P600. This also follows the study we mirror (Burkhardt, 2006).

We, therefore, expect an N400 indexing dependency formation (Burkhardt, 2006), prediction (Kuperberg, 2016; Kuperberg; Brothers; Wlotko, 2020), or lexical retrieval (Aurnhammer *et al.*, 2023) — depending on the account — followed by a P600 reflecting integration (Burkhardt, 2006, 2007; Aurnhammer *et al.*, 2023), possibly decomposable into frontal and centro-parietal components (Brothers *et al.*, 2020; Kuperberg; Brothers; Wlotko, 2020). Exploratorily, we could also expect the Nref, which is related to anaphoric ambiguity (Van Berkum, 2009; Nieuwland; Coopmans; Sommers, 2019), and/or the N300, which is sensitive to visual complexity of objects and scenes (Kumar; Federmeier; Beck, 2021), although it is not frequently reported in multimodal research and our model might not be sensitive enough to separate these overlapping negativities.

Indeed, we are also grouping together dependency formation, prediction, or lexical retrieval, because our design does not allow for strong conclusions to untangle them — although lexical retrieval would seem less likely given that we are dealing with emoji that sometimes do not even clearly refer to a single lexical item (see Chapter 6 for a more detailed analysis of this possibility).

If we do not obtain the P600, however, this could indicate that integration effort is not always indexed by this particular Event-Related Potential, and that integration through the P600

might be structure dependent. Alternatively, it could indicate that multimodal integration is fundamentally different.

More concretely, our hypotheses are as follows, for each condition:

- i) *Given*: smallest amplitudes for the N400 and the P600 potentials, reflecting effective semantic access — through prediction, dependency formation or lexical retrieval;
- ii) *Bridged*: reflecting its partially dependent, partially new nature, it should first pattern with the *given* condition, with attenuated N400 amplitudes (but larger than *given*), and then pattern with the *new* condition, with similarly large P600 amplitudes;
- iii) *New*: largest N400 and P600 amplitudes, since it is not dependent on prior context and consists of a new referent that needs to be integrated.

In essence, for the N400: $given < bridged < new$; and for the P600: $given < bridged \approx new$.

To test these hypotheses, we create a sentence–emoji corpus, and conduct two experiments: a norming procedure, to ensure each created stimulus has its intended interpretation, and the main experiment, using electroencephalography and ERP methodology. In the next section, we describe both in detail.

4 CREATING AND VALIDATING THE SENTENCE-EMOJI CORPUS

4.1 CORPUS DESIGN AND CONDITIONS

A total of 450 visual stimuli were constructed, consisting of sentence–emoji pairs distributed across three experimental conditions. Each sentence was repeated exactly three times in the corpus, followed each time by a unique emoji acting as an independent discourse unit (Grosz; Kaiser; Pierini, 2021, 2023). In the *given* condition, the image, by the use of semantic redundancy, was intended to merely refer back to the previous context, adding little to no information; in the *bridged* condition, controlled to have no semantic redundancy, it was designed to trigger a bridging inference; and in the *new* condition, it was meant to have no clear relation to the sentence.

The manipulation, therefore, operates on the degree of informational dependence between the elements. The design is modeled on Burkhardt (2006), whose investigation of definite-DP-mediated bridging established the given–bridging–new framework that the present conditions attempt to mirror.

Consider the following examples:

Item 1:

- a. The event was canceled ✖
- b. The event was canceled 🌧
- c. The event was canceled 🍌¹

In “a”², “✖” — conventionally linked to the concept of negation — would only reinforce the previous sentence about an event cancellation, without adding substantial information. In “b”, “🌧” could be interpreted as the cause of the cancellation, which would represent new information, albeit still dependent on the prior context. In “c”, the relation between the sentence and 🍌 should be harder to establish, depending on greater elaboration.

The use of “would”, “should”, and “could” throughout this description is deliberate. Production, as well as comprehension, is a phenomenon influenced by multiple factors, both cultural and idiosyncratic — what one person reads as a clear causal relation, another may find meaningless. The success in crafting a corpus with the intended gradations, therefore, can only be measured by submitting it to public evaluation.

¹ “O evento foi cancelado”, in Portuguese.

² Throughout the corpus and the scripts used, there is always a correspondence between the letters a, b, and c (sometimes capitalized) and the *given*, *bridged*, and *new* conditions, respectively.

A web-based norming experiment using the PCIBex (2026) platform was designed to verify that each stimulus elicited its intended interpretation. In line with Burkhardt’s (2006) procedure, we aimed to retain 120 of the triads of stimuli most consistently interpreted as expected (40 per condition). These would constitute the stimulus set for the main experiment.

4.2 STIMULUS CONSTRUCTION

Each stimulus was composed of between 4 and 6 words ($M = 4.43$) and between 22 and 25 characters ($M = 23.49$), with one emoji ³ after the final word (Section A.1). Length was kept short for naturalistic presentation: the stimuli should be akin to messages received on platforms such as WhatsApp or Instagram, where we seldom see full texts, but rather sequences of quick messages. It was also kept reasonably constant to minimize variation in working-memory demands in the second experiment, when sentences would be presented word by word (Rapid Serial Visual Presentation — RSVP). For the same reason of naturalistic presentation, we did not use periods to mark the end of the stimuli (see Section 5.4).

No emoji was repeated, resulting in 450 different images being used, comprising various emoji categories ⁴.

Some examples were only inspired by Grosz, Kaiser, and Pierini (2021) and Pierini (2021), while some were more directly adapted to Portuguese, like “The event was canceled 🌂” above⁵. Other examples are (Item 2a) “I admire that man 🧐”⁶, (Item 11a) “I enjoy being at home 🏠”⁷, and (Item 3a) “I trained a lot today 🏋️”⁸. Some constructions with the verbs “admire” and “impress” are also inspired by Grosz, Kaiser, and Pierini’s (2021) analysis of how they can interact with emoji.

The majority of the stimuli, nevertheless, had to be newly crafted.

³ We used Fluent UI Emoji Font for Windows 11. Emoji appearance varies substantially across platforms, so we presented them in image form. The ones that are present in this dissertation were also inserted as images. The full corpus is also available in .png in the referenced repository (Section 1.1).

⁴ Smileys; People; Animals & Nature; Food & Drink; Activity; Travel & Places; Objects; Symbols; and Flags; to cite only the main ones (as available in Emojipedia (2026)).

⁵ “The party will not 🌂 be cancelled tomorrow 🌂” — (Pasternak & Tieu, 2020, as cited in Grosz; Kaiser; Pierini, 2021, p. 13).

⁶ “Eu admiro aquele homem 🧐”, from a tweet analyzed by Grosz, Kaiser, and Pierini (2021, p. 345): “I really admired this man 🧐”.

⁷ “Eu gosto de ficar em casa 🏠”, also from a tweet in Grosz, Kaiser, and Pierini (2021, p. 345): “So glad I stayed home today 🎮👍”.

⁸ “Treinei muito ontem 🏋️”, adapted from recurrent examples in Pierini (2021) such as “John trained today 🏋️” (p. 725).

Based on Grosz, Kaiser, and Pierini’s (2021) observation that face emoji are preferentially interpreted as first-person indexes, i.e., as reflecting speaker stance, we only paired face emoji with sentences that were already in the first person, e.g.: (Item 21b) “I can’t stand winter 🙄”⁹.

The context sentences varied greatly in tense, voice, and complexity, but they were always the same for each condition of their triad. A full syntactic parsing of each of them is available in the referenced repository (Section 1.1).

4.3 DESIGN TRADE-OFFS

Two methodological decisions warrant explicit justification. First, contrary to Burkhardt (2006), the manipulations fell onto the target, not the context.

While manipulating the context rather than the target has the advantage of minimizing the effects of, e.g., the variations in frequency and visual complexity, it also has the great disadvantage that the processing up until the target will be different for each sentence, and in that case controlling the variation among the sentences would be imperative¹⁰. Given that pairing different texts with the same image while trying to reduce the variation among the sentences proved to be less productive than pairing different images with the same text, this was the chosen approach.

Second, no emoji was repeated across the corpus. This was a decision that derived from the one above. Theoretical perfection would be attained if every emoji could appear exactly once, counterbalanced (Latin square) in each condition. In practice, this constraint proved too rigid: attempting to match sentences at this level of combinatorial precision was unproductive, and the sentences it did yield became progressively less natural.

If we could not match every item, we opted to match none. This, on the one hand, carries the advantage of generalization. Many experimental approaches, when analyzing emoji, have restricted themselves to a very narrow set, generally composed of face emoji¹¹. Using multiple categories allows us to investigate the multimodality phenomenon through a wider spectrum. On the other hand, as mentioned above, it does introduce a degree of variability that, even when not strictly controlled, we attempted to address systematically.

⁹ “Eu não suporto o inverno”, in Portuguese.

¹⁰ Having the context unchanged between conditions allowed us, e.g., to do the baseline check described in the next chapter (Section 5.2.2).

¹¹ See, e.g., Gantiva *et al.* (2020), Liao *et al.* (2021), Tang, Chen, *et al.* (2020), Tang, Zhao, *et al.* (2021) and Tang, Chen, *et al.* (2024), and Yuasa, Saito, and Mukawa (2011) (the latter for face emoticons).

The first two questions used in the norming study sought to minimize this problem by evaluating whether each target emoji was visually comprehensible and made sense with the sentence.

4.4 DISCOURSE RELATIONS AND REFERENCE TYPE IN THE *GIVEN* CONDITION

An attempt was made to annotate the discourse relations between emoji and sentence, following the taxonomy in Grosz, Kaiser, and Pierini (2021) (as mentioned in Chapter 2):

- Explanation (event β explains event α): e.g., (Item 10b) “He’s not drinking today 🚗” (“He’s not drinking today because he’s driving”);
- Elaboration (event β is part of event α): e.g., (Item 6b) “The smell is too strong 🧦” (where the socks elaborate on the source of the smell);
- Narration (event β temporally follows event α): e.g., (Item 126b) “I dropped by the food-truck 🌭” (“I dropped by the food-truck and bought a hot dog”);
- Result (event β is a result/consequence of event α): e.g., (Item 16b) “The worker got distracted 🔥” (“The worker got distracted, causing a fire”);
- Background (event β indicates the background where α takes place): e.g., (Item 31b) “The wind is good now 🪁” (said in a context of flying a kite).¹²

As they also observed in their article, however, these categories are not exhaustive and many examples can fit in more than one. In “I’ve decided to cheat on my diet 🍦”, for instance, one could interpret the relation between the ice cream and the sentence as being one of Elaboration: “I’ve decided to cheat on my diet and ice cream is a part of this escapade”; or Background: “I’ve decided to cheat on my diet [said while eating ice cream]”; among others.

We therefore took note of what seemed to us the most salient semantic relation in each case (Section A.3), but did not attempt to control this aspect.

It is also worth returning to a particularity of the *given* condition: how the reference to the previous sentence is made, and to which term it refers. As introduced in Section 3.2, an image can refer back either by iconicity (direct visual resemblance) or by semantic association.

Using only iconicity in the *given* condition — the closest option to what was done with pure text — would be extremely difficult, but we did annotate which stimuli referred back to the sentence through iconicity and which did so through semantic association (Section A.2). We also

¹² In Portuguese: “Ele não vai beber hoje 🚗”; “O cheiro está fortíssimo 🧦”; “Eu passei no food-truck 🌭”; “O operário se distraiu 🔥”; “O vento está bom agora 🪁”.

noted to which elements the reference was made (Section A.4), albeit this point was less clear, as it falls into some of the same complexities as in the interaction between language and gestures¹³:

- (Item 25a) “The food was awful 🤢” — Does the emoji reinforce the negative tone of the whole sentence? Does it refer only to awful?
- (Item 33a) “My head is hurting 🤯” — Does the emoji refer only to head? Or does it encapsulate and reinforce the whole sentence?¹⁴

4.5 PSEUDORANDOMIZATION

To prevent sentences with the same topic from following each other during the experiments — which could cause an effect of narrative continuity between stimuli and influence comprehension — we annotated each sentence and emoji with thematic tags. These tags were then used to feed a Python script for semi-automated pseudorandomization. Examples include [EMO]: emotions; [EVT]: events (parties, games, etc.); [ESP]: sports; and [REL]: interpersonal relations¹⁵.

There were also markers for sentence valence (that were kept simple): [NEU] (neutral), [POS] (positive), and [NEG] (negative). E.g.:

- (Item 39a) “I’ve brought you a present 📺”¹⁶ — tags (sentence): [EVT], [REL], [POS]; tags (emoji): [EVT].

The Python script detected the tags and sorted the stimuli into 10 different counterbalanced lists, each containing 45 stimuli (15 per condition). The thematic emoji tags were defined in the script as priority constraints¹⁷ (at least two intervening stimuli between any two items sharing a strict tag) with a heavy penalty for any unavoidable violation, while the valence ones were defined as “soft”. There was also a secondary flexible constraint, regarding the maximum number of items of the same condition that could appear in a row — no more than three items of the same condition could appear consecutively.

¹³ See Tieu, Schlenker, and Chemla (2019), Pierini (2021).

¹⁴ In Portuguese: “A comida estava terrível 🤢”; “Minha cabeça está doendo 🤯”.

¹⁵ See Section 1.1.

¹⁶ “Eu te trouxe um presente 📺”, in Portuguese.

¹⁷ A silent bug (column name mismatch) prevented the sentence tags from being read by the script. It was corrected in the EEG pseudorandomization and also in the script that is made public. The visual inspection, however, was the safeguard.

The outcome of the automated process was inspected visually to check if there still remained sequences that could be perceived as a narrative continuum. The few cases were manually corrected.

4.6 THE NORMING STUDY

With the lists ready, we designed a web-based norming experiment, using the PCIBex platform (2026), with the objective of verifying that each sentence had its intended interpretation. Participants were presented with three questions to which they could respond through three 7-point Likert scales:

1. “To what extent is the emoji visually identifiable?” (From “Impossible to identify” to “Perfectly identifiable”);
2. “To what extent do the emoji and the sentence make sense together?” (From “They make no sense at all” to “They make perfect sense”);
3. “To what extent does the emoji add information to the sentence?” (From “It does not add any information” to “It adds a lot of information”).

The first question sought to rule out emoji that were difficult to visually decode or unfamiliar, like “🤖” (to which 77.27% of people attributed ratings of 4 or lower).

The second question intended to separate the *given* and *bridged* conditions, which needed to make sense together (albeit for different reasons), from *new*, which was not supposed to make sense at all. Also, because we knew that sentence–emoji pairs in the *bridged* condition contained no semantic redundancy, if a participant rated it as making sense, the coherence would have to be mediated by a non-explicit inferential link: indirect evidence of successful bridging.

Informativeness was nevertheless also prompted by the third question, which sought to further refine *given* from *bridged* by tackling their key difference (information addition). This data would be especially relevant in the analysis of the second experiment.

The questions were also an attempt to minimize the effects of using a varied array of stimuli. By removing the triads most prone to visual error and by probing how strongly the remaining sentences had the desired interpretation, we optimized the corpus as best we could.

4.7 RECRUITMENT AND INCLUSION/EXCLUSION CRITERIA

Recruitment was conducted through link sharing on social media (WhatsApp and Instagram) and by advertisements in the Faculty of Letters of the Federal University of Rio de Janeiro. Inclusion required participants to be between 18 and 40 years old; to have familiarity with emoji, GIFs or other digital imagery; and to provide minimal identification (a valid email).

By limiting the valid participants to the window of 18 to 40 years, we attempted to select a relatively homogeneous demographic while also choosing those who grew up during the internet age, learning its rules implicitly. This was also done having the second experiment in mind, which, being in person, would likely draw participants primarily from the Faculty of Letters, whose student body falls predominantly within this range.

The participants, while filling out their information, also had to report their familiarity with emoji: “I use and receive them daily”; “I use and receive them a few times a week”; “I use and receive them sometimes”; “I rarely or never use them, but I receive them daily”; “I rarely or never use them, but I receive them a few times a week”; “I rarely or never use them, but I receive them sometimes”; and “I never receive or use them”.

Initially, only those who answered “I use and receive them daily” and “I use and receive them a few times a week” were to be kept (see Section 4.12.1 below). This was done because the participants had to be familiar with decoding sentence–emoji pairs, so they could be reliable evaluators. Also, previous research with visual narratives has mentioned the effects of “visual literacy” on inference generation (Cohn and Kutas (2015); Nakazawa (2005) as cited therein).

Providing minimal identification was established with the reasoning that it would discourage uncommitted participants (such as internet trolls).

4.8 PARTICIPANTS

In total, 246 people had completed the experiment when the first analysis was processed (see Section 4.11 below), of whom 230 were considered valid after applying the exclusion criteria¹⁸. Statistics for age: $M = 24.03$, $SD = 4.73$, $\min = 18$, $\max = 40$; for gender: 162 Female (70.43%), 61 Male (26.52%), 7 Other (3.04%); and for education: 34 Complete High School (14.78%), 74 Complete Undergraduate Studies (32.17%), 1 Incomplete Lower Education (0.43%), 121 Incomplete Undergraduate Studies (52.61%).

¹⁸ Section B.1 shows their distribution per list.

Figure 5 – Likert 2 (Coherence).

Brazil was defeated 🇧🇷

To what extent do the emoji and the sentence make sense together?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
 They make They make
 no sense at all perfect sense

Figure 6 – Likert 3 (Information Addition).

Brazil was defeated 🇧🇷

To what extent does the emoji add information to the sentence?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
 It doesn't add It adds a lot
 any information of information

An explanation and a training section preceded the experiment and people could navigate freely during the first one. This was designed mainly to clarify what was meant by the questions the participants had to answer. While the question about whether emoji and sentence made sense together was straightforward, the other two merited more context.

Because visual clarity was not our primary focus, but was only being used as a control measure, we attempted to be very clear about what was intended in the first question (“To what

extent is the emoji visually identifiable?”). We used “🍏” to illustrate the case where the emoji depicts the object it refers to in a reasonably typical way, which most people would rate 7 or close to 7 (perfectly identifiable), and “🌀” as an example of one that would probably generate confusion and be rated close to 1 (impossible to identify).

Regarding the third question, we were faced with a seeming contradiction, as it concerned the core point of our general analysis. On the one hand, “To what extent does the emoji add information to the sentence?” could be very metacognitive if left unexplained. On the other, explaining how to identify something we were submitting to the public with the very objective of assessing their natural interpretation seems like circular reasoning.

However, even if people knew all the design rules (i.e., that the *given* condition items are supposed to add little to no information, while *bridged* items are supposed to be informative), there would still be variation as to how to apply them. “The event was canceled 🍌” to an astronomer could instantly make connections with astronomy-related events arise, and because people were also prompted to answer with what first came to mind, the astronomer would likely be inclined to rate closer to “they make perfect sense” and to “it adds a lot of information”.

We therefore decided to keep it as minimal as possible, with one representative example for the *given* condition (“I took my car to the mechanic 🔧”), one for *bridged* (“I cheered a lot for him today 🍌”), and none for *new*. We did not say that either of the examples was a perfect 7 or 1, but only that most people would probably gravitate toward those ends in each case.

In this manner, we sought to “calibrate” and show what was meant by “it doesn’t add any information” and “it adds a lot of information”, but not to restrain the possibilities by showing what nonsense should look like.

Note that the example for the *given* condition used semantic association (a screwdriver as a symbol for mechanic), and not iconicity. This was done with the explicit objective of preventing people from forming the strategy of “iconicity equals one in the Likert scale, everything else rates higher”.

Finally, for the second scale (“To what extent does the emoji add information to the sentence?”), the instructions were limited to a reminder that some pairs can make more sense together than others and that this perception will vary person to person. No examples were given.

4.10 PRACTICE PHASE

The practice phase was designed only to conclude the instructions and get participants used to the experiment and how the questions and Likert scales would appear on the screen. The sentences that appeared for the practice were not used in the main experiment. They were: “I’ve left the door open 🚪” and “I didn’t finish the project 🧶”¹⁹.

Crucially, multiple answers were possible. Even during setup, there was a certain degree of divergence: although we had built “I’ve left the door open 🚪” as *given*, with the emoji only referring back to “door”, someone remarked that the emoji is working opposite to the context, as the door is actually shown closed.

In the same manner, with “I didn’t finish the project 🧶”, some could have understood “knitting project”, while others could have not figured out the emoji at all or what it represented in the sentence.

No feedback was given, as we did not want to risk further influencing the answers.

The mean answers in Table 2 show that there was indeed variation as early as this point:

Table 2 – Practice phase averages and standard deviations.

Training	Scale	Mean	SD
Door	Likert 1 (Visual Clarity)	6.65	0.84
Door	Likert 2 (Coherence)	6.05	1.58
Door	Likert 3 (Information Addition)	2.92	2.42
Wool	Likert 1 (Visual Clarity)	6.25	1.48
Wool	Likert 2 (Coherence)	4.84	2.20
Wool	Likert 3 (Information Addition)	5.34	2.28

¹⁹ In Portuguese, respectively: “Deixei a porta aberta 🚪” and “Não terminei o projeto 🧶”

4.11 RESULTS AND DISCUSSION

Two separate analyses of the PCIBex data were conducted²⁰: one shortly after sufficient data were collected, which culminated in the selection of the stimuli for the second experiment; and one post hoc, done while verifying data for usage in the statistical analysis of the second experiment’s results.

The reanalysis was deemed necessary in view of new participants completing the experiment before it was taken offline (and so a reanalysis could improve the robustness of the results), and of the need to address certain methodological limitations. These new results did not ultimately influence the selection of the stimuli, but they are important in order to accurately characterize what we did obtain from the PCIBex experiment, and they did influence subsequent data treatment. To avoid confusion, the first analysis is referred to as “stimulus selection”; the second as “post hoc”.

All mentions regarding the PCIBex experiment in further chapters refer to the post hoc analysis, unless otherwise specified.

4.11.1 Stimulus selection

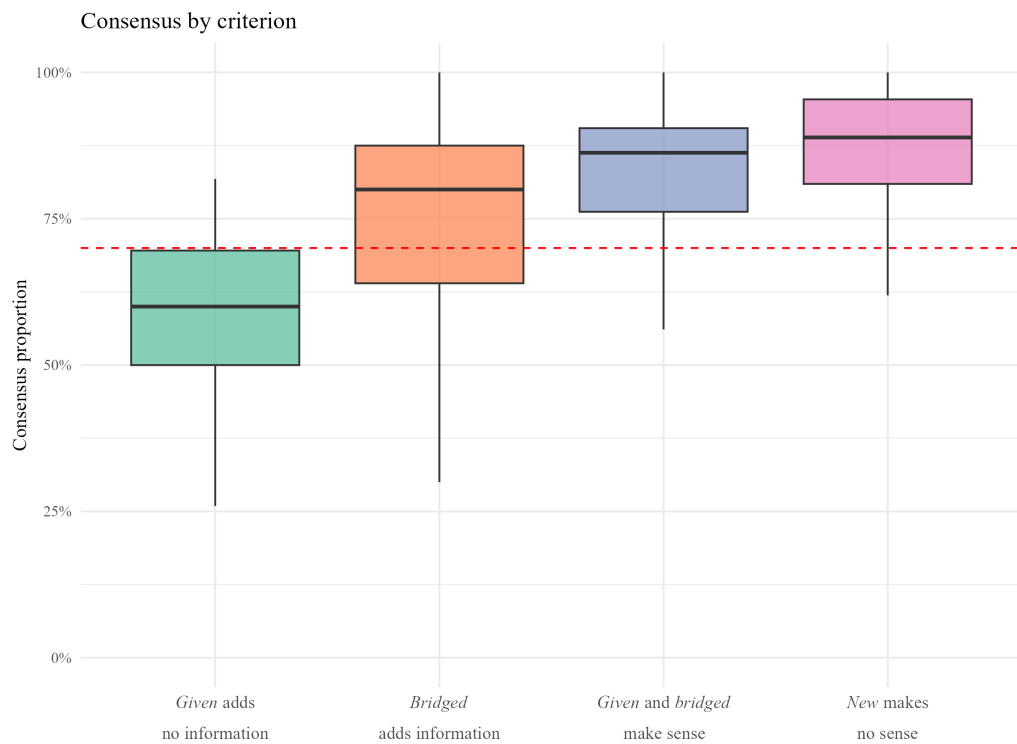
For each condition, we analyzed the Likert rating percentages for the questions that most define them: the *given* condition had to be high on “making sense”, but not “adding information”; the *bridged* condition had to be high on both “making sense” and “adding information”; and the *new* condition had to be low on “making sense”.

We had initially set the reasonable threshold of 70% of agreement for scores greater than 4 for the *given* and *bridged* conditions, and less than 3 for the *new* condition in the coherence scale; and scores less than 4 for the *given* condition, and greater than 4 for the *bridged* condition on the information scale.

A primary caveat, however, emerged: participants were less in agreement regarding the *given* condition adding no information, making it difficult to adhere to such strict rules. Figure 7 presents the consensus by criterion (“*given* adds no information”; “*bridged* adds information”; “*given* and *bridged* make sense”; “*new* makes no sense”). The dashed line indicates the threshold of 70%:

²⁰ Both analyses were conducted in R (R Core Team, 2026) using the tidyverse (Wickham; Averick, *et al.*, 2019) for data processing and visualization, readxl (Wickham; Bryan, 2025) for data import, and kableExtra (Zhu, 2024) for the tables in this dissertation.

Figure 7 – Consensus by criterion. The dashed red line indicates the target threshold of 70%.

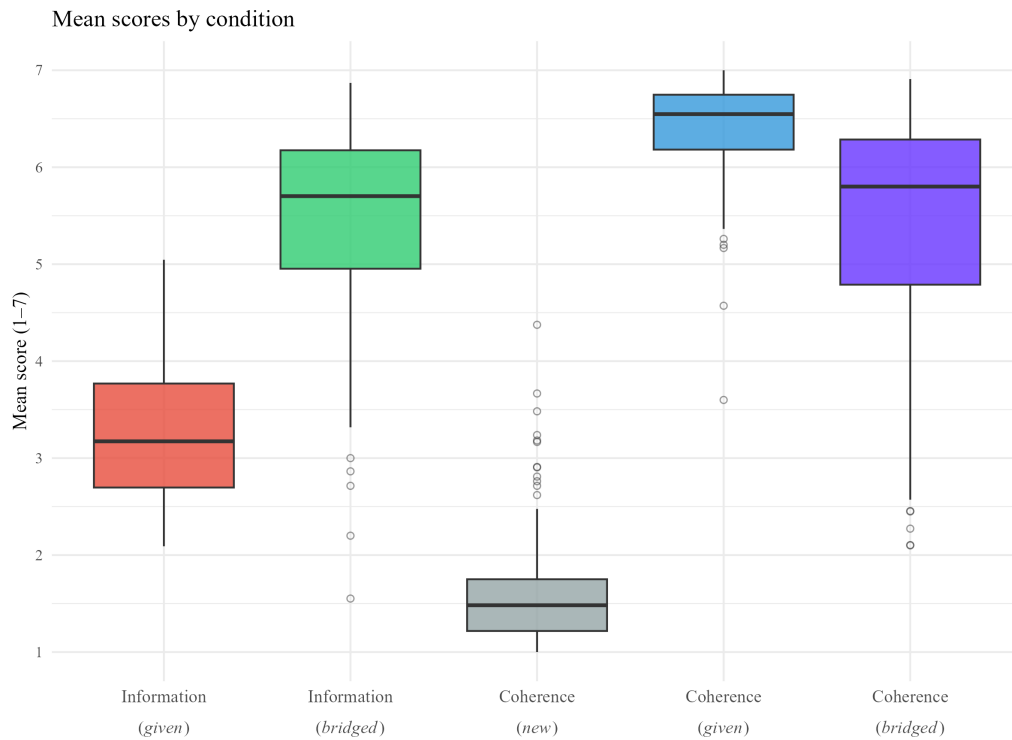


This greater variability was expected and can be attributed to at least two reasons. The first, as noted above, is the fact that the Likert scale for information addition could potentially be too metacognitive. We were, after all, very conservative in the instructions, and information addition is an abstract concept to evaluate.

The second is that participants might have been treating subtle pragmatic or emotional nuance as information in itself. This is corroborated by the mean scores (Figure 8 below). On average, the *given* condition pairs are perceived as significantly less informative than the *bridged* ones, yet participants are reluctant to give them a rating of 1 (even in cases of iconic representation), while they readily attribute “no sense at all” to *new* stimuli. The mean for *given* hovers around 2 (the item with the lowest average for the *given* condition in this analysis is item 76, with 2.09²¹):

²¹ “I ran out of the building 🏢” — “Saí correndo do prédio”, in Portuguese.

Figure 8 – Mean scores by condition.



Our strategy, then, was to rely on the results for the coherence question (“To what extent do the emoji and the sentence make sense together?”), for which there was much more agreement.

As mentioned above, being coherent in the absence of a clear relation would provide evidence of bridging, albeit indirect. Other studies on bridging, such as Burkhardt (2006, 2007), only control their stimuli indirectly, by measuring relatedness between the words used.

Consensus regarding the coherence of the *given* and *bridged* conditions, and the lack thereof in the *new* condition, was therefore used to select the stimuli. Agreement concerning the informativeness of *bridged*, or its absence in *given*, was considered only secondarily. Furthermore, we used the mean difference in informativeness scores between *given* and *bridged* (termed “Information Contrast”) to rank the valid items from most to least optimal.

Two additional experimental caveats warrant explicit attention. First, there is the fact that the items are not independent, but were judged independently: there was no way of separating the conditions of a triad. If, for instance, *given* and *bridged* items were rated solidly as expected, but the rejection of the meaning in *new* items or their visual clarity was very low, the whole triad had to be eliminated. In order to incorporate a candidate item into our selected stimulus set, the item had to be judged as expected in all three conditions.

Second, the results of the Likert scales themselves are not completely independent. Not making sense, for example, impacts emoji identification and vice versa. Take the aforementioned emoji “🐚”. One would be much more inclined to say that 🐚 is identifiable in “I found something by the sea 🐚” than in “I went to work by bike 🐚”. At a minimum, the category becomes clearer (something one would find by the sea) when the emoji is not detached from context. Conversely, even though the meaning becomes clearer in the first case, participants would still probably be hesitant to give it higher ratings. This co-dependence between visual decoding and context is known to neuroscience²². The *new* condition, therefore, suffered more from low ratings of visual clarity than the other two.

Not making sense and visual clarity also influence the category of adding information. The questions were presented in the same order for each stimulus and no instructions were given about what to do in the third scale if an emoji was deemed unidentifiable or nonsensical — people had to form their own strategies. It seems nevertheless reasonable that many thought that what is not identifiable cannot make sense or add information, and that what does not make sense cannot be too informative (*new* averages in this regard were indeed around one).

Table 3 – Pearson correlations among the three norming scales.

	Visual clarity	Coherence	Information
Visual clarity	1.00	0.31	0.20
Coherence	0.31	1.00	0.58
Information	0.20	0.58	1.00

The Pearson correlations (Table 3) demonstrate this mutual influence clearly²³.

Given these limitations, we needed a selection mechanism that operated at the level of triads rather than judging items in isolation. We therefore decided to treat the sense of *given* and *bridged* as a single category, while visual clarity ratings were averaged per item, and triads for which more than 30% of ratings were 4 or lower were excluded (if a person rated 4, it would indicate visual ambiguity).

²² Bar (2004), for example, demonstrates the neural underpinnings of contextual scene perception.

²³ This shows the correlations of the three scales at the triad level, across all responses — it is different from Table 8, which shows the correlations among the three rERP *predictors*, i.e., the by-item mean ratings of the 120 items used.

With visual clarity addressed and *given* and *bridged* treated as a single category, we applied different thresholds and tested how many items were left in each case, finally setting the reduced threshold of 60% of agreement that *given* and *bridged* make sense (scores greater than 4) and that *new* makes no sense (scores less than 3) — relaxing the threshold of consensus while keeping the rating criteria themselves intact.

Following this practice, 115 sentences were automatically selected and categorized into three tiers of quality. Unsurprisingly, the best-rated item was this triad:

Item 7:

- a. Brazil was defeated 🇧🇷
- b. Brazil was defeated ⚽
- c. Brazil was defeated 🍒²⁴

To achieve the desired 120 triads, five items that were only marginally below the threshold were selected and added to the third tier: Items 6; 61; 2; 19; and 66 (Section B.2 shows details, including the reasons why some gap numbers were not selected).

The full obtained list is made public (Section 1.1) and constituted the list used for the construction of the second experiment.

4.12 POST HOC ANALYSIS

The post hoc evaluation revisited three points, in order: participant inclusion/exclusion criteria; the threshold parameters used to determine coherence; and limitations in the first analysis concerning 1) the consequences of grouping the sense of *given* and *bridged*; 2) the consequences of grouping visual clarity; 3) the reproducibility of the script used for selection.

The reanalysis did not substantially alter the stimulus selection: a core of 112 stimuli remains unchanged between the two analyses. It does, however, characterize our results more faithfully.

At the end of this section (Section 4.12.5), we show an inferential statistical analysis of the whole 150 set of triads.

4.12.1 Participant inclusion/exclusion criteria

First, we became aware that while the filtering for age was working properly in the first analysis, the filter for emoji familiarity had been applied unevenly: only excluding those who

²⁴ “O Brasil foi derrotado”, in Portuguese.

answered “I use and receive them sometimes” and “I rarely or never use them, but I receive them daily”. Also, the filter for valid identification was only really applied to one person, rendering it of minimal impact.

This made us reconsider the exclusion parameters as a whole.

Although it was justified that participants should have minimal familiarity with emoji and other internet media, keeping only those who answered that they “use and receive them daily” or “use and receive them a few times a week” would essentially make valid only two of seven different answers participants could give regarding the frequency of usage. This was too conservative, especially when we consider that the age window was already conceived to select people who grew up during the internet age and therefore reduce confounds of unfamiliarity with internet media.

Additionally, the phrasing of the question and answers presented a small interpretation gap: the question was “do you have familiarity with emoji/stickers/GIFs?”, separating “GIFs” from “stickers”, which is not always a clear distinction for people. The answers did not specify whether they were talking about GIFs, stickers, or emoji, referring to the whole as “them”: a participant who, e.g., has a lot of contact with internet GIFs, but never uses them in conversation in the form of moving stickers, would probably answer low on the scale and be removed.

We therefore decided to filter in the reanalysis only those who said they “never receive or use” emoji (that is, no one), and to remove the valid identification filter. This produced the new valid N of 241 people ²⁵ (from a total of 248).

4.12.2 Coherence judgments

The coherence judgments were also revised following the same logic.

We had initially required ratings greater than 4 for making sense in *given* and *bridged*, and less than 3 for rejecting the sense in *new*. The asymmetry was viewed as appropriate because higher ratings for *new* are more harmful to the design: if it gets 4, for example, it means that it is perceived as ambiguous, which would approximate it to *bridged*. Also, people were less hesitant concerning the *new* condition, and so we could be more strict to ensure the control condition’s integrity.

²⁵ Revised statistics for age: $M = 24$, $SD = 4.83$, $\min = 18$, $\max = 40$; for gender: 168 Female (69.71%), 66 Male (27.39%), 7 Other (2.9%); and for education: 36 Complete High School (14.94%), 77 Complete Undergraduate Studies (31.95%), 1 Incomplete Lower Education (0.41%), 127 Incomplete Undergraduate Studies (52.7%). Section B.1 shows per list distribution.

However, when we established that making sense is obtained only when participants rate higher than 4, we excluded the middle ground. This was motivated by a desire to select only the very best stimuli, but it was also too conservative. If a sentence–emoji pair is being rated a 4 out of 7, it does not automatically mean it does not make sense, but rather the person sees something in the pair that prevents them from confidently judging either way. This point was therefore loosened to include greater than 3 ratings for *given* and *bridged*, while holding *new* at ratings lower than 3. Information addition measurements were left untouched.

Figure 9 and Figure 10 show the updated consensus and mean scores under the revised criteria.

Figure 9 – Revised consensus measures including additional participants and corrected criteria, cf. Figure 7 and Figure 8.

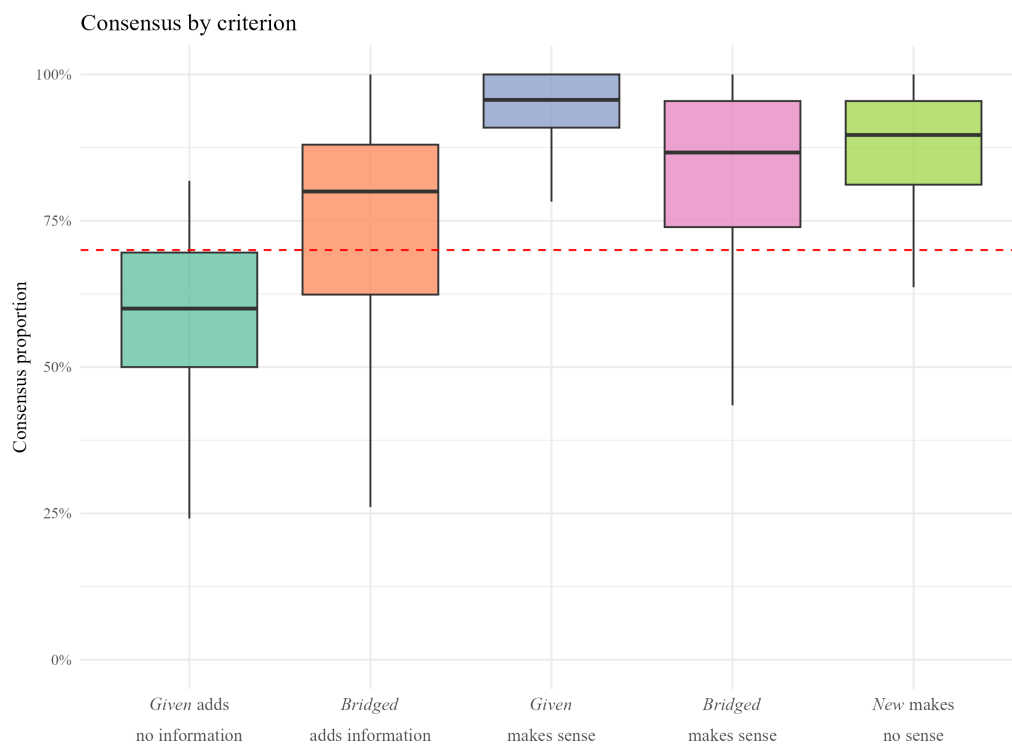
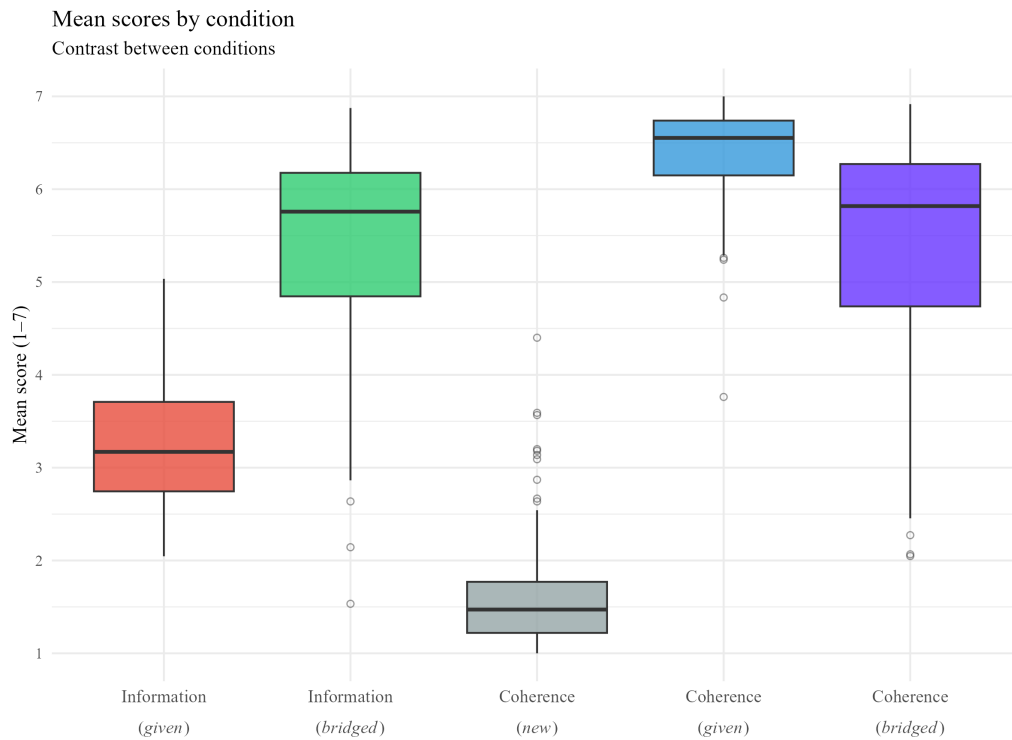


Figure 10 – Revised averages including additional participants and corrected criteria, cf. Figure 7 and Figure 8.



4.12.3 Acknowledged limitations in the selection of the stimuli

4.12.3.1 Grouping the sense of *given* and *bridged*

Figure 10, splitting the sense of *given* from the sense of *bridged*, illustrates the central finding. While at the time it appeared to be the best methodological approach given the constraints observed above, grouping *given* and *bridged* together had the effect of masking variance in a few cases, with one inflating the other. This concerns some of the items that were automatically selected (62, 102, and 149), but mostly regards the ones which were manually chosen to complete the set of 120.

Applying the revised criteria for coherence and demanding agreement of more than 60% for *given* and *bridged* independently, 112 items are now automatically selected, and not 115. Items 62, 102, and 149 are left out. Items 102 and 149 remain within margin, with 54.55% and 59.09% of agreement for meaning in *bridged*, respectively. Item 62, however, alongside 2 and 61, becomes suboptimal — only 47.62% agreement for the sense in *bridged* in items 2 and 62, and only 42.86% agreement for meaning in *given* in item 61:

Item 2:

- a. I admire that man 🧑 (83.33% agree that it makes sense)
- b. I admire that man 🦋 (47.62% agree that it makes sense)
- c. I admire that man 📦 (86.36% agree that it does not make sense)

Item 62:

- a. I've left the pan uncovered 🍲 (100.00% agree that it makes sense)
- b. I've left the pan uncovered 🦋 (47.62% agree that it makes sense)
- c. I've left the pan uncovered 🍷 (100.00% agree that it does not make sense)

Item 61:

- a. The sea is too quiet 🤖 (42.86% agree that it makes sense)
- b. The sea is too quiet 🧑 (81.82% agree that it makes sense)
- c. The sea is too quiet 🏠 (95.45% agree that it does not make sense)²⁶

In this new light, at least one of those stimuli would not have been chosen (item 58, for example, now presents itself as a better choice).

To address this, we ran two categorical analyses for the EEG data: one including 2, 61 and 62, and one without them. The first one is presented in the Results section; the other is presented in the appendix (Section C.2.2). It was, however, not necessary to do so with the continuous analysis, as discussed in Section 5.1.5.

4.12.3.2 Grouping visual clarity

Averaging visual clarity at the triad level (i.e., averaging for a score per item) had similar consequences: it was lenient toward low clarity in the *new* condition when its *given* and *bridged* counterparts were otherwise clear. In 10 cases, more than 50% of the *new* item's ratings were low-clarity (≤ 4), despite the triad's average remaining below the 30% threshold. E.g. (the full list is in Section B.3):

Item 102:

- a. I crashed the car on the trip 🌟 (19.05% of ratings ≤ 4 regarding visual clarity)

²⁶ In Portuguese, respectively: “Eu admiro aquele homem”; “Esqueci a panela aberta”; “O mar está muito quieto”.

- b. I crashed the car on the trip 🚗 (4.55% of ratings ≤ 4 regarding visual clarity)
- c. I crashed the car on the trip 🚗 (54.55% of ratings ≤ 4 regarding visual clarity)²⁷

Given the interaction between identification and interpretation discussed above, however, as well as the minimal error in the corresponding *given* and *bridged* conditions, averaging visual clarity still was a necessary trade-off. Separating the conditions for the purposes of visual-clarity screening would have broken the triad-level design and hindered selection. Also, as we will see, the EEG experiment used a magnified emoji image, which further helped reduce visual ambiguity.

Visual clarity was also inserted as a predictor in the rERP analysis, with results showing that it did not individually influence the obtained waveforms.

4.12.4 Reproducibility

Finally, regarding code reproducibility: “Semantic Contrast” (the mean difference in coherence between conditions *given* and *bridged* relative to the *new* condition) was added as a secondary sorting parameter. This addition prevents algorithmic ties; otherwise, the *dplyr* package would default to sorting by item ID whenever items shared identical consensus and Information Contrast scores.

4.12.5 Inferential validation of the conditions

Regardless of the selection-level caveats discussed above, the conditions themselves are strongly distinct. After revising the criteria, we used the post hoc results to fit a cumulative link mixed model to each Likert scale, treating data as ordinal. We fitted it over the full 150 set of triads, rather than the 120 selected ones. This probes the validity of the whole experiment construction, while diminishing the circularity of statistically testing items that were already selected based on their difference. To the extent that the norming procedure was an experiment in its own right, it warrants its own inferential analysis.

As Table 4 illustrates, both *given* and *bridged* stimuli were judged more coherent than *new*, with *given* rated reliably more coherent than *bridged* (*given* > *bridged* > *new*); *bridged* stimuli were judged more informative than *given* and *new* (*bridged* > *given* > *new*); and visual clarity was highest for *given*, followed closely by *bridged* and then *new*.

²⁷ “Eu bati o carro na viagem”, in Portuguese.

Table 4 – Condition contrasts per norming scale (cumulative link mixed models, scale treated as ordinal; Bonferroni-corrected).

Scale	Contrast	Estimate	SE	z	p	$\chi^2(2)$	LRT p
Visual clarity	<i>given – bridged</i>	0.50	0.075	6.62	< 1e-04	978.7	< 0.001
Visual clarity	<i>given – new</i>	1.98	0.072	27.41	< 1e-04	978.7	< 0.001
Visual clarity	<i>bridged – new</i>	1.49	0.066	22.61	< 1e-04	978.7	< 0.001
Coherence	<i>given – bridged</i>	1.36	0.054	25.13	< 1e-04	9135.2	< 0.001
Coherence	<i>given – new</i>	5.63	0.079	71.67	< 1e-04	9135.2	< 0.001
Coherence	<i>bridged – new</i>	4.28	0.067	63.84	< 1e-04	9135.2	< 0.001
Information	<i>given – bridged</i>	-1.89	0.049	-38.99	< 1e-04	4840.4	< 0.001
Information	<i>given – new</i>	1.71	0.053	32.23	< 1e-04	4840.4	< 0.001
Information	<i>bridged – new</i>	3.61	0.059	60.68	< 1e-04	4840.4	< 0.001

A numeric mixed model, lmer, treating the scale as continuous, yields similar ordering and significance (Table 5), confirming the overall soundness of the corpus. It also shows that the results do not depend on the ordinal treatment. This matters especially because we treated the data as continuous in the rERP analysis (Section 5.3)

Table 5 – Condition contrasts per norming scale (linear mixed models, scale treated as numeric; Bonferroni-corrected).

Scale	Contrast	Estimate	SE	z	p
Visual clarity	<i>given – bridged</i>	0.14	0.028	5.08	< 1e-04
Visual clarity	<i>given – new</i>	0.89	0.028	31.77	< 1e-04
Visual clarity	<i>bridged – new</i>	0.75	0.028	26.73	< 1e-04
Coherence	<i>given – bridged</i>	0.93	0.035	26.86	< 1e-04
Coherence	<i>given – new</i>	4.80	0.035	137.98	< 1e-04
Coherence	<i>bridged – new</i>	3.86	0.035	111.31	< 1e-04
Information	<i>given – bridged</i>	-2.17	0.045	-48.02	< 1e-04
Information	<i>given – new</i>	1.59	0.045	35.06	< 1e-04
Information	<i>bridged – new</i>	3.76	0.045	83.12	< 1e-04

5 THE NEURAL CORRELATES OF MULTIMODAL BRIDGING

The norming experiment produced a selected corpus of 120 sentence–emoji pairs, organized into three experimental conditions where the emoji accompanied the sentence as an independent discourse unit. In the *given* condition, the emoji simply referred back to the prior context, with little to no information addition; in the *bridged* condition, it triggered a bridging inference; and in the *new* condition, it had no clear relation to the sentence. E.g.:

Item 140:

- a. I like having company 😊
- b. I like having company 🐶
- c. I like having company 🚗

The second experiment, using this corpus, compared the neural responses to the different conditions, seeking to investigate the neural correlates of bridging and the dynamics of information processing in the brain. Consistent with previous research, we expected a negativity in the N400 window (graded as *given* < *bridged* < *new*) possibly accompanied by overlapping anterior components such as the N300 or the Nref, followed by a positive shift (*given* < *bridged* $\approx$ *new*).

5.1 MATERIALS

Because we used Rapid Serial Visual Presentation (RSVP), i.e., the selected stimuli were broken into units and presented in sequence to the participants while they had their neural activity recorded, there was a methodological issue to be addressed. With stimuli following each other without any sort of interruption, participants would be quick to notice that certain sentence–emoji pairs demanded more effort to interpret, and also that they should expect all emoji to be placed in the final position.

The first could potentially incentivize reasonings along the lines of “some stimuli are nonsensical, I will not even try” — which would be reflected in the recordings and hinder our objectives, given that we wanted to analyze how interpretation unfolds. The latter could generate a confound of preparation for the change of modality.

In order to prevent participants from forming interpretation strategies based on the experimental design, as well as to discourage positional expectations regarding the emoji, we therefore added 120 fillers to the selected stimuli. They consisted of sentences where the emoji replaced one or more phrase elements:

Fillers:

- F_2) We saw a 🐘 in the zoo
- F_3) The 🕯️ was blown out by the wind
- F_4) She loves 📺 of mystery

This format was chosen because, despite using emoji and text, it does so in a fundamentally different relation to the one used in the targets. Also, it serves the purpose of distracting people from the experimental design without establishing a harsh contrast between filler and target that could induce surprisal effects or other confounds (as would happen if sentences with no emoji were used).

The replaced element was not systematically chosen, and it varied from the first up to the last in the sentence. Sentences had between 22 and 28 characters ($M = 23.51$), and 4 and 7 words ($M = 5.41$), counting the emoji. More variation in size was allowed, but we tried to keep them in a length comparable to the stimuli, following the same rule that it should look like a short message received through social media. They also covered many different subjects and had the same tags from the PCIBex experiment applied to them. This would feed a new pseudorandomization procedure, described below (Section 5.1.1).

Originally, 160 fillers had been created, but after a pilot experiment that ran too long, 120 were deemed to be sufficient. This received informal confirmation via participants' responses after each session. Most of them commented that they thought the experiment was about the fillers and about emoji substitution rather than bridging, also remarking that they frequently thought they had to “mentally pronounce” the words to understand the message.

This brings us to the fact that emoji substitution is an interesting phenomenon in itself (see Tang, Chen, *et al.* (2024) or Cohn, Roijackers, *et al.* (2018)). Unfortunately, our design allows for limited findings in that regard, as every pseudorandomized list used the same set of fillers.

It does allow, however, for some tentative investigations regarding differences in processing depending on which sentence element is substituted, on what is the sentence theme, or the differences in processing homonyms versus non-homonyms (see Scheffler *et al.* (2022)), for instance:

- F_144) Fiquei de 🛌 em casa hoje (“I 🛌 at home today”) — to be read “I ‘lazier’ at home today”, where the “🛌” is a homonym with “preguiça” (“lazyness”) in Portuguese.

Future research should explore these relations.

5.1.1 Pseudorandomization

Whereas the PCIBex experiment used 10 different counterbalanced lists, each with 45 items, the EEG experiment used only three, each containing 120 stimuli plus 120 fillers. This was done because of the time-consuming nature of EEG experimentation, which would not allow for as many participants as we used in the web-based test, as well as the number of observations required per condition, per participant. We therefore had to maximize the number of items to which each participant was exposed.

The lists were pseudorandomized using a Python script in much the same manner as in the PCIBex norming test, and with the same intent, i.e., to prevent stimuli from forming a narrative continuum that could influence interpretation in non-predictable ways. The difference was that the script had to account for the fillers, and that here both sentence and emoji thematic tags constrained the ordering — excluding the most frequent object, animal, nature, and consumable categories, which were too common to space and also varied enough to not form a narrative.

Both targets and fillers shared thematic tags. Those were then read by the script and distributed among three different lists. Unlike the previous iteration, this one allowed for a shorter interval between stimuli bearing the same thematic tags (at least one intervening item was required between strict-tag matches). This was done both because there was greater variability across themes and so narrative continuity was rarer (the fillers were much easier to construct and to vary), and because keeping the same distance would yield no valid outputs. There was also a new constraint limiting consecutive items of the same type (target or filler) to three, while the cap on consecutive items of the same condition was relaxed from three to four (with the same reasoning of making valid outcomes possible); runs of fillers were tolerated.

Fillers did not contain valence tags; those were already a soft constraint in the other pseudorandomization and were deemed unnecessary here given the other constraints and the variability.

The output was again visually inspected and corrected manually in the few cases where narrative continuity was still perceived despite the shuffling. Three pseudorandomized lists were obtained. To counterbalance effects of fatigue, we used those three lists plus their reversed counterparts in the stimulus presentation, for a total of 6 lists (4 is the reverse of 3, 5 of 2, 6 of 1).

5.1.2 Participants

Thirty native Portuguese speakers¹ from the student body of the Federal University of Rio de Janeiro (UFRJ) participated, newly recruited through advertisements in the Faculty corridors, social media, and official university channels. None had taken part in the previous PCIBex norming experiment. All reported familiarity with emoji, GIFs, and other digital imagery and were within the same age range as in the first experiment (18–40): $M = 24.03$, $SD = 4.4$, $min = 19$, $max = 36$. Twenty-seven were right-handed, two left-handed, and one ambidextrous. All had normal or corrected-to-normal vision and reported no language-related disorders or difficulties.

While it is typical practice to control for handedness in neurocognitive studies, based on empirical data, we understand that language lateralization only correlates relatively weakly with handedness, such that we chose to accept an incidental number of left-handers (2) and ambidextrous (1) people in our participant group. As justification, we may cite Szaflarski *et al.* (2002), who found 22% of non-typical lateralization — including bilateral and right hemisphere — for left-handed participants vs 4-6% for right-handed participants. Mazoyer *et al.* (2014) cite “typical” dominant left hemisphere activation for language for 88% of right-handed participants and 78% of left-handed participants, while a study by Packheiser *et al.* (2020) concluded that “only 0.4% of the variance in language lateralization were explained by handedness” (p. 1).

One participant (uncounted, see Section 5.2.1) was excluded during setup because it was impossible to establish minimal impedance values (due to an ill-fitting cap); another (Participant 3) was excluded during pre-processing due to too many artifacts in the signal. This yielded a final sample of 28 subjects.

5.1.3 Procedure

The experiment was conducted in the Syntactic Access Lab (Acesin), room D103 of the Faculty of Letters of the UFRJ. Participants sat comfortably in a chair while they viewed the stimuli through RSVP and had their neurophysiological responses recorded through active electroencephalogram electrodes placed with the aid of a cap. They were all instructed to read for comprehension, as if they had received the messages on WhatsApp or seen them on X or other social media.

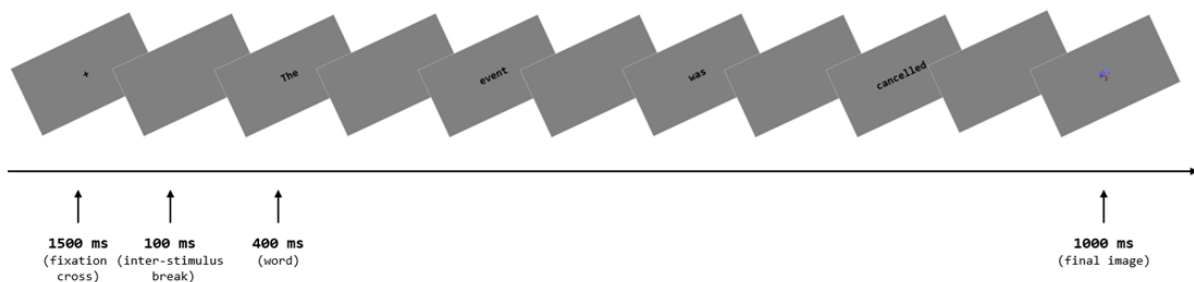
¹ Participants self-identified their gender as: 16 Female (53.33%), 13 Male (43.33%), 1 Non-binary (3.33%).

Stimuli were presented on a 16-inch monitor with 165 Hz refresh rate against a medium gray (128, 128, 128) canvas ■. White would risk introducing pupil dilation confounds every time the screen refreshed; black blended with some images that were partially black; other colors would give an unnatural look.

Text was programmed (E-Prime) to appear word by word in black at the center of the screen in Consolas bold size 40. Image size was kept at 275×227 pixels. Not so big as to make participants' eyes move, not so small as to not be identified given monitor distance — participants sat at approximately 90 cm from it. Magnifying the emoji also served the purpose of countering the discussed visual ambiguity of certain *new* condition stimuli. Fluent UI Emoji Font for Windows 11 was used (made available in the referenced repository (Section 1.1) in image form to avoid differences in rendering across platforms).

A fixation cross preceded each stimulus. All non-final sentence elements were presented for 400 ms with a 100 ms inter-stimulus interval (gray canvas), including mid-sentence filler images; all final items, including words in some fillers, were shown for 1000 ms. This prevented participants from associating longer presentation times solely with final images, which could again have exposed experimental design. Figure 11 displays the procedure:

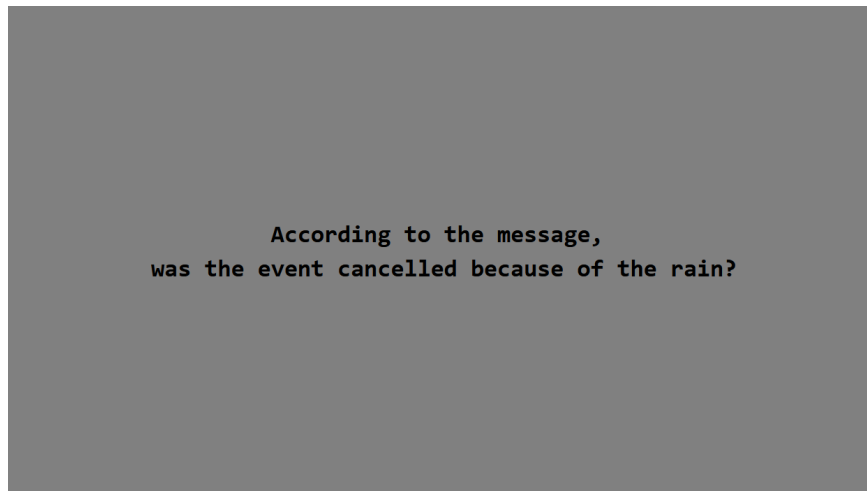
Figure 11 – Stimuli presentation



In order to prevent effects of anticipation regarding the beginning of each trial, a variable inter-trial-interval (ITI) followed the final element of each trial (a fixed sequence ranging from 875 ms to 1200 ms²). It also ensured that any lingering processing of the last stimulus would not be time-locked and would therefore average out. A comprehension question followed the jittered interval in 25% of trials to further incentivize reading for comprehension (Figure 12):

² The exact sequence: 952 ms, 1000 ms, 1123 ms, 875 ms, 1200 ms.

Figure 12 – The question panel. It stayed on the screen until participants answered “Yes” (blue) or “No” (red) using Chronos.



The experiment lasted for approximately 30 minutes and included one pause every 40 stimuli (for a total of 6), with a bigger interval proposed mid-experiment (after 120 stimuli).

Instructions were kept minimal and concerned only experiment structure and essential recommendations for proper data collection (e.g., avoiding surprisal effects on the signal, avoiding movement artifacts). They were as follows:

Slide 1

In this experiment you will see a sequence of sentences with emoji presented in serial form while your neural activity is measured by EEG equipment.

Slide 2

The elements will appear in the center of the screen. A fixation cross will momentarily appear before the first element of each sentence, so you know where to look.

Slide 3

Randomly, after some sentences, a comprehension question will appear, alongside the answer options: YES and NO. If you want to answer YES, press the blue button. If you want to answer NO, press the red button.

Slide 4

During the experiment, uncross your legs and fingers and avoid moving, including blinking. Try to blink between sentences, when the fixation point appears on the screen. We will have pauses so you can rest.

A practice phase followed, consisting of the sentence “I did a quick stop 🛑” followed by the question “Was the stop long?”. No feedback was provided, and the real trials started after a second confirmation screen that the participant dismissed when ready by pressing any button.

Participants used a Chronos device (Psychology Software Tools) to navigate the instructions and to answer the comprehension questions. The button responses (blue at the far left, red at the far right) were not counterbalanced, as the brain responses to them were not central to our analysis.

Comprehension questions were shown after approximately 25% of the trials (61).

They could probe either the emoji:

- The woman almost fainted 🤯 — “Did the message show a parrot?”

The sentence:

- She sleeps with the lights on 💡 — “Does the message say she likes sleeping early?”

Or the conjoint meaning:

- She saw a 🦄 in the cartoon (filler) — “According to the message, did she see a fantastic creature?”
- The event was canceled 🌧️ (*bridged*) — “According to the message, was the event canceled because of the rain?”

When it covered the conjoint meaning, it had the secondary purpose of verifying that participants were indeed making the inference in the *bridged* condition. Other research (see, e.g., Nieuwland, Coopmans, and Sommers (2019)) on anaphora resolution has more explicitly probed this, by having participants say if a person or object had been referred to or not. As we were not disclosing experimental design, and we did not want to influence natural interpretation too heavily, we preferred to probe this only incidentally. From the 31 questions for *bridged* items, only 13 were of this sort, distributed across the lists (four in lists 1 and 6 and in lists 2 and 5, and five in lists 3 and 4; see Section C.1 and the referenced repository (Section 1.1) for the full list). “According to the message” was added to avert participants from drawing the answers from their own conclusions (also inspired in Burkhardt (2006)).

We also tried to counterbalance the questions across the stimuli. The expected responses, however, were not counterbalanced, due to the difficulty of establishing varying responses for the same question throughout three different lists. A 500 ms blank screen followed each question before the cross reappeared marking the beginning of the next trial.

All electrodes were connected to an actiCHamp amplifier (model 5001, Rev. 2, Brain Products GmbH, Gilching, Germany)⁴. Preparation lasted around 30 minutes, and the whole procedure lasted around an hour. Most recordings were done under natural light, except for three recorded at night with the lights off, illuminated only by the monitor (participants 10, 15, and 29).

The experiment was programmed using E-Prime (Psychology Software Tools, 2016). Upon the presentation of every target or filler image, the Chronos device sent a corresponding event trigger to the continuous EEG recording, acquired via PyCorder V1.0.9. The sampling frequency was 500 Hz and the data were not filtered online except for hardware anti-aliasing (250 Hz, half the sampling rate).

5.1.4 Preprocessing

MNE (Gramfort *et al.*, 2013) was used for analysis. For preprocessing, we closely followed Aaron Newman’s procedure (2026), adapting it where needed.

First, offline filters were applied to the raw data to attenuate artifacts outside the range of interest — highpass: 0.1 Hz, lowpass: 30 Hz.

Then, artifact handling proceeded in two complementary stages: Independent Component Analysis (ICA) to remove ocular components, and Autoreject (applied twice — once solely to flag noisy data for ICA, once to clean the final epochs) to handle remaining artifacts.

We used the experiment parameters to align (using an automated script) the EEG data with the E-Prime sequence and obtain the exact windows of active experimental blocks — excluding pauses and initial setup. This was done because if Autoreject were applied over the whole raw data, including pauses where participants were encouraged to move, speak, or rest, the analysis could potentially be skewed. Intervals were therefore marked as “BAD” (a marking that makes MNE automatically ignore these data segments). The alignment also served to attach item metadata — so the rERP analysis would be possible.

Because ICA performs best on segmented data with low-frequency drift removed, we applied a second highpass filter at 1 Hz (leaving the lowpass unchanged) to the marked data and divided it into one-second epochs.

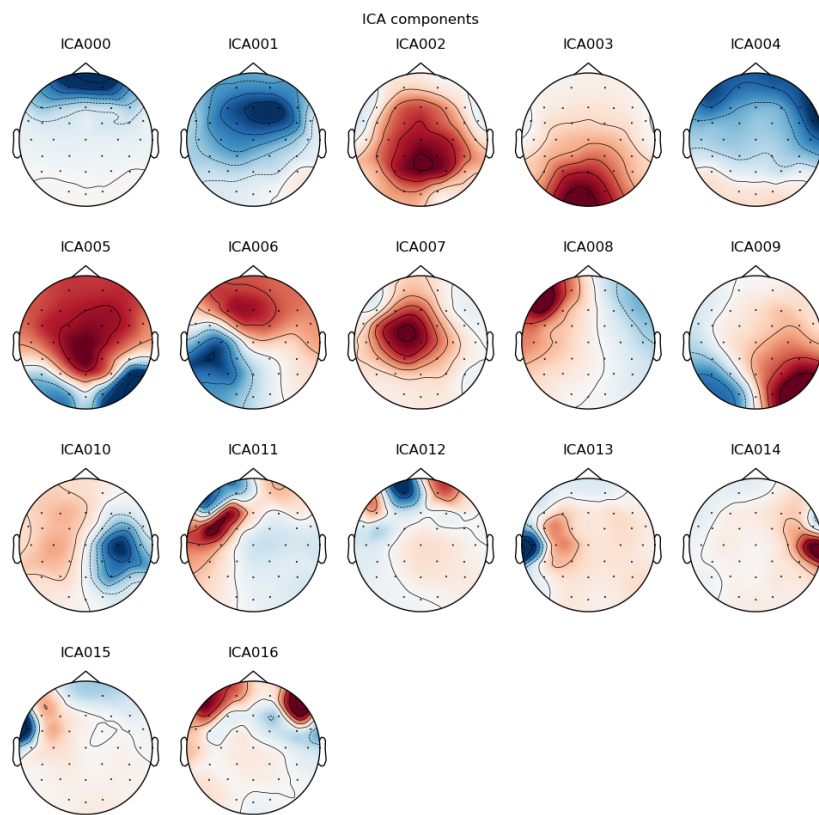
Autoreject was applied to further mark (“BAD”) any channels and epochs that exceeded automatically calculated thresholds, categorizing them as noise; it also marked channels for interpolation when possible. These data were then used to fit the ICA, which identifies independent

⁴ Firmware versions: DLL_25.14.08.04, DRV_03.04.01.146, CTRL_05.16.02.09, FPGA_44.00.00.00, DSP_06.11.05.25.

components in the remaining signal after ignoring the “BAD” epochs flagged both manually (by the aligning script) and by Autoreject.

Without a limit, ICA would derive as many components as there are channels, which, as Newman remarks, is rarely useful. We therefore set the threshold to 0.99, retaining only as many components as were needed to account for 99% of the signal’s variance. For reproducibility we fixed `random_state = 42`, so that the same outcome would be produced on every run (see Section 6.1).

Figure 14 – ICA for Participant 2.

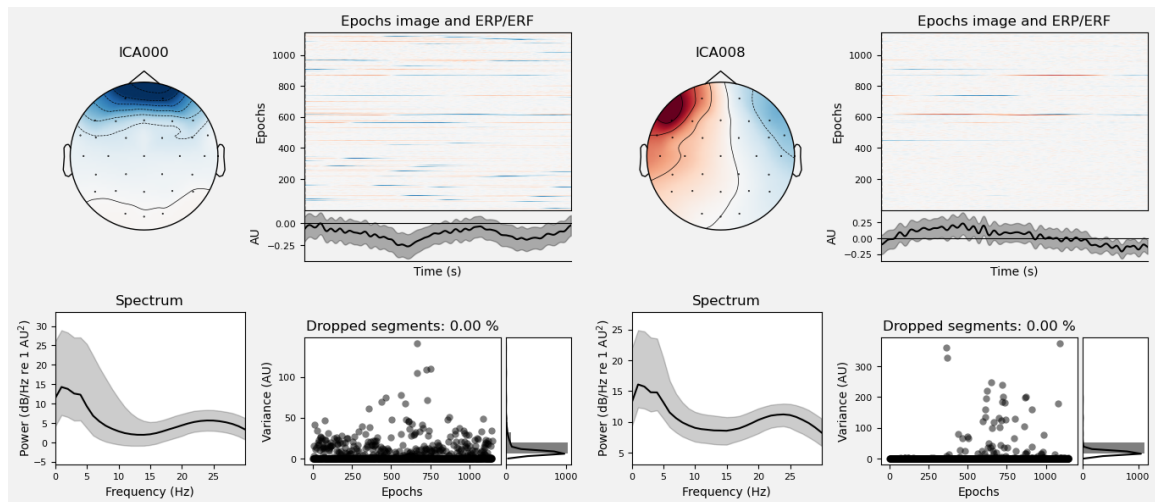


After fitting the ICA, we ran the automatic exclusion routine, which iteratively lowers a correlation threshold (from 3.5, in increments of 0.05) until at least two frontally distributed components are flagged, expected to be the blink and horizontal-eye-movement components.

Because dedicated EOG electrodes were not used, frontal channels (Fp1, Fp2, F7, F8) were used as proxies, as they are situated closest to the eyes. This led, in a few cases (participants 4, 18, 19, and 25), to the removal of more than two components. Visual inspection showed that the additional flagged components correlated with other frontally distributed artifacts.

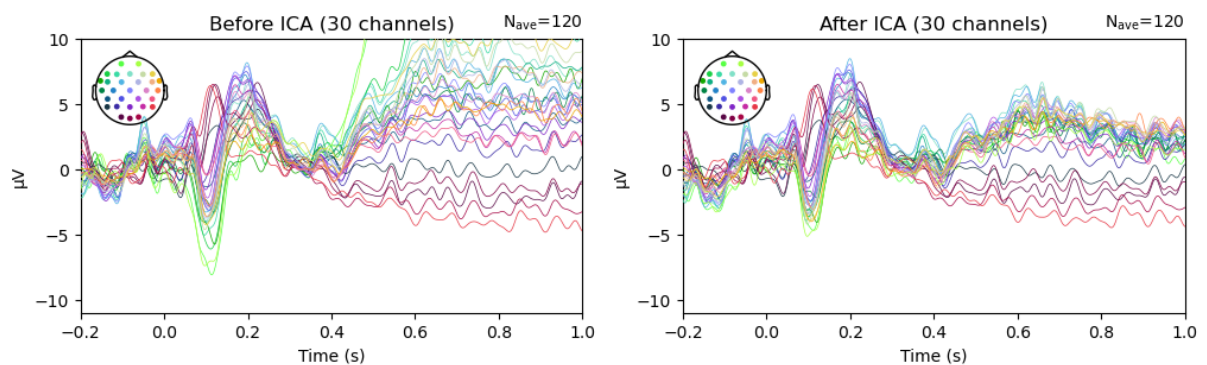
The ICA solution was then saved to be applied to the first filtered dataset (highpass 0.1 Hz, lowpass 30 Hz). For Participant 2, e.g., the eliminated ICs were numbers 000 and 008, as Figure 15 shows.

Figure 15 – ICA components removed from Participant 2's data. The detailed view shows a raster plot with rapid streaks and a topoplot showing greater activity near the eyes, characteristic of rapid eye movements or blinks.



Returning to the initially filtered data, the triggers relating to the events of interest were found, and the data were divided into epochs constructed around them, extending from -0.2 s to 1 s, as the ERPs of interest generally peak between ~ 250 – 350 ms (early negativities such as the N300 or the Nref), ~ 300 – 500 ms (N400) and ~ 500 – 1000 ms (P600). After epoching, we applied the ICA corrections to them, as Figure 16 shows:

Figure 16 – Before and after applying ICA



Autoreject was then applied one more time. Unlike the initial pass — where data were merely annotated as “BAD” to ensure the ICA was fitted only to the clean signal — this final application rejected any remaining artifact-contaminated epochs and performed the necessary spatial interpolations, i.e., substitutions of bad channels by the signal captured by their neighbors. Figure 17 shows Autoreject effects for one participant.

Figure 17 – Autoreject example from Participant 2.

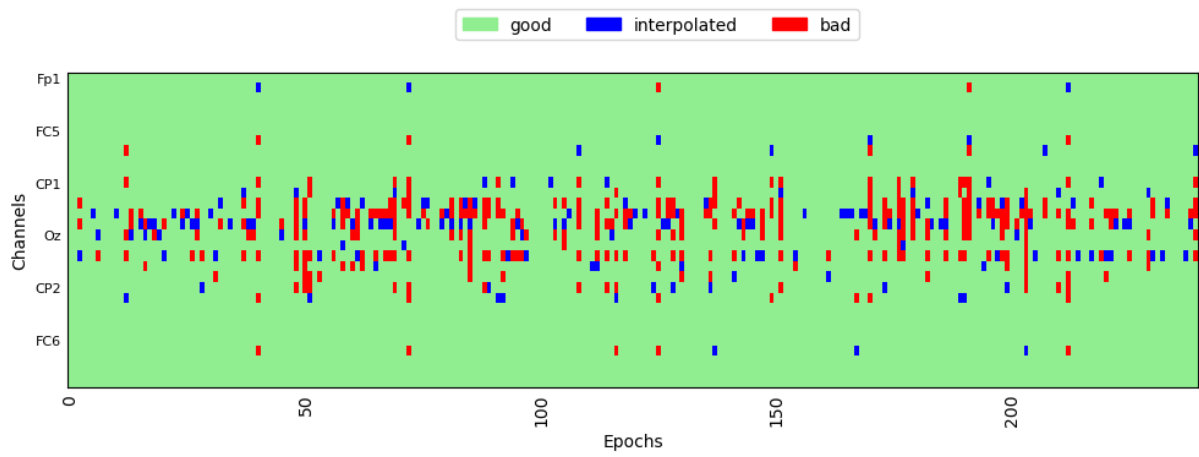
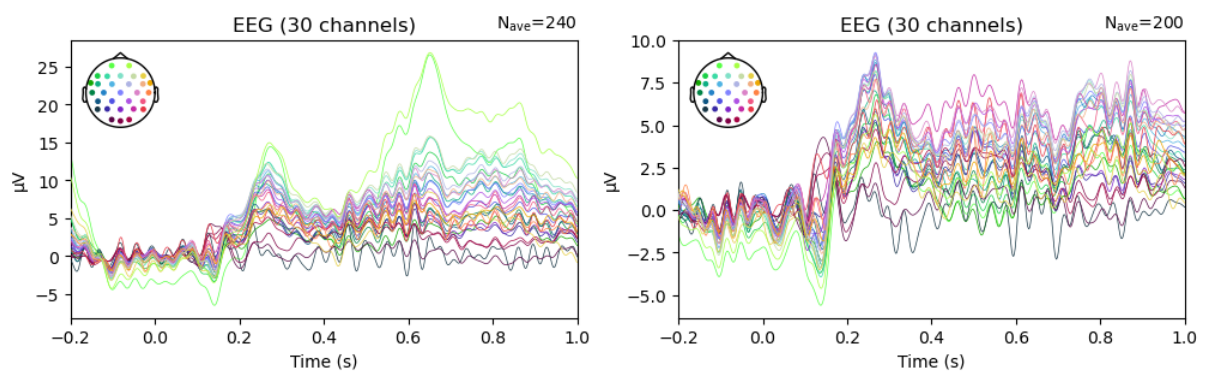


Figure 18 shows the effect on the signal:

Figure 18 – Final Autoreject applied



Recording quality varied among participants, but Autoreject and ICA were sufficient to recover usable data whenever frequent eye movements or elevated impedances were present. Participant 3, however, had to be removed. The signal-to-noise ratio improves by the square root

of the number of trials. At 3 trials for the *bridged* condition, 4 for the *new* condition, and 5 for the *given* condition, too few trials remained to reliably suppress noise.

Statistics for the retained trials after the exclusion are presented in Table 7:

Table 7 – Trial retention per condition after excluding Participant 3.

Condition	Mean (before exclusion)	SD (before exclusion)	Mean (after exclusion)	SD (after exclusion)
<i>Given</i>	34.38	7.78	35.43	5.45
<i>Bridged</i>	34.52	7.87	35.64	5.12
<i>New</i>	34.69	7.64	35.79	4.95

Finally, the surviving epochs were averaged within each participant to produce one evoked response per condition — *given*, *bridged*, and *new*. These were then combined into grand averages across participants and compared through difference waves.⁵ Because we had already used both mastoids as online references, we did not rereference.

5.1.5 Analysis

As we were following Newman’s procedure, Spatio-Temporal Cluster-Based Permutation presented itself as the natural analytical path. It approaches the problem of multiple comparisons rather elegantly, by using the maximal statistic method (through permutations), and clustering. Also, given that it tests a defined contrast specified by prior hypothesis, it lies at the ideal midway point between exploratory and hypothesis-driven analysis (Maris; Oostenveld, 2007).

The procedure conducts multiple *t*-tests on the data from each individual electrode while permuting the labels. If there is no significant difference between the compared elements, then permuting the labels will not affect the statistical distribution — if there is, however, there will be significantly higher values for *t* using the correct labels.

Furthermore, it acknowledges that any given electrical activity is rarely registered by a single electrode or constrained to a single millisecond, resolving this through clustering significant data points into spatio-temporal clusters. This mitigates the risk of inflating the false positive rate by comparing a single phenomenon multiple times.

⁵ Additional Python packages used: NumPy (Harris *et al.*, 2020), SciPy (Virtanen *et al.*, 2020), Pandas (McKinney, 2010), and Matplotlib (Hunter, 2007)

The only caveat is that the method cannot reliably indicate component topography and latency (Sassenhagen; Draschkow, 2019), as it only shows the activated significant clusters. To work around this, it has to be interpreted alongside ERP plots and topoplots.

Our analysis applied 50,000 permutations on the data, using channel adjacency derived from the montage, with a cluster-defining threshold at the default and clusters considered significant at $p < .05$. Three contrasts were tested: *bridged – given*, *new – given*, and *new – bridged*.

However, this test would show only if there was a categorical difference in processing between the conditions, but the PCIBex norming experiment had also provided exactly the type of continuous data⁶ we would need to further explore any categorical differences we would observe. So we also conducted a second analysis, using regression-based Event-Related Potential methodology (rERP; Smith and Kutas (2015a,b)).

The rERP⁷ framework involves computing a separate regression model for each participant at each electrode and time sample (Aurnhammer *et al.* (2023)), and then comparing within-participants. It allows us to observe the effect of parameters set as predictors, in our case: Information Addition — the mean information scores from the PCIBex experiment (Likert 3); Coherence — the mean values for “making sense” (Likert 2); and Visual Clarity — the means for Visual Clarity (Likert 1).

The first analysis would therefore demonstrate whether there is a difference in the amplitudes between conditions; the second, whether this difference varies continuously (i.e., correlates) with the ratings for Information Addition, Coherence, or is rather explained by Visual Clarity.

The coefficients were estimated by ordinary least squares (OLS). Our regression model can be expressed by the formula: $y = \beta_0 + \beta_1 \cdot \text{Coherence} + \beta_2 \cdot \text{Info} + \beta_3 \cdot \text{VClarity} + \varepsilon$, where, for a given electrode and time point, y is the EEG amplitude, β is the vector of the coefficients estimated at that electrode and time point (β_0 being the intercept), and ε is the residual. All predictors were z -scored⁸. The multiple comparisons problem is here corrected through the Benjamini-Hochberg FDR (False Discovery Rate) correction. Analysis was done by fitting a model per participant and then by pooling all participants’ data as if from a single participant (following Aurnhammer *et al.* (2023)).

We adapted the rERP analysis pipeline from Aurnhammer *et al.* (2023).

⁶ As mentioned in Section 4.12.5, for the purposes of this analysis, we treated ordinal data from the Likert scales as continuous, as treating it both as ordinal and continuous data yielded similar results.

⁷ Conventional ERP averaging is a special case of rERP: with a single categorical predictor coded per condition, the rERP estimates reduce to the by-condition averages. rERP is better suited for continuous data.

⁸ Normalized by the standard deviation.

As reported above, in Section 4.12.3.1, the post hoc analysis showed that items 2, 61 and 62, which had been selected after the norming experiment, were suboptimal. We therefore ran the Spatio-Temporal Cluster-Based Permutation test twice — once on the full set and once excluding items 2, 61, and 62. The pattern of significance was roughly the same for the *new* – *given* negativity ($p = 0.002$), and the *new* – *bridged* contrast continued to be nonsignificant (lowest $p = 0.11$). The bridged – *given* distinction, however, fell almost exactly at the threshold of significance we had established ($p = 0.053$). Given that this result lies at the margin, and that, as discussed below, at least item 2b could be justified for retention, we opted to report and discuss the results for the full item set in the main text, choosing to privilege trial retention over strict stimulus purity. The results obtained upon excluding items 2, 61, and 62 are reported in the appendix (Section C.2.2), and the complete alternative pipeline is available in the referenced repository (Section 1.1).

The rERP, by contrast, regresses on each item’s continuous ratings rather than its categorical label, so a marginal item contributes at its measured value rather than distorting a condition average; those items were therefore retained there.

5.2 RESULTS

5.2.1 Comprehension questions

We have included Participant 3, as their E-prime answers were recorded (they were only removed from the experiment in preprocessing). The uncounted participant removed for the ill-fitting cap, however, was also excluded here.

No participant performed below 85% accuracy ($M = 94\%$, $\min = 85\%$, $SD = 3\%$) in the comprehension questions, and so none was removed based on this criterion.

There were also no apparent differences in accuracy across conditions, with correct responses distributed as *given* — 279/299 (20 mistakes); *bridged* — 276/302 (26 mistakes); *new* — 278/298 (20 mistakes). Out of 870 filler responses, 832 were correct (38 mistakes).

The aforementioned suboptimal item 2b — “I admire that man 🤘” — had 90% correct responses to its question, namely “Does the author say they admire the man’s musical abilities?”. As the question probes a conjoint interpretation of text and emoji, this allows us to infer that the bridging was successful in this case.

On the other end of the spectrum, “She’s going to get dressed 🎩” had only 30% correct responses to the question “Does the message specify that she will wear fancy clothes?”. However, the drop in accuracy could reflect ambiguities in the question wording — “message”, here, could be read as referring only to the text, and “specify” reinforces that interpretation — rather than failure to establish a bridging connection.

As pointed out above, the questions were not strictly controlled (some were more difficult than others). Any other result regarding them would therefore be only exploratory.

5.2.2 Baseline check

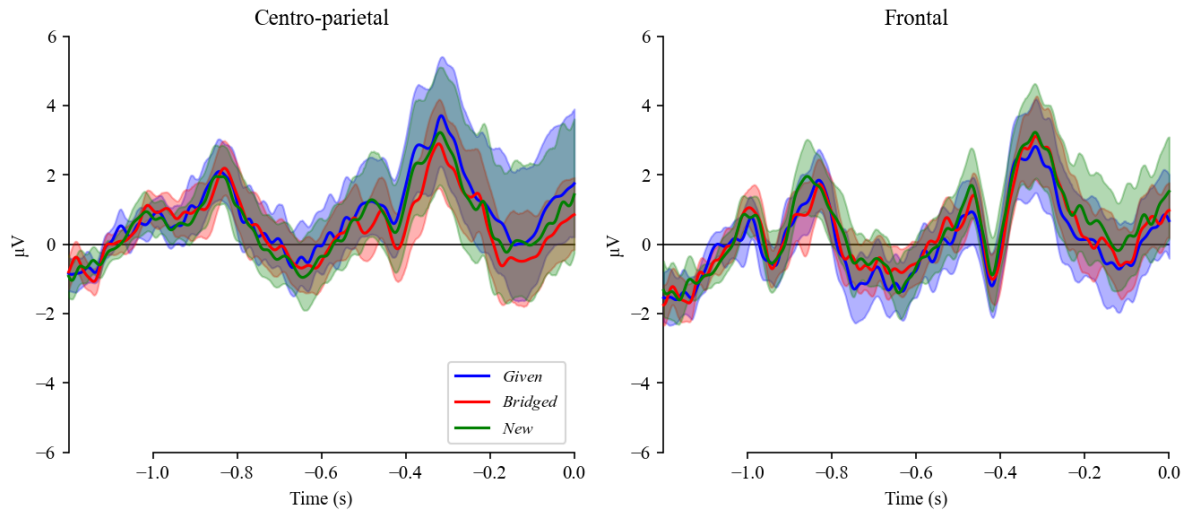
A baseline verification was conducted comparing ERP responses to the sentences up until the moment the emoji was shown. Because they were exactly the same, there should not be any relevant differences in their processing. Confirming that would add robustness to our main analysis.

We resegmented the continuous EEG signal from 1.2 seconds before the trigger — which marks the beginning of the emoji presentation —, constituting new epochs; preprocessing was otherwise identical. Not all targets had the same number of words, but no pre-emoji text lasted less than 1 second; it therefore seemed a reasonable window to test (400 ms per word, plus 100 ms of interstimulus interval: two words totaling 1000 ms before the emoji).

To maintain consistency with the main analysis, Participant 3 was left out, although they did preserve more epochs here, after ICA and Autoreject: 20 for the *given* pre-condition, 13 for the *bridged* pre-condition, and 15 for the *new* pre-condition.

Figure 19 shows the expected alignment of all conditions before the final image.

Figure 19 – Pre-emoji grand-average ERPs by condition at centro-parietal and frontal ROIs, with 95% confidence intervals. *Given* is shown in blue, *bridged* in red, and *new* in green. On the x -axis, 0.0 is the trigger (image onset); these epochs span 1.2 s before emoji appearance. The -1.2 to -1.0 s interval was used for baseline correction, so the inspectable window runs from -1.0 to 0.0 s. Shaded areas indicate the confidence intervals.



To be certain, we also ran the Spatio-Temporal Cluster-Based Permutation test, with no significant differences found between the conditions. It is in the appendix to this dissertation (Section C.2.1) and also made public via the referenced repository (Section 1.1).

5.2.3 Observed ERPs

We used the centro-parietal (C3, Cz, C4, P3, Pz, P4) and frontal (F3, Fz, F4, Fp1, Fp2) electrodes as regions of interest (ROIs) for the plots below, following our hypotheses regarding the time windows of the N400, the P600, and early negativities (such as the N300 or the Nref), as well as their canonical distributions. Figure 20 shows the grand-averaged ERPs by condition at those sites. Negative voltage is plotted upward to match convention.

Visual inspection reveals a distinct peak in early processing (~ 150 ms) where all conditions overlap. Then, there is a negativity peaking at around 400 ms, visible both in frontal and centro-parietal electrodes, stronger for the *bridged* and *new* conditions. After that, the conditions shift to more positive waves and seem to merge again, with no apparent differences in later time windows for the frontal electrodes, and a small difference for centro-parietal. Interestingly, this positive shift persists throughout the duration of the image presentation (see Section 5.4); also, the *given* condition seems to be more positive than the others.

Figure 20 – *Given* is shown in blue, while *bridged* is red and *new* is green. The trigger (image onset time) is indicated by the dashed line at 0.0 s. The interval between -0.2 and 0.0 s was used for baseline correction, so the inspectable window runs from 0.0 to 1.0 s. Shaded areas indicate the confidence intervals.

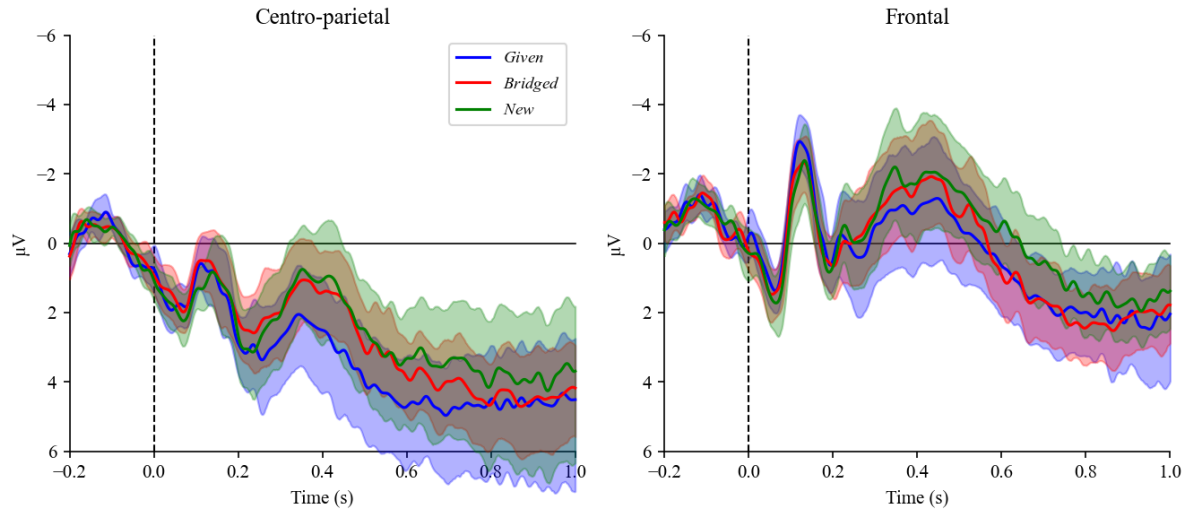


Figure 21 presents the voltage scalp distribution across time through topoplots. Here, the three conditions show a positivity that starts as soon as the image appears (0.0 s) and increases steadily, first at posterior, then at central regions.

Figure 21 – Grand-average scalp distributions by condition across time. Red is positive, blue is negative. $N = 28$.

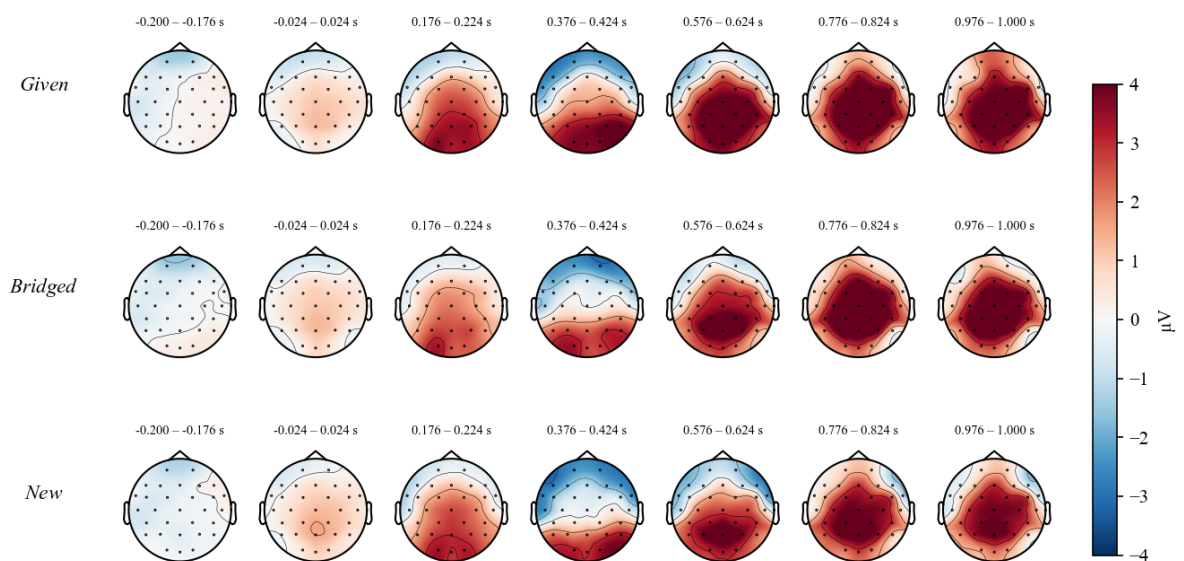
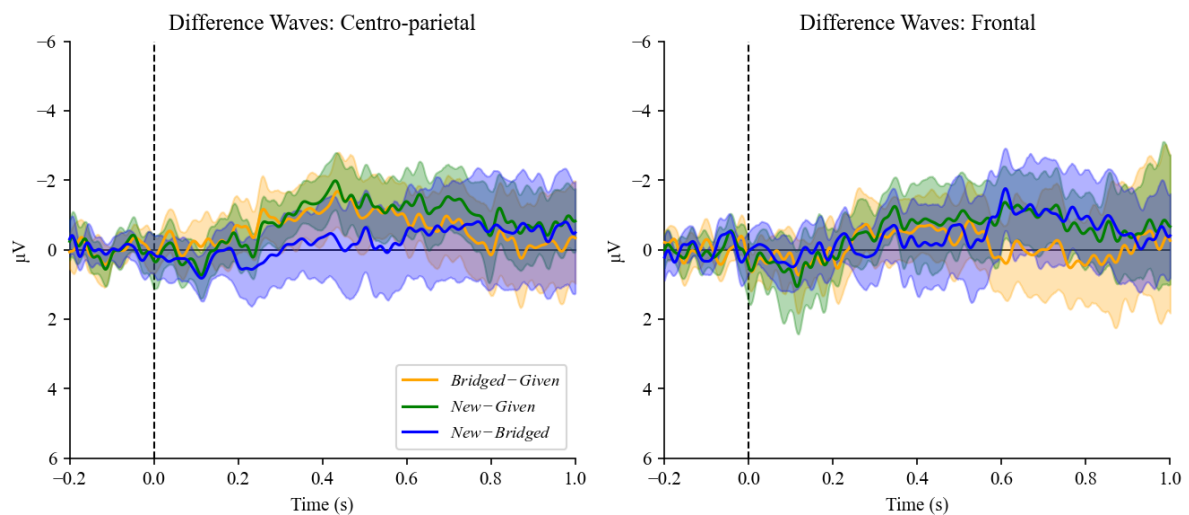


Figure 22 shows the difference waves, i.e., the subtraction between the averaged signal obtained by all the electrodes within a Region of Interest — each average takes the shape of a waveform, which is then subtracted in all its combinations. It reveals that the difference wave between *new* and *bridged*, in blue, runs very close to 0 in the centro-parietal electrodes, while both the difference between *bridged* and *given* (in orange) and between *new* and *given* (in green) take the form of a negative deflection that peaks at around 400 ms. In the frontal electrodes, *bridged* – *given* loses contrast (runs closer to 0), while a negative difference between *new* and *bridged* appears to emerge in later windows, running close to *new* – *given*; the distinction is, however, less clear.

Figure 22 – Difference waves (*bridged* – *given*, *new* – *given*, *new* – *bridged*) at centro-parietal and frontal ROIs, with 95% confidence intervals.



Now that we have presented the visual inspection of the data, focusing on frontal and parietal ROIs, we proceed with the statistical analysis. As we have mentioned, it is an analysis guided by previously established contrasts based on hypotheses (*bridged* – *given*, *new* – *given*, *new* – *bridged*), but it also takes into account all channels and timepoints — it is therefore partially exploratory.

5.2.4 Spatio-Temporal Cluster-Based Permutation Test

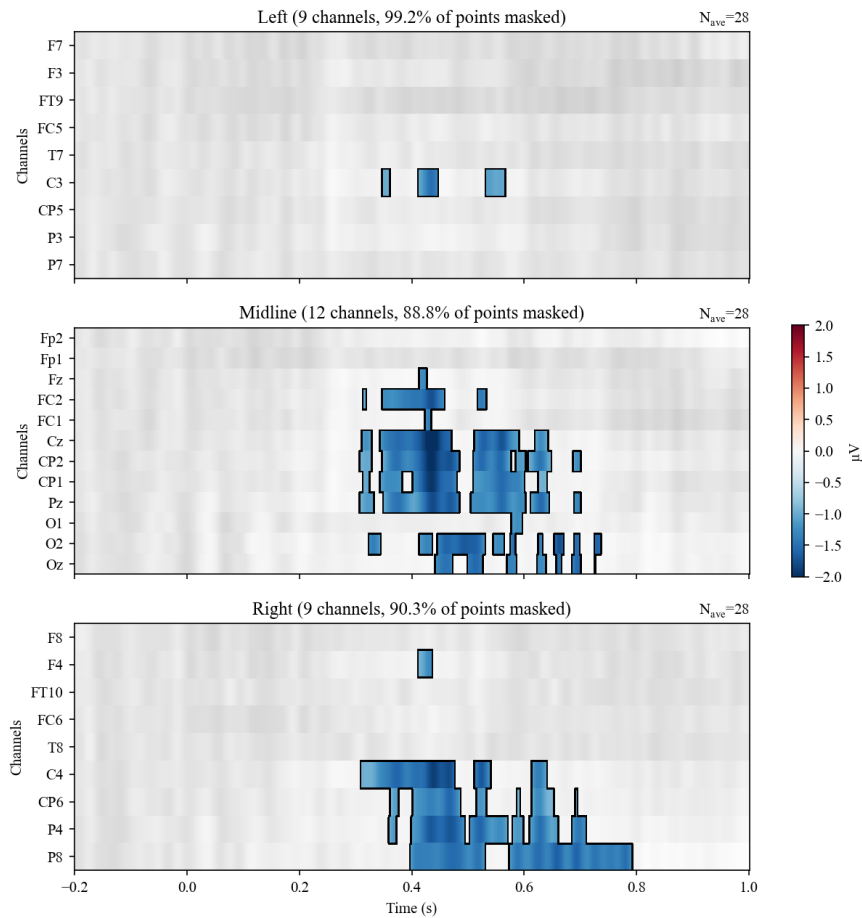
As mentioned above, the Spatio-Temporal Cluster-Based Permutation Test establishes whether the conditions differ, but it does not license, by itself, claims about effect onset, offset,

or scalp location (Sassenhagen; Draschkow, 2019). We therefore treat the existence and direction of each contrast as the inferential result, interpreting the cluster raster plots alongside their associated topoplots.

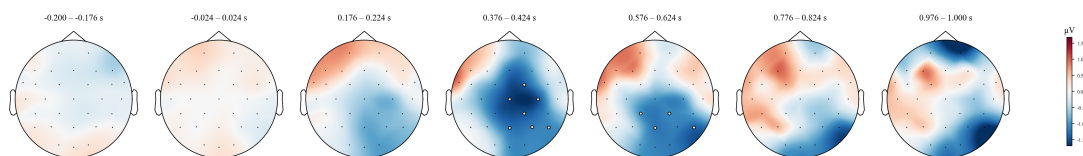
The test reveals a statistically significant ($p = 0.04$) negative difference between *bridged* and *given*, with *bridged* eliciting more negative waves than *given*. The raster plot in Figure 23a shows broadly distributed negative clusters for *bridged* relative to *given*, predominantly visible on the midline (frontal, central and parietal, but also occipital) and right electrodes (stronger in central and parietal), starting at around 250 ms and staying negative until around 700 ms on the midline and 800 ms on the right. Figure 23b, presenting the evolution through topoplots, displays the same pattern of centro-parietal/frontal activation, within the same time window (although it seems to fade later than the raster plot shows).

Figure 23 – Analysis of the *bridged* – *given* contrast.

- (a) Raster plot dividing the head into three sections (top to bottom): Left, with 9 channels (F7, F3, FT9, FC5, T7, C3, CP5, P3, P7); Midline, with 12 channels covering fronto-central, central, and occipital sites (Fp2, Fp1, Fz, FC2, FC1, Cz, CP2, CP1, Pz, O1, O2, Oz); and Right, with 9 channels (F8, F4, FT10, FC6, T8, C4, CP6, P4, P8). Coloured points mark statistical significance for the *bridged* – *given* difference ($p < .05$); on the colour scale, positive voltages are red and negative blue. The x -axis marks time in seconds; the y -axis marks the electrodes.



(b) Topography evolution of the difference wave across time.

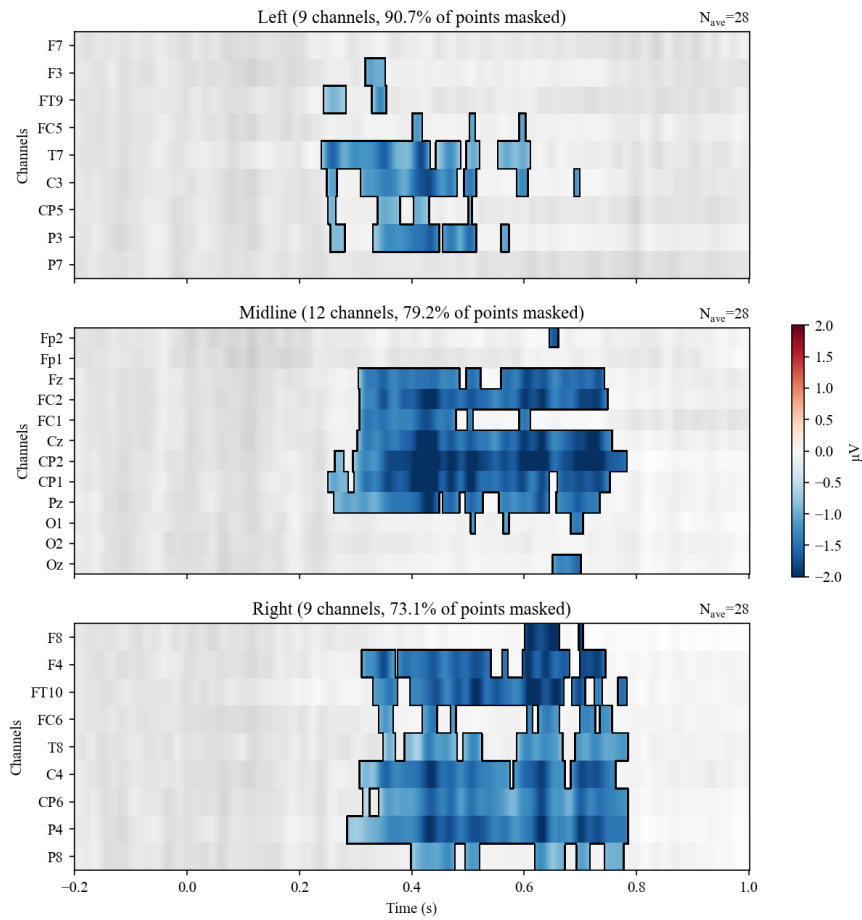


The contrast between *new* and *given* (in Figure 24a) also yielded a significant negative difference ($p = 0.002$), with *new* being more negative than *given*, and more cluster activation over left-hemisphere electrodes compared to the *bridged* – *given* contrast. There was also a broader

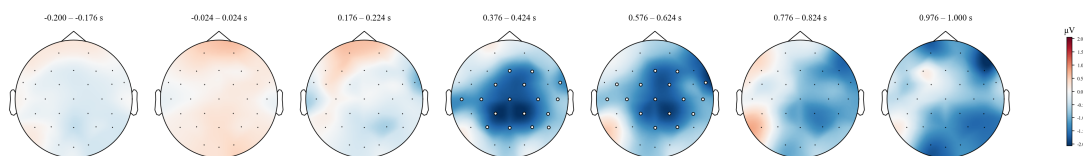
distribution of the negativity through midline and right electrodes. On the left, it starts earlier, at around 200 ms, while on the right and on the midline, it starts at around 250 ms (like in *bridged – given*). It ends at around 600 ms on the left, and — as with *bridged – given* — at around 800 ms on the midline and right. Figure 24b, presenting the topoplot evolution of the difference waves, shows the same pattern of widely distributed activation (again, appearing to end later than the cluster plot shows).

Figure 24 – Analysis of the *new – given* contrast.

- (a) Raster plot dividing the head into three sections (top to bottom): Left, with 9 channels (F7, F3, FT9, FC5, T7, C3, CP5, P3, P7); Midline, with 12 channels covering fronto-central, central, and occipital sites (Fp2, Fp1, Fz, FC2, FC1, Cz, CP2, CP1, Pz, O1, O2, Oz); and Right, with 9 channels (F8, F4, FT10, FC6, T8, C4, CP6, P4, P8). Coloured points mark statistical significance for the *new – given* difference ($p < .05$); on the colour scale, positive voltages are red and negative blue. The x -axis marks time in seconds; the y -axis marks the electrodes.



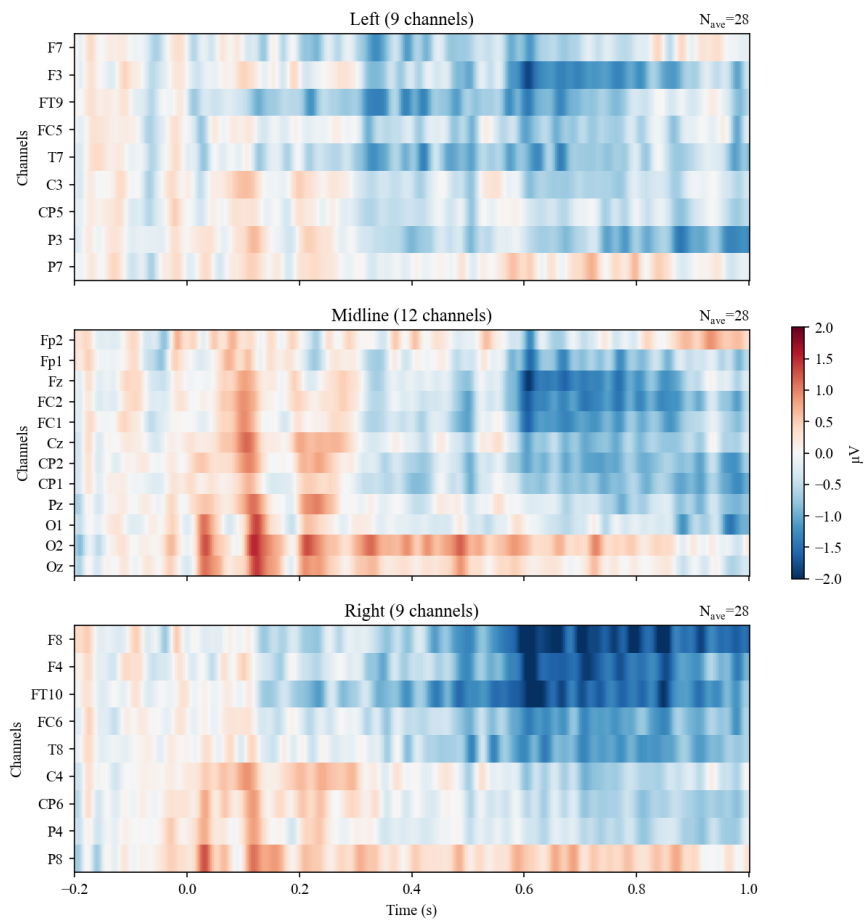
(b) Topography evolution of the difference wave across time.



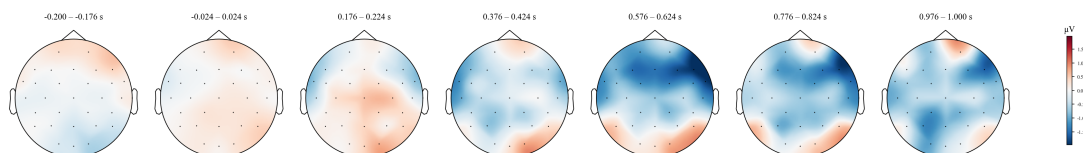
The topoplots in Figure 25b seem to indicate a frontally distributed negativity for *new – bridged*, but no significant clusters were found (lowest $p = 0.11$), as Figure 25a presents.

Figure 25 – Analysis of the *new – bridged* contrast.

- (a) Raster plot dividing the head into three sections (top to bottom): Left, with 9 channels (F7, F3, FT9, FC5, T7, C3, CP5, P3, P7); Midline, with 12 channels covering fronto-central, central, and occipital sites (Fp2, Fp1, Fz, FC2, FC1, Cz, CP2, CP1, Pz, O1, O2, Oz); and Right, with 9 channels (F8, F4, FT10, FC6, T8, C4, CP6, P4, P8). Unlike the previous contrasts, the panel is fully coloured: there were no significant cluster points for the *new – bridged* contrast, so only general tendencies are shown. On the colour scale, positive voltages are red and negative blue. The *x*-axis marks time in seconds; the *y*-axis marks the electrodes.



(b) Topography evolution of the difference wave across time.



Across both contrasts, the significant clusters indicate that both *bridged* and *new* are more negative than *given*. The topographies suggest this negativity may extend somewhat beyond the cluster windows, though this did not reach significance. No positive significant clusters were found.

5.3 rERP

To prepare for the rERP analysis, we used the epoched files from the previous preprocessing, realigned the EEG and E-Prime trial data⁹, and labeled each of the remaining epochs — after ICA and Autoreject were applied — with their respective trial information (item Id and Condition). This allowed us to relate the EEG data to the scores from the PCIBex experiment: Likert 1 — Visual Clarity; Likert 2 — Coherence; Likert 3 — Information Addition¹⁰, and to save it to a single file to fit the rERP.

We used Aurnhammer et al.’s (2023) Julia language¹¹ script for conducting the regression, as well as their R code as the basis for the plot functions, adapting the Julia code very little and the R plotting code where our design required.

To verify predictor multicollinearity, we examined the Pearson correlations and variance inflation factors among the predictors. These show some expected relations that were already mentioned above, when we were discussing the methods used for stimuli selection (Section 4.11.1). Crucially, however, the Variance Inflation Factors are well below 5, the threshold above which collinearity is commonly considered problematic (see Kim, 2019), with the highest value (the least independent predictor) being Coherence (see Table 8).

Collinearity would inflate variance and make it more difficult to determine which predictor is influencing the waveform. However, the fact that a predictor reaches significance despite this mathematical penalty would demonstrate the robustness of the effect. Table 8 shows the Pearson correlations¹²:

⁹ The triggers only sent information about condition, not about which individual item was being presented — this information had to be retrieved by aligning the two data.

¹⁰ As a recall, the questions were:

1. “To what extent is the emoji visually identifiable?” (From “Impossible to identify” to “Perfectly identifiable”);
2. “To what extent do the emoji and the sentence make sense together?” (From “They make no sense at all” to “They make perfect sense”);
3. “To what extent does the emoji add information to the sentence?” (From “It does not add any information” to “It adds a lot of information”).

¹¹ Bezanson *et al.* (2017), with the DataFrames.jl (Bouchet-Valat; Kamiński, 2023) and Distributions.jl (Besançon *et al.*, 2021) packages, alongside CSV.jl, StatsBase.jl, and Combinatorics.jl. Plotting and data handling in R (R Core Team, 2026) used ggplot2 (Wickham, 2016), data.table, gridExtra, and scales.

¹² By-item mean ratings, unlike Table 3, which correlates the three scales at the triad-level, across all responses.

Table 8 – Pearson correlations and VIF among the rERP predictors.

X	Coherence	Information Addition	Visual Clarity	VIF
Coherence	1.00	0.71	0.67	2.72
Information Addition	0.71	1.00	0.49	2.01
Visual Clarity	0.67	0.49	1.00	1.80

Figure 26 shows the density plots for the responses to the Likert scales on Coherence and Information Addition from the PCIBex norming study. It is another way of looking at the distribution of norming study answers, as compared to what we did in Section 4.6, and it further demonstrates that even though there is some overlap, there is a distinct distribution for the different conditions. The lines indicate the per condition averages:

Figure 26 – Density plots: norming rating distributions by condition.

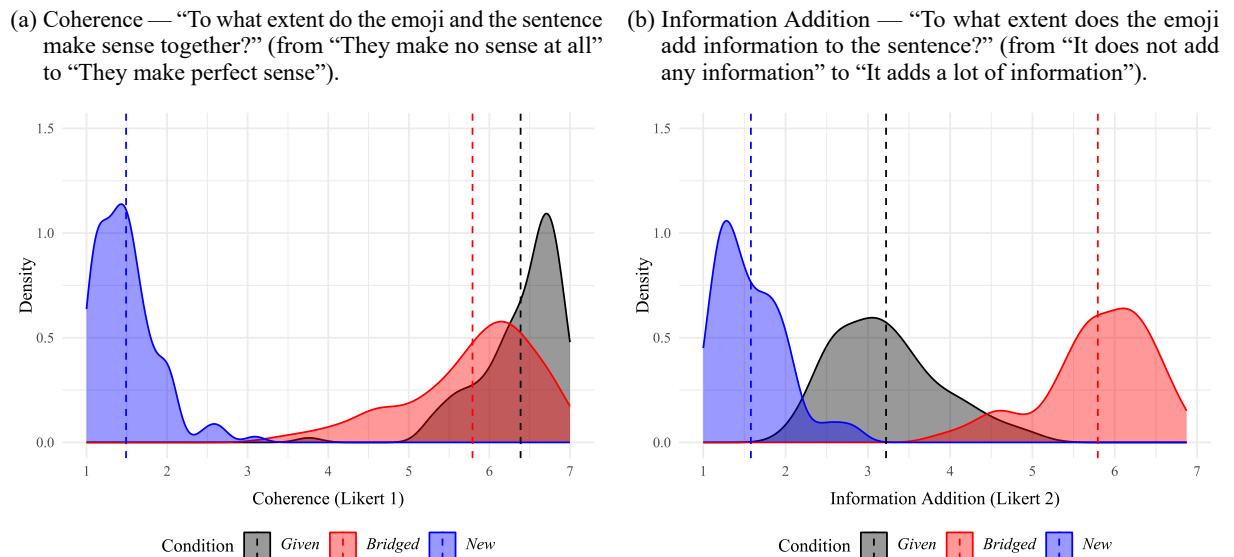


Figure 27 shows the regression estimates on electrode Pz and the residuals, i.e., the actual measured ERPs minus the estimated waveforms. We can see that the residuals are close to 0, which means the predictors do account for the observed data.

Figure 27 – Estimated waveforms and residuals at Pz. On the left, the solid lines are the mean ERP values estimated by the regression model and the shaded areas are the confidence intervals; on the right, the residuals, i.e., the difference between measured and estimated ERPs.

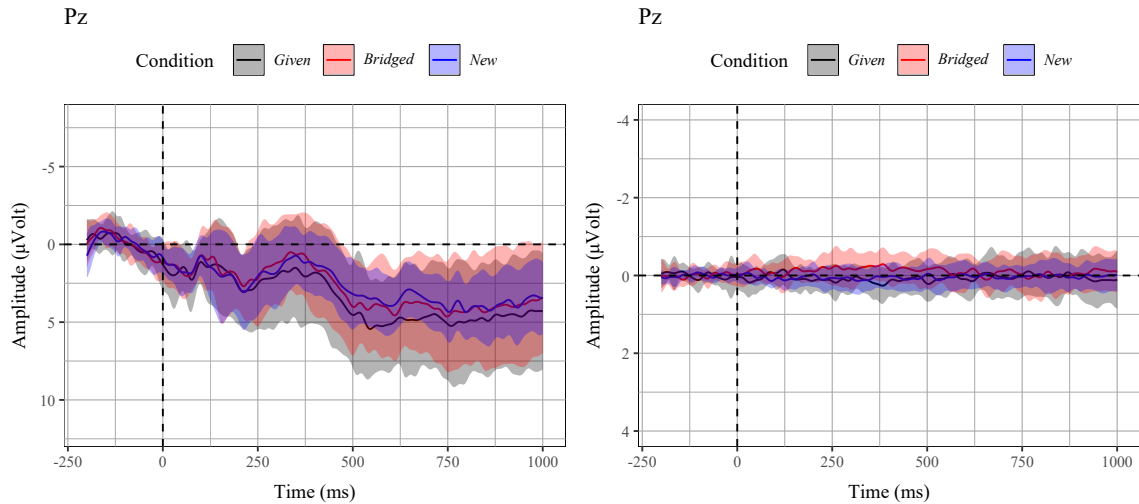
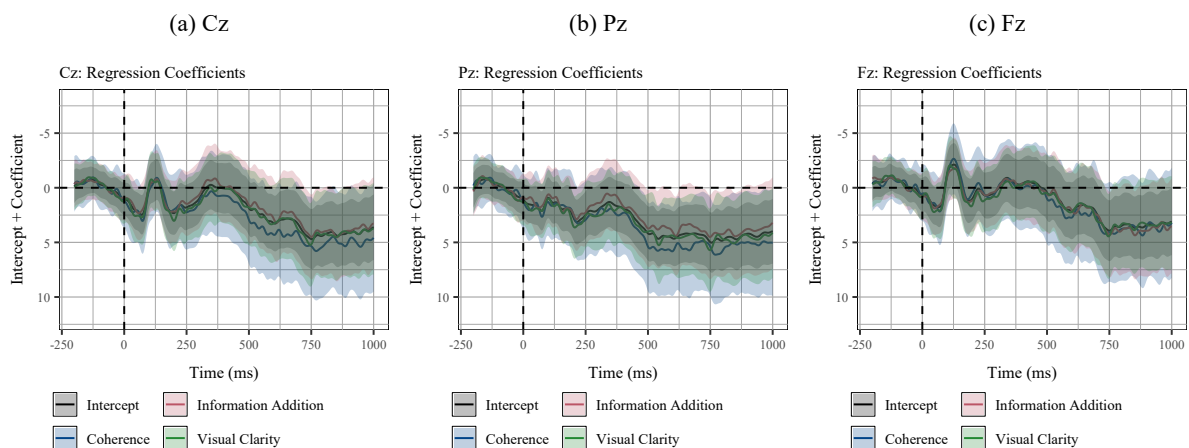


Figure 28 shows the rERP coefficients — Coherence (blue), Information Addition (red), and Visual Clarity (green) — each added to the intercept (black), at electrodes Cz and Pz. The intercept is the predicted waveform when every predictor is held at its average value (zero, since the predictors were z-scored). Each remaining waveform is obtained by adding one predictor's coefficient to this intercept, so that it can be read as an estimated ERP rather than as an abstract slope around zero, and the effects in terms of amplitude variation (x -axis) over time (y -axis) can be more easily visualized. The shadows represent the confidence intervals.

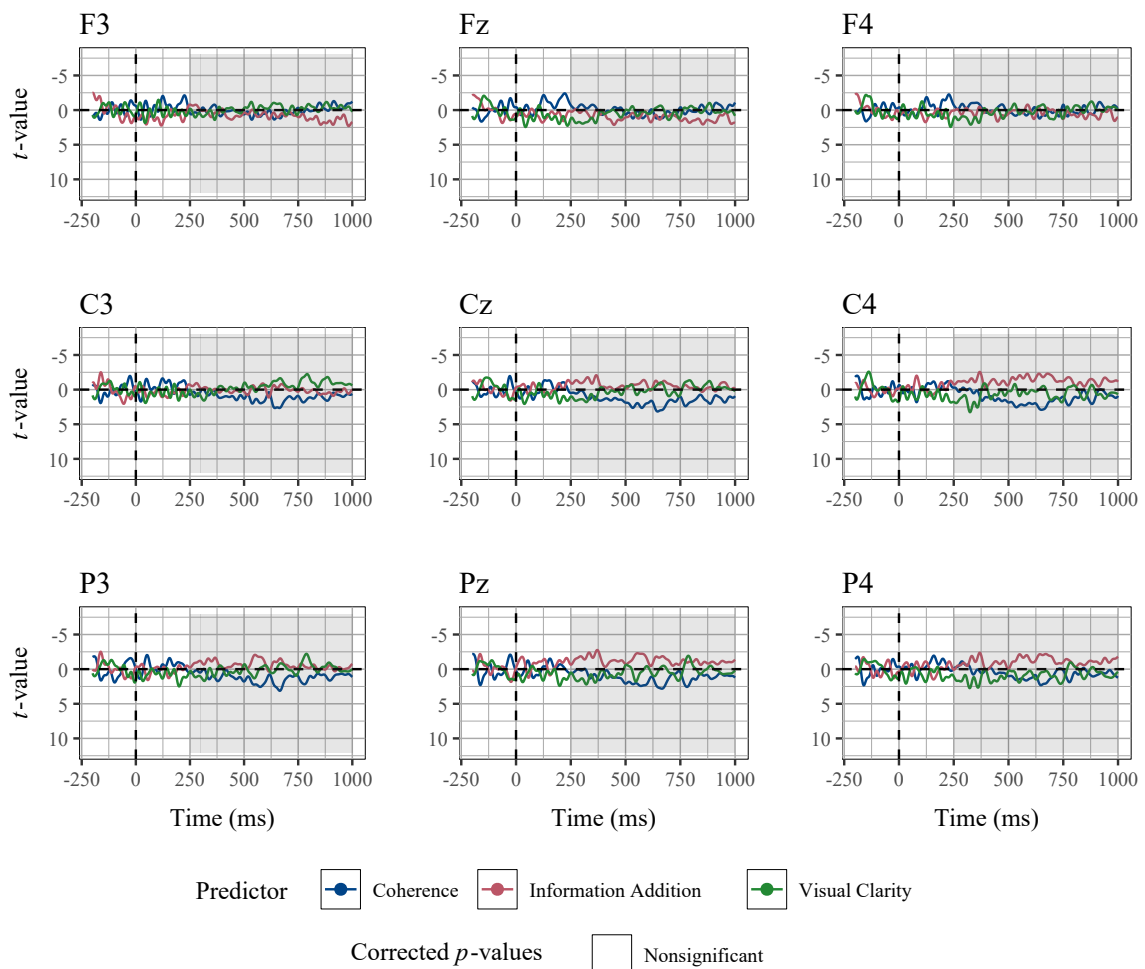
Figure 28 – Regression coefficients added to their intercept.



Finally, Figure 29 shows the inferential statistics. The x -axis is time, while the y -axis contains the t -values. Each predictor was t -tested at each time point: if the line is close to 0, it means this particular predictor is not influencing wave amplitude at that moment; if it is far from 0 and positive, it indicates the predictor increases wave amplitude; if it is negative, that it decreases it.

Figure 29 – Inferential statistics shown with FDR (False Discovery Rate) corrected t -values. Coherence is shown in blue, Information Addition in red, and Visual Clarity in green. Significant points, had there been any, would appear as bars in the predictor's colour below the waves, covering the time windows of significance. The grey areas are our time windows of interest (250–350 ms for earlier negativities; 300–500 ms for the classic N400; 500–1000 ms for the P600).

Inferential statistics (FDR-corrected t -values)



Because of the multiple comparisons problem, a t -test can be misleading. We therefore applied the Benjamini-Hochberg correction for False Discovery Rate (FDR) control. The statistically significant ($p < .05$) influence of one of the predictors would show as a bar of the same color of the predictor below the wave plot. In our case, however, no independent effects of Coherence, Visual Clarity or Information Addition were observed.

Two results depart from our predictions and frame the discussion that follows. Instead of an N400 followed by a P600, we found a negativity sustained through the late window — conditions *bridged* and *new* more negative than *given* throughout, with no positivity where a P600 was expected — and the rERP revealed no reliable effect of any continuous predictor. Together, they point to a distinction that is categorical (*given* vs. not-*given*) rather than graded, which is what we develop below and in the following chapter.

5.4 DISCUSSION

The purpose of the present study was to assess how multimodal bridging unfolds in the brain via Event-Related Potentials, and whether it patterns with previous findings for purely linguistic (Burkhardt, 2006) or purely visual bridging (Cohn; Kutas, 2015). Following the literature, we expected a negativity in the N400 window — indexing dependency formation (Burkhardt, 2006), lexical access (Aurnhammer *et al.*, 2023) or prediction (Kuperberg, 2016; Kuperberg; Brothers; Wlotko, 2020), depending on the account — possibly accompanied by overlapping anterior components such as the N300, given the pictorial stimulus (Kumar; Federmeier; Beck, 2021; Kutas; Federmeier, 2011), or the Nref (Van Berkum, 2009), followed by a positive shift indexing integration or the establishment of a new referent (Burkhardt, 2006, 2007; Brouwer *et al.*, 2017; Aurnhammer *et al.*, 2023).

Our hypotheses were that *given* would correlate with attenuated N400 and P600 amplitudes; *bridged* with an intermediate N400 and a P600 comparable to *new*; and that *new* would show the largest amplitudes for both the N400 and the P600.

The Spatio-Temporal Cluster-Based Permutation test showed broadly distributed clusters of sustained negative difference between *new* and *given* ($p = 0.002$) and between *bridged* and *given* ($p = 0.04$) — being more negative for *new* and *bridged*, with the *new* – *given* difference being more broadly distributed than *bridged* – *given*. The *new* – *bridged* contrast was not significant (lowest $p = 0.11$). Interpreted with the topoplots, it indicates a negativity that starts early, peaks at around 400 ms and is sustained until around 800 ms. The rERP analysis, in turn, shows no

reliable effects for any of our predictors (Coherence, Information Addition and Visual Clarity). These results seem to stand as a hybrid of prior findings.

In what follows, we first turn to the baseline check, which provides evidence that our findings are not artifacts. We then interpret the negativity as an N400, setting our results against Burkhardt's (2006) purely linguistic and Cohn and Kutas's (2015) purely visual bridging. Finally, we address the absence of a P600 and the possibility that component overlap masks additional potentials. This discussion is extended and conjugated with the first experiment's results in the next chapter (Chapter 6).

5.4.1 The baseline verification

As mentioned above, a baseline check, made possible by the target manipulation of our design, was conducted in order to ensure that all differences obtained after the trigger stemmed from the processing of the emoji in the context of the sentence, and that there were no prior effects in sentence processing itself. As its results show (Section 5.2.2, Section C.2.1), processing pre-trigger overlapped as expected. Visual Clarity was also not found to independently modulate the waveforms in the rERP analysis.

5.4.2 The N400

As discussed in Chapter 2, Burkhardt (2006), working with definite-DP-mediated bridging in a purely linguistic environment, found that given DPs showed attenuated N400 and P600 amplitudes compared to bridged and new DPs, and that bridged DPs patterned first with given DPs, showing an attenuated N400, and then with new DPs, showing similarly enhanced P600 amplitudes. She then proposed that the N400 reflected the dependency formation between the DP and its prior context, while the P600 reflected its integration, or the updating of the mental model.

Cohn and Kutas (2015), working with a purely visual context — i.e., Peanuts comic strips — found a graded P600 that correlated with integration effort, but no classic N400. In its place, they found a sustained frontal negativity, which they attributed to the working-memory demands of omitting event information. They argue its anterior distribution was incompatible with a true N400 — unlike the centro-parietal negativity they obtained at the final panel of their Expected condition, where a genuine semantic clash occurred (the critical panel shows Snoopy retrieving the thrown ball, but the final panel reveals Linus with it instead, as illustrated in Section 3.3).

Here, as in Burkhardt (2006), we interpret our findings as showing an N400 effect, followed by a sustained negativity. The cluster-based test (interpreted alongside the ERPs and topoplots) shows a predominantly centro-parietal negative effect with a right-side bias (Kutas; Federmeier, 2011, p. 3) both for *new* and for *bridged* relative to *given*, although more broadly distributed for *new* – *given*. It activates frontal sites, which could indicate component overlap (commented below, on Section 5.4.3) but is also in accordance with research on the N400 for pictures, which frequently reports more frontally distributed N400s (Kutas; Federmeier, 2011, p. 8; and also Tang; Chen, *et al.*, 2020, with a similar context using emoji). More crucially, however, it is consistent with the component’s established functional sensitivity (Kutas; Federmeier, 2011): it had its smallest amplitude for the condition which had the smallest semantic gap between sentence and emoji, and it had larger amplitudes in the conditions in which the gap was bigger, in line with previous research that obtained N400 modulations by manipulating semantic priming, discourse constraints, or expectations, as well as the widely accepted view that it indexes the access to semantic features of incoming stimuli (Nieuwland; Van Berkum, 2006; Franklin *et al.*, 2007; Van Berkum, 2009; Kutas; Federmeier, 2011; Wang; Schumacher, 2013; Kuperberg, 2016).

Even conflicting models, such as the Multi-Stream and Retrieval-Integration, would predict an N400 in this particular situation, the difference being which internal processes or level of representation they would interpret the N400 to be indexing — as well as what would be happening in the subsequent time window. Aurnhammer *et al.* (2023), for example, made sure to maximally prime both their targets and distractors in their design, to demonstrate that an N400 effect does not correlate with integration processes, but rather that it encodes lexical retrieval. As our *bridged* and *new* conditions do not explicitly lexically prime the subsequent emoji, this could potentially explain obtaining an N400 effect regardless of theoretical affiliations — although priming through semantic relation is known in the literature (Kutas; Federmeier, 2011).

Cohn and Kutas’s (2015), on the other hand, did not obtain a classic N400 effect correlated with the inference manipulations, but only a frontal negativity. Only when the following panel was incongruent with the preceding panel did they obtain the N400 — accompanied by an N300 — (Expected condition, between the critical and final panels): Charlie Brown throws the ball [panel 2]; Snoopy is shown retrieving it [critical panel]; and then Linus retrieves it instead [final panel].

The difference between our results might stem from a difference in the structure of our stimuli. Albeit also visual, their stimuli are embedded in a structure — an event sequence — while ours is of a different nature entirely. We have known for some time that the processing of scenes and visual objects differ, as the brain has different subsystems dedicated to them (Epstein;

Kanwisher, 1998). There is still not much research directed at ERPs related to unfolding visual scenes — and even less so for visual inferences — but those that do exist have found a somewhat more frontally distributed N400, not always preceded by an N300, as they frequently overlap (Kutas; Federmeier, 2011; Sitnikova *et al.*, 2008; Federmeier; Kutas, 2001).

The word–picture paradigm of Willems, Özyürek, and Hagoort (2008) (Section 3.5) is perhaps more relevant here as a comparison: they obtained an N400 of similar latency and amplitude across every mismatch condition, including the double mismatch. To the degree that our conditions *given* and *new* resemble, respectively, their “The man gave his wife a nice flower that evening 🌹” and “The man gave his wife a nice flower that evening 🍒”, it seems natural that we obtained comparable results.

The paper additionally points to an alternative explanation to component overlap for not obtaining an N300, which is reported to precede an N400 and also preceded it in Cohn and Kutas’s (2015) final panel for the Expected condition. Remarking that an N300 was also not found in Ganis, Kutas, and Sereno (1996), they argue that, at least in sentence context, the N300 does not seem to arise. Other studies with emoji also do not mention it (Tang; Chen, *et al.*, 2020, 2024; Weissman; Tanner, 2018; Yu *et al.*, 2022). It has been argued that differences in the N400 distribution might be due to overlapping and nonoverlapping semantic representations (Özyürek *et al.* (2007)).

5.4.3 The absence of a P600 effect and component overlap

Unlike both Cohn and Kutas (2015) and Burkhardt (2006), we did not obtain a P600 effect in the following time window. Instead, the data indicate a negativity that begins around 250 ms and is sustained up to 800 ms, which we are interpreting as an N400 followed by a sustained negativity.

As noted in Section 3.4, multimodal ERP components have proven elusive: sometimes mirroring language, sometimes departing from it. The positional account described in Tang, Chen, *et al.* (2024) (Section 3.4) bears directly on this. They propose that sentence-final emoji, with a weaker syntactic role and functioning like a comment or isolated word, elicit an N400 but no P600, whereas sentence-intermediate emoji — their syntax and semantics being more interwoven — elicit a P600. Our emoji were not only sentence-final but independent discourse units, standing further still apart from the syntactic structure. They interact with the previous context, as well as with the situation model that it initiates, at a highly conceptual, but not structural, level. On

Tang et al's (2024) account, the N400-without-P600 pattern we obtained is exactly what should be expected.

Another alternative answer, as indicated before, could be component overlap. There are clusters of frontal sites, and the topoplots show them as well. Because we only see a negative difference, both in *bridged* – *given* and in *new* – *given*, this could be masking other negative potentials we could not identify, such as some frontally distributed components — the Nref (Van Berkum, 2009; Nieuwland; Coopmans; Sommers, 2019), posited to index referent search; or the N300, which, sensitive to visual complexity, has been reported to co-occur with the N400 with visual stimuli (Kumar; Federmeier; Beck, 2021). It could also be masking an P600, as Delogu, Brouwer, and Crocker (2021) argue.

If we look at the ERPs (Section 5.2.3), we see that after a negative deflection, the wave goes back to being more positive (with *given* being most positive). If this is just the tail of an N400 or the beginning of a P600, it is unclear. A P600 reflecting integration has been strongly argued by recent research (Brouwer *et al.*, 2017; Delogu; Brouwer; Crocker, 2019; Aurnhammer *et al.*, 2023), so it would make sense. Additionally, strong negativities have been reported to mask the P600 before, as they often overlap temporally and spatially¹³. Entertaining this possibility, one explanation for the *given* condition being the most positive in later windows would be that, upon encountering *given*, the least negative condition during the previous time window, the participant accesses its semantic features rapidly, as they are more readily primed, and integrates it faster than for *bridged* and *new*. Delogu, Brouwer, and Crocker (2021) argue that a rERP analysis could potentially untangle the N400 and the P600, but we obtained a null result in ours.

This, however, does not seem to account for other cases where the N400 coexisted with the P600, such as Burkhardt (2006). She also obtained an N400, her *given* condition was the least negative in the N400 window, like ours, but this did not mean that it was still more positive in the subsequent P600.

The explanation that includes fewer assumptions remains, therefore, that of an N400 followed by a sustained negativity. As Cohn and Kutas (2015, p. 15) state, frontal negativities have been posited to reflect working memory demands, as well as the cost of “computing or maintaining inferred event information within or between sentences”. Willems, Özyürek, and Hagoort (2008) also obtained a negativity that lingered after the N400 window (p. 1244):

¹³ Possibly, this is the reason why the Expected condition in Cohn and Kutas (2015) had the lowest P600 amplitudes for its final panel — because it had a “true” N400, this attenuated the subsequent positivity.

The stronger negativities to the mismatch conditions in the late time window resemble the findings in the N400 time window, although only the language mismatch condition differed significantly from the correct condition. However, no differences were observed between the experimental conditions. Therefore, these late effects are best explained as a carryover effect of the strong N400 effects.

This lingering negativity could also be attributed to our design decision not to include a period after the stimulus to preserve naturalistic presentation (see Section 4.2). Not knowing whether the sentence has ended could put additional strain on working memory. However, every final element was exhibited longer than other sentence elements, including filler images – more than twice the time and enough to otherwise indicate sentence ending.

Having interpreted the EEG results on their own terms, we now draw the two experiments together. The following chapter synthesizes the norming and EEG findings, draws a possible reinterpretation of our conditions suggested by the data, and sets out the study's limitations and directions for future work.

6 GENERAL DISCUSSION

This dissertation set out from an observation drawn from Grosz, Kaiser, and Pierini (2021, 2023): if activity emoji can act as free adjuncts with anaphoric effect, then the emoji–text relation can be treated, within the given–new paradigm (Clark; Haviland, 1977), as a bridging inference — one whose neural correlates could be probed by manipulating the informational dependence between an emoji and its host clause. The preceding chapters reported two experiments to that end; here we interpret their results together.

We mirrored Burkhardt’s given–bridged–new experimental conditions and hypothesized that we would obtain results similar to hers: an N400 reflecting dependency formation (Burkhardt, 2006), prediction (Kuperberg, 2016; Kuperberg; Brothers; Wlotko, 2020), or lexical retrieval (Aurnhammer *et al.*, 2023), followed by a P600 reflecting integration effort (Burkhardt, 2006, 2007; Brouwer *et al.*, 2017; Aurnhammer *et al.*, 2023), possibly decomposable into frontal and centro-parietal components (Brothers *et al.*, 2020; Kuperberg; Brothers; Wlotko, 2020). In accordance with the discourse-comprehension literature, we also allowed for the Nref, related to anaphoric ambiguity, and/or the N300, sensitive to the visual complexity of objects and scenes (Kumar; Federmeier; Beck, 2021), though not frequently reported in multimodal research.

As a reminder, our design was as follows:

Item 7:

- a. Brazil was defeated 🇧🇷 — *given*;
- b. Brazil was defeated 🌐 — *bridged*;
- c. Brazil was defeated 🍒¹ — *new*.

To test our hypotheses, we conducted two experiments: a norming procedure, to verify whether the constructed stimuli behaved as expected, and a main experiment, using EEG and the ERP methodology.

The first experiment asked participants to reply, through three separate 7-point Likert scales:

1. “To what extent is the emoji visually identifiable?” (From “Impossible to identify” to “Perfectly identifiable”) — with the intention of removing unfamiliar or badly designed stimuli;
2. “To what extent do the emoji and the sentence make sense together?” (From “They make no sense at all” to “They make perfect sense”) — to separate *given* and *bridged* stimuli,

¹ “O Brasil foi derrotado”, in Portuguese.

which should make sense with the sentence, from *new*, which should not have a clear relation with it;

3. “To what extent does the emoji add information to the sentence?” (From “It does not add any information” to “It adds a lot of information”) — to separate *given*, which should not add information, from *bridged*, which should.

Fitting a cumulative link mixed model — reported in Section 4.12.5 — to each Likert scale (treating the data as ordinal), we found that: both *given* and *bridged* stimuli were judged more coherent than *new*, with *given* rated reliably more coherent than *bridged* ($\text{given} > \text{bridged} > \text{new}$); that *bridged* stimuli were judged more informative than *given* and *new* ($\text{bridged} > \text{given} > \text{new}$); and that Visual Clarity was highest for *given*, followed closely by *bridged* and then *new*. A linear mixed model, treating the data as continuous, yielded the same ordering, confirming the overall soundness of the corpus.

The ERP experiment, however, did not draw the same three-way distinction. Partially against our hypotheses and unlike Burkhardt (2006), our Spatio-Temporal Cluster-Based Permutation test, interpreted alongside the ERP waveforms, revealed that the *bridged* and *new* conditions were negative compared to *given*. This negativity, starting at around 250 ms and peaking at around 400 ms, was sustained until later time windows (~ 800 ms). The *bridged* and *new* conditions did not differ from each other, but only in their relation towards *given*, with the negativity in *new* being more broadly distributed.

The fact that *given* is consistently perceived as different from *bridged* and *new* is especially interesting given a worry that we had previously discussed (Section 4.3), namely the problem of the *given* condition’s heterogeneity regarding its semantic redundancy — through iconicity or semantic association. These two types of reference, especially given the fact that semantic association items were a majority, could have blurred the line between the *given* and *bridged* conditions, but they did not.

The main difference with Burkhardt’s (2006) findings for the N400 is that, while she found graded amplitudes (“given” < “bridged” < “new”), our results did not show a significant difference between *new* and *bridged*, despite their differences in distribution relative to *given*. This null result could indicate two different scenarios. First, since absence of evidence is not evidence of absence, it could simply be that a difference exists but that, through limited statistical power or limited sensitivity of our design, we failed to detect it. Alternatively, the difference may be genuinely absent. On this reading, three explanations could be entertained — none

asserted at this moment, but each weighed in what follows: that 1) neither *bridged* nor *new* receives a felicitous interpretation, and hence the leveled N400 reflects a generalized semantic difficulty — the classic N400 towards difficulty in semantic processing; that 2) both *bridged* and *new* are interpreted felicitously within the same time window, requiring similar cognitive effort; or that 3) our design differs from Burkhardt’s in a way that justifies similarly greater processing demands for both *bridged* and *new*, whether because our context–target relations differ or because linguistic and multimodal processing are fundamentally distinct. We take these hypotheses in turn.

The first and second hypotheses are unlikely for reasons that mirror each other. The first — that neither *bridged* nor *new* receives a felicitous interpretation — is undermined by the norming study (Section 4.6), whose Coherence scale was the main criterion for stimuli selection. Except for two items (2 and 62), every single *bridged* item had more than 52.17% agreement for ratings greater than 3 in the Coherence scale ($M = 87.54\%$, $SD = 11.32\%$, $\max = 100\%$). Additionally, the accuracy for the questions that probed for conjoint meaning was very high, as shown in (Section 5.2.1).

The second — that both *bridged* and *new* are interpreted felicitously, alike — fails for the opposite reason: the norming was especially rigorous toward the *new* condition, requiring over 60% consensus for ratings below 3. No *new* item was retained with less than 58.33% agreement that it made no sense — ratings lower than 3 ($M = 88.47\%$, $SD = 9.66\%$, $\max = 100\%$).

Even if we believe that, because of the symbolic and inherently fluid nature of images, one could always eventually establish a connection between them and text, this would require many assumptions. It seems implausible that this integration for *new* would be reflected at the same time window as for *bridged*.

Burkhardt’s (2006) paper mentions she used salient roles of the type “conductor-concert” for her stimuli — “As a consequence, the critical DPs in the BRIDGING condition represent very salient roles (in relation to the introductory context sentence)”, (p. 162). A bridging inference through salient roles works especially well because, beyond priming semantically related words, a word such as “concert” prompts the entire situation model (Dijk; Kintsch, 1983) to which a “conductor” belongs, facilitating processing. This might be a better explanation for the difference between her results and ours, since our *bridged* condition included a very varied array of discourse relations between sentence and emoji, and not only salient roles relations.

Pulling that thread, even when we used the discourse relation Elaboration (Section 4.3), which seems closest, we can see that the constraints seem to be somewhat looser with emoji

than they are with definite DPs. For example, in “Brazil was defeated 🏆”, the ball completes the situation by indicating in which context Brazil was defeated. To a Brazilian, one might expect that the context of soccer might become immediately activated, but this was not the case: many participants pointed out that they thought of elections before seeing the emoji. Even if the Brazil–football connection is strong, there are several other possible completions that can be thought of. The same occurs with “The event was canceled 🌧️”: the umbrella emoji fits the context and completes it, but the text does not necessarily or immediately prime it — many possible explanations would be fitting, other than rain. Even in “I took the chocolate one 🍪”², an example that offers a pronoun (in Portuguese, it uses a null object) to be matched by the anaphoric emoji, we can see that it is also neither necessarily nor immediately primed — one could think of other chocolate things, and a donut might not be the immediate choice for the Brazilian context.

The anchor, in Burkhardt’s (2006) terms, is there, but it is perhaps fainter: the context accepts the emoji, but does not necessarily prompt it.

On the other hand, there is a substantially different relation in the *new* condition. The context is unwelcoming to the emoji, and its occurrence breaks discourse in a manner that is difficult to recover — people were not afraid to rate 1 in the norming experiment whenever they thought it did not make sense. The reader struggles to attend to the Gricean Maxim of Cooperation and to the given–new contract (Clark; Haviland, 1977) in much the same manner as it does with Burkhardt’s stimulus example, because it seems to introduce a completely detached situation. Comparing “Tobias talked to Nina. He said that the conductor was very impressive.” (Burkhardt, 2006, p. 161) and “Brazil was defeated 🍒”, we can see that both include a rupture that, although resolvable, requires a lot more construction to make sense of it.

If despite those intrinsic differences, they are still processed similarly when it comes to that time window, this might call for a reinterpretation of our conditions — one suggested by the pattern of results rather than predicted in advance, and so requiring independent confirmation.

Indeed, while our *new* condition maps well to Burkhardt’s (2006) “new”, her “bridged” condition seems to be better mirrored not by our *bridged*, but by the semantic-association *given* condition. This relates to participants showing hesitation, in the norming study, towards rating *given* stimuli with a 1 for their informativeness. In “He’s not drinking today 🍸”, for instance, it is true that there is not much information added (it does not seem likely that the reader will understand “he’ll not drink today, but if he did, he would drink a martini”), but 🍸 is still

² “Eu peguei o de chocolate 🍪”, in Portuguese.

a different linguistic sign from “drinking” and could be interpreted as a partially new entity. The 🍸 is nevertheless primed more efficiently than it is in *bridged* and *new*, regardless of the level of representation that we take into account, whether semantic — because the semantic aspects of 🍸 are being more effectively primed by “drinking” — or situational — the martini glass being, at the same time, a salient entity (an alcoholic beverage), and a new, more specific entity (a martini glass) in hyponymy.

This is also consistent with the null rERP results. As we can see in (Section 5.3), there was no influence of Information, Coherence, or Visual Clarity in either the early window or the latter. As a null result, it could mean that they did exert influence that our model could not identify, but it is also consistent with this interpretation that both *bridged* and *new* lacked sufficient constraints or were equally unexpected. Coherence and Information Addition are at least partially post hoc judgments, i.e., made after encountering the apparent break in discourse and creating the bridge; the bridging effect — also sometimes called backwards inference — happens in reverse. Even highly coherent stimuli can be unexpected. If the *bridged* and *new* conditions match in terms of processing, therefore, this would point to the unprimed, unconstrained and unexpected state in which both are, when compared to *given*. This would redefine our comparison in terms of context constraints:

- a. Given/highly constrained;
- b. Unconstrained, compatible;
- c. Unconstrained, incompatible.

In an informal conversation with Professor Matthew Crocker, we discussed the possibility that the N400 here reflects pure lexical access: since both *bridged* and *new* feature emoji representing unprimed lexical items, both would elicit an N400. For example, in “He walked through the city 🚶”³, there would possibly be priming for “He walked”; but in both “He walked through the city 🗽” and “He walked through the city 🍪”, there would be no priming for the Statue of Liberty or the cookie, respectively — these words would have to be accessed, which would be reflected in similarly large N400 amplitudes.

Treating the N400 as an index of lexical access, however, assumes that lexical retrieval is in fact being executed, which is not a certainty. There is no guarantee that the participant proceeds from 🗽 to “Statue of Liberty” to “New York” before integrating (or that they go from

³ “Ele caminhou pela cidade”, in Portuguese.

🗽 to “New York” or “America”), rather than just updating the mental model without necessarily passing through the lexicon. There are some considerations that are worth weighing:

1. Participants would presumably still have to integrate the retrieved lexical item, and if the negativity reflects pure lexical retrieval, we did not find the integration-related P600;
2. “City” conceivably primes the Statue of Liberty more efficiently than it primes “cookie”, as “city” is an underspecified entity, and the listener/reader might be actively searching for its resolution;⁴
3. Many emoji in the corpus do not so readily prompt a single lexical item.

This alternative, rather than explaining our data, therefore opens more interesting research paths, such as the one concerning whether the interpretation of emoji necessarily activates a lexical item. This seems to have been the case for the fillers (as a reminder, people often reported they had to “mentally read” the emoji to understand the message), which replaced sentence elements, but would require confirmation for the emoji as discourse units.

As Professor Elsi Kaiser also pointed out, there are many intriguing differences between words and images that could help explain why emoji depart from processing with words, such as the possibility that images force us to commit to a specific mental image, or the possibility that, like words, they can prompt a whole category (upon seeing 🍷, do people access the item “martini”? Or do they more generally interpret “drink”?).

On a more exploratory note, however, we could also entertain the possibility that there is an effect and that we either did not achieve statistical power to find it, or that our design was not sensitive enough for it. The topoplots do show a more distributed negativity for *new* relative to *given* than for *bridged* relative to *given*. There were also more clusters activated for *new* relative to *given*. This could indicate that there is indeed a difference in processing and that *new* recruits more cognitive effort. It is possible that our model and number of participants were enough to detect the difference between *bridged* and *new* relative to *given*, which was stronger, but that *bridged* and *new* differ more subtly from each other, requiring more data.

For the P600, on the other hand, the fact that we did not find the effect while Burkhardt (2006) obtained it might point to the differences in the medium used. As we have observed before in Section 3.7, the bridging Burkhardt (2006) investigated works within the linguistic structure: in “Natalie went to a wedding in Italy. **The groom** talked about a recent trip to Boston.”, “the groom” is embedded into a sentence and follows a structural continuum with the preceding context. Our

⁴ Again, as we have pointed out before, the amount of predictive inference one does is still debatable.

stimuli, on the other hand, are structurally isolated, and their relation is more semantically based — they are not necessarily knitted together — similar to what Gunter and Weinbrenner (2017) argue happens for iconic gestures.

Research on emoji (Tang; Chen, *et al.*, 2024) has previously pointed out that obtaining a P600 or not might be related to how syntactically integrated the emoji is, with final word replacement of adjectives eliciting the N400, but not the P600, which was only found when occupying more structural positions such as verbs. Willems, Özyürek, and Hagoort (2008), working in a paradigm closer to ours, albeit in a different medium (audio and image), also did not find P600 effects for “The man gave his wife a nice flower that evening 🌹” and “The man gave his wife a nice flower that evening 🍒” — which parallel, respectively, our *given* and *new* conditions — finding only a sustained N400.

Research on gestures also points to integration through the N400 (Özyürek *et al.*, 2007; Özyürek, 2014). Gunter and Weinbrenner (2017) argue that most research on gestures — with violation or semantic disambiguation — that reports an N400 dwells on iconicity, which is highly intertwined with context, both influencing each other on a semantic level (Gunter; Weinbrenner, 2017, p. 9).

Our difference with Cohn and Kutas’ (2015) results might also be explained by this. They found P600 effects and, in the inference-requiring conditions, a frontal negativity rather than a classic N400, but they were also working with structurally bound sequences.

Weissman and Tanner (2018) did find the P600 while working with ironic emoji as independent units, but their irony experiment prompted participants to reanalyze the whole mental model upon encountering the ironic emoji — provided they perceived the irony. Our stimuli, functioning mostly like free adjuncts, did not prompt such a reanalysis — at least not systematically.

Gunter and Weinbrenner (2017) also found a P600 effect while working with gestures, which they attributed to cue-validity monitoring. However, they were working with abstract pointing — a gesture with no inherent semantic content, oriented toward organizing referents in discourse rather than carrying meaning iconically. On their own account, it is this discourse-structural character that separates pointing (P600) from iconic gesture (N400 only).

Additionally, Brothers *et al.* (2020) found that the P600 appears to be dependent on stimulus length, semantic richness and global constraints. In a series of experiments manipulating these variables, they observed that short, impoverished contexts did not elicit either a frontal or parietal P600, but only an N400.

One might therefore read all these P600 accounts as arising from a structural or reanalysis demand that our stimuli, functioning as loosely-bound free adjuncts, simply do not prompt.

This hypothesis, attempting to reconcile different research findings on multimodality, however, is tentative — as all hypotheses based on a null are — and hangs on parallels traced from an overarching interpretation of the argument in Tang, Chen, *et al.* (2024) for their own data. What appears to be clear, nevertheless, is that which Event-Related Potentials will be obtained in emoji research depends heavily on the emoji’s relation to its accompanying text.

6.1 LIMITATIONS AND FURTHER RESEARCH

First, our model could have benefited from more statistical power. At 28 participants and ~35 trials per condition, it leans towards the lower side for ERP research, and more data would have made us more confident when drawing inferences about the absence of the P600 or of a difference between the *bridged* and *new* conditions. This is probably especially true for the rERP analysis, which came back with no reliable individual influence for any of our predictors.

An earlier iteration of the rERP analysis, including slightly more trials, showed a timid influence of Coherence for the P600 time window, indicating that the more a stimulus was perceived as coherent, the more the wave was positive within that window, with condition *given* being more positive — which would make even less sense with a P600. However, when correcting the scripts for publication in the repository, we found that, although we were using Autoreject with `random_state = 42` to ensure reproducibility, we were using `n_jobs = 4`, i.e., four parallel processes — and simultaneous processing is inherently non-reproducible. This overrode `random_state = 42`, causing each iteration to give slightly different results. After we corrected to `n_jobs = 1` to guarantee reproducibility, some marginal epochs were lost, affecting the rERP results, but not the cluster test. As the effect relied on just a few epochs, was already timid, and was not reproducible, we did not report it in the discussion, but it reinforces the idea that more data would be beneficial.

A second group of limitations concerns stimulus design. As mentioned, the *given* condition is a heterogeneous baseline, as it includes items that refer back through iconicity and what we called semantic association. This did not impact its difference from the *bridged* and *new* conditions, but future research should nevertheless attempt to better isolate and compare iconic versus semantic association representations in the *given* condition — perhaps introducing more semiotic granularity to the umbrella term “semantic association” we used — and to explore

whether any multimodal reference could indeed be considered *given*. It should also attempt to control the constraints and expectations for the inferences in the *bridged* condition, which would perhaps work better with longer contexts — perhaps a context such as in Burkhardt (2007), where she manipulated constraints from necessary to probable and inducible (Section 3.2). A longer context, although less naturalistic, could also address the point made above related to richness and length of context in order to obtain the P600 (Brothers *et al.*, 2020). Using larger contexts might provide evidence of whether our findings are derived from stimulus size.

Comparing discourse relations within the *bridged* category would also be an interesting endeavor, although it suffers from the caveats described in Section 4.3, namely that one could interpret a sentence–emoji pair in multiple different ways. Perhaps this would demand a more extensive and different norming procedure, investigating converging interpretations of each pair, whether it is more consistently perceived as Explanation, Elaboration, Irony, etc.

Emoji variability is also a point that could have been better controlled. We tried to control this through the Visual Clarity scale, and using a varied set was the calculated price of generalization — a point discussed in Section 4.3 — but it does, nevertheless, introduce a possible confound. Emoji interpretation is, moreover, culturally specific, which limits generalization.

A third group concerns what our design cannot resolve. As we discussed in Section 3.7 and Section 5.4.3, although we could speculate on their existence, our design cannot reliably separate overlapping components within the sustained negativity we obtained. Future research could attempt to bypass this by exploring the data in its oscillatory dimensions, parallel to what was done in Nieuwland, Coopmans, and Sommers (2019). Relatedly, the reinterpretation given in this chapter of our conditions is useful insofar as it is a possible explanation for the absence of a difference between the *bridged* and *new* conditions, but it is also post hoc and would need independent confirmation. It is a point that might also become clearer by inspecting iconicity versus semantic association in a single experimental design, comparing whether they are processed similarly.

Finally, there are procedural caveats. There are the already mentioned ones in the selection of stimuli, discussed in Section 4.12 — namely the interdependence of the Likert scales, the consequences of grouping together *given* and *bridged* for selection (which initially hid the suboptimal agreement for a few items), the averaging of Visual Clarity, which was lenient towards the *new* condition, and the metacognitive nature of Likert scale 3 (“To what extent does the emoji add information to the sentence?”). There is also the decision of not using a period for naturalistic presentation, which could have impacted sentence wrap-up processes.

The comprehension questions, too, were not strictly controlled, and had varied difficulty, which limited the inferences we could make through them; we could have made them more systematic in order to provide evidence of successful bridging. The fact that the fillers are all the same across lists also limits investigation possibilities, although separating them in more or less homogeneous categories for comparison might be possible in future research. Looking beyond discourse integration, investigating direct lexical access through emoji also remains a highly promising path for future exploration.

7 CONCLUSION

We set out to investigate how the given–new dynamics are computed when discourse unfolds across two modalities — text and emoji — and what the neural correlates of multimodal bridging might be. Inspired by Grosz, Kaiser and Pierini’s (2021) idea that emoji can function as independent discourse units and mirroring Burkhardt’s (2006) given–bridged–new design, we built and normed a sentence–emoji corpus where we controlled the informational dependence between emoji and their host clauses, measuring the ERP responses to this manipulation.

Our results were only partly predicted by our hypotheses. In the norming study, participants reliably separated *given*, *bridged*, and *new*. In the main experiment, however, this distinction collapsed into a two-way contrast — a sustained N400-like negativity for both *bridged* and *new* relative to *given*, with no reliable difference between *bridged* and *new*, and no subsequent P600. Multimodal bridging, therefore, patterns with its purely linguistic counterpart (Burkhardt, 2006) in eliciting an N400-like negativity, but not in eliciting the integration positivity Burkhardt (2006) obtained. On the other hand, the results do not pattern with those reported by Cohn and Kutas (2015) for visual scenes.

The absence of a difference between *bridged* and *new* could indicate, we argued, that what most sharply divides the conditions may be not the given–bridged–new gradient but rather the degree to which the prior context constrains the emoji — *given* would stand as highly constrained and predictable, while *bridged* and *new* would both be equally unconstrained and, therefore, equally taxing. Additionally, we propose, following the interpretation in Tang, Chen, *et al.* (2024) for their own paradoxical results, that obtaining a P600 effect might depend on how tightly the emoji is bound to the text: as our emoji function similarly to free adjuncts and were not structurally tied to the sentence, interacting with it at a more semantic level, we only obtained an N400, but no P600.

Methodologically, the present work carries the given–new contract (Clark; Haviland, 1977) and the study of bridging into text–emoji multimodality — a level of analysis that, insofar as emoji research has concentrated mostly on how the image integrates within the sentence rather than within the discourse, seems to have remained largely unexplored. It leaves behind a normed sentence–emoji corpus and an analysis pipeline on which future research can build.

These conclusions, however, are drawn cautiously. They rest partially on null results; and the reinterpretation of our conditions, useful as it is as a possible explanation, remains post hoc and would require independent confirmation.

Even so, to the extent that emoji can behave as independent discourse units, the dynamics between given and new information — so extensively studied in purely linguistic and purely visual settings — seem to extend to text–emoji multimodality as well, a possibility that we believe warrants the continued attention of Neurolinguistics.

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APPENDICES

Appendices

APPENDIX A – CREATING AND VALIDATING THE SENTENCE-EMOJI CORPUS

A.1 SCRIPTS AND LARGE TABLES

Due to space constraints, all analysis scripts and the complete stimuli tables — including word/character counts, semantic and syntactic analyses, tag legends, applied tags and selected stimuli (original and rerun) — have been omitted from this appendix and are only available in the repository: https://github.com/ghapoliveira/multimodal_inferences.

A.2 ICONICITY X SEMANTIC ASSOCIATION COUNT (BEFORE AND AFTER SELECTION)

Table 9 – Reference form count: Iconicity x Semantic Association

Reference Form	Total	Selected
Semantic Association	91	71
Iconicity	59	49

A.3 SEMANTIC RELATION COUNT (BEFORE AND AFTER SELECTION)

Table 10 – Semantic relation count

Discourse Relation	Total	Selected
Elaboration	63	54
Explanation	49	40
Background	26	19
Multiple	4	4
Narration	4	2
Irony	2	0
Result	2	1

A.4 SYNTACTIC ROLE OF THE REFERRED ELEMENT (BEFORE AND AFTER SELECTION)

Table 11 – Syntactic role of the referred element (before and after selection)

Referred Role	Total	Selected
Adjunct of the infinitive verb	1	1
Adjunct of the main verb	23	20
Adnominal adjunct of the direct object	1	0
Direct object of the infinitive verb	5	2
Direct object of the main verb	22	17
Gerund	1	1
Indirect object of the infinitive verb	1	1
Indirect object of the main verb	2	2
Infinitive verb	4	3
Main verb	11	9
Noun complement of the direct object	2	2
Noun complement of the main verb's adjunct	1	1
Noun complement of the subject complement	1	0
Participle verb	3	3
Subject	23	19
Subject complement	9	7
Whole sentence	40	32

APPENDIX B – THE NORMING STUDY

B.1 PARTICIPANT PER LIST DISTRIBUTION (ORIGINAL AND POST HOC)

Table 12 – Participants per group, original vs. post-hoc selection

Group	N (original)	N (post hoc)
1	21	22
10	20	21
2	21	22
3	29	30
4	22	23
5	22	22
6	23	23
7	24	25
8	27	29
9	21	24
Total	230	241

B.2 MANUALLY ADDED ITEMS

As the automatic criteria only selected 115 stimuli, a few more items were manually selected to complete the desired 120. They are (percentages according to the first analysis):

- Item 6 — good *given* and *bridged* sense consensus (86.27%), rejection of the sense in *new* marginally below the threshold (57.14%);
- Item 61 — *given* and *bridged* sense consensus marginally below the threshold (58.54%), but very high agreement for the rejection of sense in *new* (95.24%);

- Item 2 — follows item 61 in the ranking and is identical in *given* and *bridged* sense consensus statistics (58.54%); high agreement for the rejection of sense in *new* (85.71%);
- Item 19 — follows item 2 in the ranking; high *given* and *bridged* consensus (78.05%) marginally below the threshold for the rejection of sense in *new* (57.14%);
- Item 66 — solid *given* and *bridged* sense consensus (68.63%), high agreement for the rejection of sense in *new* (81.82%), marginally low on information contrast (0.91).

The gap items in the ranking were not selected for the following reasons:

- Item 87 — Low rejection of sense in *new* (44.83%);
- Item 88 — Very low rejection of sense in *new* (25.00%);
- Item 58 — Low rejection of sense in *new* (52.38%);
- Item 114 — Low rejection of sense in *new* (41.67%);
- Item 55 — Very low rejection of sense in *new* (38.10%); additionally, low agreement on the informativeness of *bridged* (40.00%);
- Item 22 — Low agreement on the sense of *AB* (57.14%); also low agreement on the informativeness of *bridged* (47.62%);

Numbers further down the list were not considered, as their Information Contrast (the difference between the mean informativeness of *given* and *bridged*) was well below one.

B.3 VISUAL CLARITY AFFECTED ITEMS (BASED ON THE POST HOC ANALYSIS)

Item 102:

- a. I crashed the car on the trip 🌟 (19.05% of ratings ≤ 4 regarding visual clarity)
- b. I crashed the car on the trip 🐼 (4.55% of ratings ≤ 4 regarding visual clarity)
- c. I crashed the car on the trip 🍷 (54.55% of ratings ≤ 4 regarding visual clarity)

Item 23:

- a. The boy broke his tooth 🦷 (13.64% of ratings ≤ 4 regarding visual clarity)
- b. The boy broke his tooth 🍬 (4.35% of ratings ≤ 4 regarding visual clarity)

- c. The boy broke his tooth 🦷 (56% of ratings ≤ 4 regarding visual clarity)

Item 100:

- a. He drove all day long 🚗 (0% of ratings ≤ 4 regarding visual clarity)
 b. He drove all day long 🚚 (4.55% of ratings ≤ 4 regarding visual clarity)
 c. He drove all day long 🍏 (56.52% of ratings ≤ 4 regarding visual clarity)

Item 117:

- a. I was sweating a lot 💧 (0% of ratings ≤ 4 regarding visual clarity)
 b. I was sweating a lot 🧴 (6.9% of ratings ≤ 4 regarding visual clarity)
 c. I was sweating a lot 🧴 (58.33% of ratings ≤ 4 regarding visual clarity)

Item 97:

- a. We have some news 🗣️ (0% of ratings ≤ 4 regarding visual clarity)
 b. We have some news 🧑 (0% of ratings ≤ 4 regarding visual clarity)
 c. We have some news 🌀 (59.09% of ratings ≤ 4 regarding visual clarity)

Item 4:

- a. I was so hot 🥵 (0% of ratings ≤ 4 regarding visual clarity)
 b. I was so hot 🍌 (0% of ratings ≤ 4 regarding visual clarity)
 c. I was so hot 🍌 (59.09% of ratings ≤ 4 regarding visual clarity)

Item 73:

- a. The cheese is filled with holes 🧀 (4.55% of ratings ≤ 4 regarding visual clarity)
 b. The cheese is filled with holes 🐹 (4.35% of ratings ≤ 4 regarding visual clarity)
 c. The cheese is filled with holes 🧀 (60% of ratings ≤ 4 regarding visual clarity)

Item 141:

- a. I'm scratched all over 🙄 (9.09% of ratings ≤ 4 regarding visual clarity)
 b. I'm scratched all over 🐱 (4.35% of ratings ≤ 4 regarding visual clarity)
 c. I'm scratched all over 🏠 (60% of ratings ≤ 4 regarding visual clarity)

APPENDIX C – THE NEURAL CORRELATES OF MULTIMODAL BRIDGING

C.1 ALL QUESTIONS THAT PROBED CONJOINT MEANING FOR THE *BRIDGED* CONDITION (WITH RESPECTIVE ACCURACIES)

- The loud noise woke me up 🎸
 - According to the message, was the author woken up by loud music? (Yes)
 - 100% accuracy.
- He's not drinking today 🚗
 - Does the message say he won't drink because he doesn't like it? (No)
 - 100% accuracy.
- We have some news 🧑
 - In the message, is the news a new car? (No)
 - 100% accuracy.
- They've entered the grocery store 🛒
 - According to the message, did they go in armed? (Yes)
 - 100% accuracy.
- He told a story 😞
 - Does the message show that the story was false? (Yes)
 - 100% accuracy.
- I admire that man 🎸
 - Does the author say they admire the man's musical abilities? (Yes)
 - 90% accuracy.
- Brazil was defeated 🏆
 - Does the message say that Brazil was defeated in a soccer match? (Yes)
 - 90% accuracy.
- The event was canceled 🌧️
 - According to the message, was the event canceled because of the rain? (Yes)

- 88.89% accuracy.
- He slipped on the stairs/ladder* 🪜
 - According to the message, did the slip involve playing sports? (Yes)
 - 80% accuracy.
- We have no internet 🌐
 - Was the internet interrupted for no apparent reason in the message? (No)
 - 80% accuracy.
- We went to the art gallery 🧑🏻‍🎨
 - According to the message, did they go to the gallery as a couple? (Yes)
 - 66.67% accuracy.
- I can't leave right now 🗨️
 - Does the author give reasons for not being able to go out? (Yes)
 - 55.56% accuracy.
- She's going to get dressed 👗
 - Does the message specify that she will wear fancy clothes? (Yes)
 - 30% accuracy.

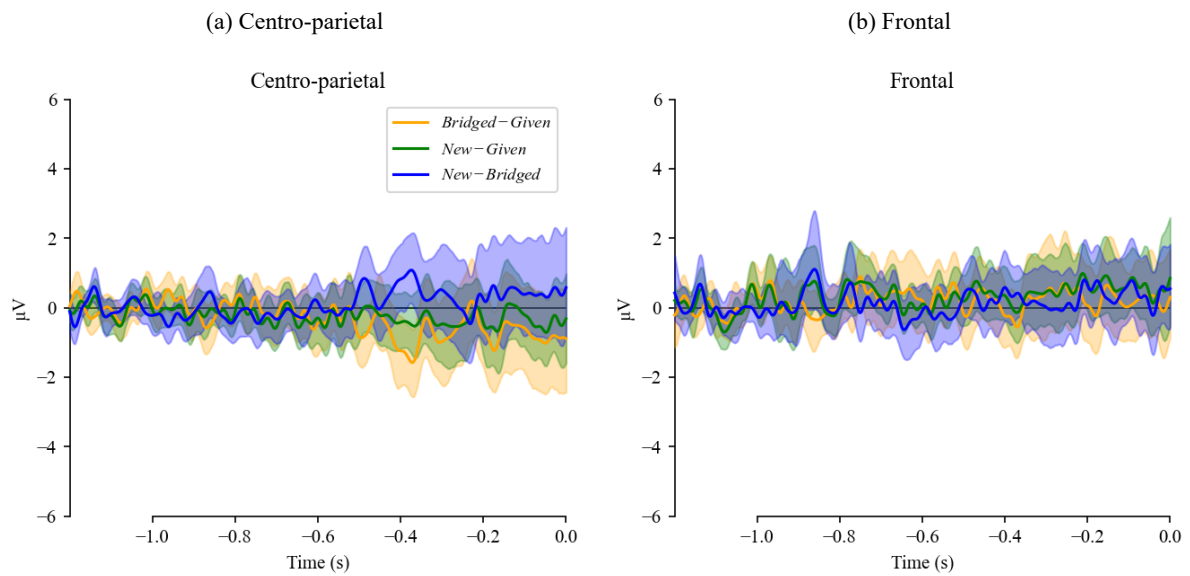
C.2 SECONDARY PIPELINES

To avoid using different parameters for the same type of plots, we established $vlim = 2, -2$ as the color contrast for all the difference waves topoplots (including the baseline check). We also used the same ROIs as in the main analysis: Centro-parietal (C3, Cz, C4, P3, Pz, P4) and Frontal (F3, Fz, F4, Fp1, Fp2).

C.2.1 Baseline check difference waves and cluster-test

Data were epoched using a time window starting 1.2 seconds before the trigger. Baseline correction was applied to the -1.2 to -1.0 s interval, leaving the -1.0 to 0 s window for analysis. As shown in Figure 30, the differences between the intervals preceding the triggers were close to zero.

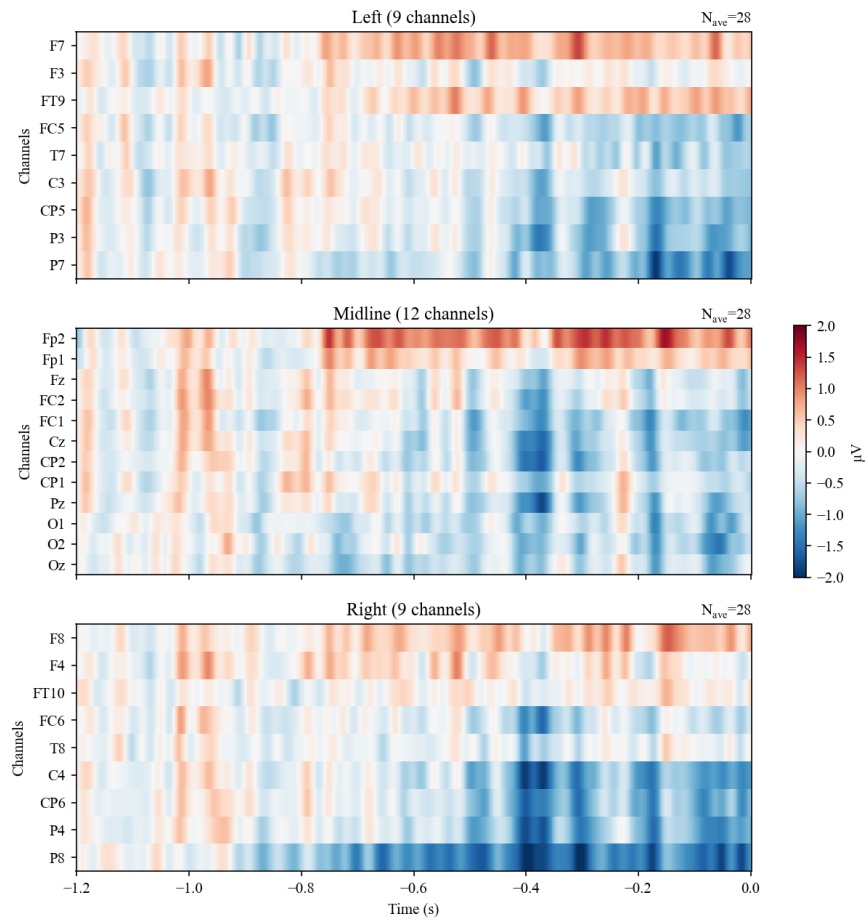
Figure 30 – Baseline check — difference waves with 95% CI



To test for significant differences between conditions, we also performed a Spatio-Temporal Cluster-Based Permutation Test. No significant clusters were observed. Figure 31 shows the *bridged – given* contrast:

Figure 31 – Baseline pipeline — cluster-permutation contrast *bridged – given*.

(a) Image plot



(b) Topography evolution

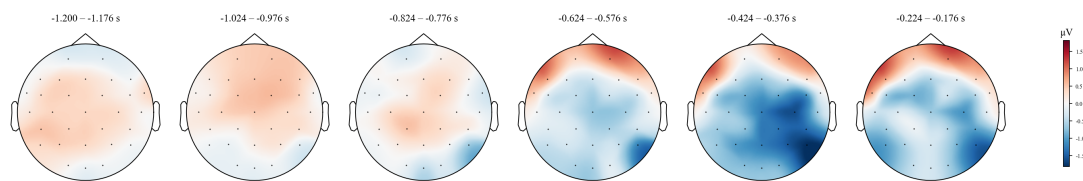
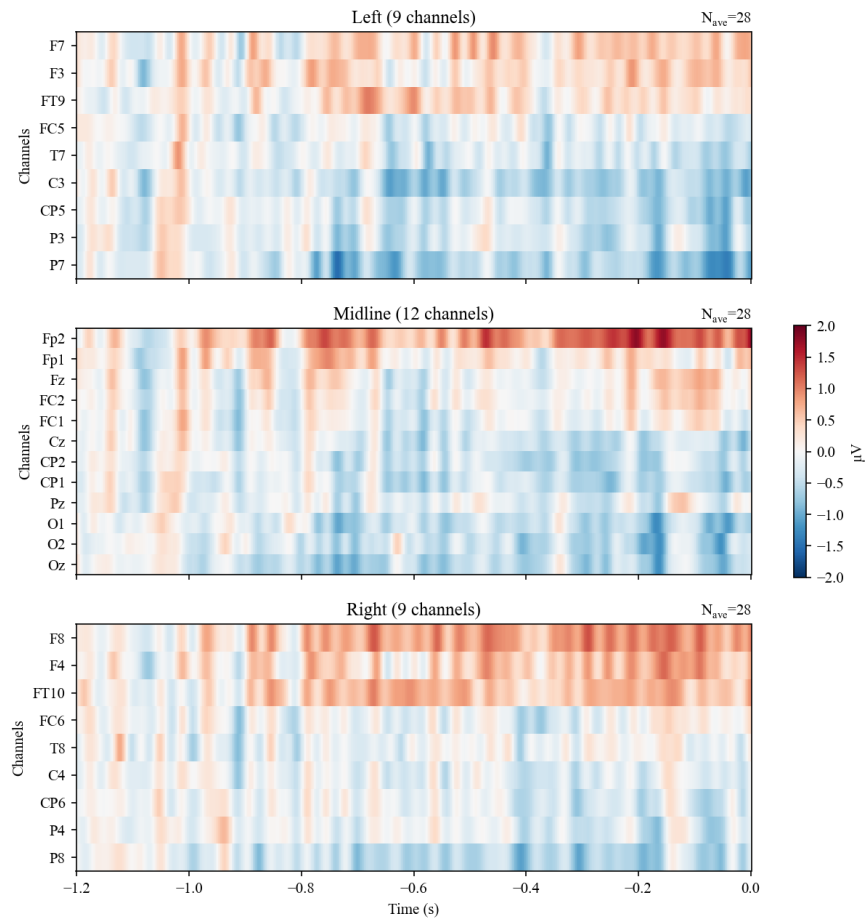
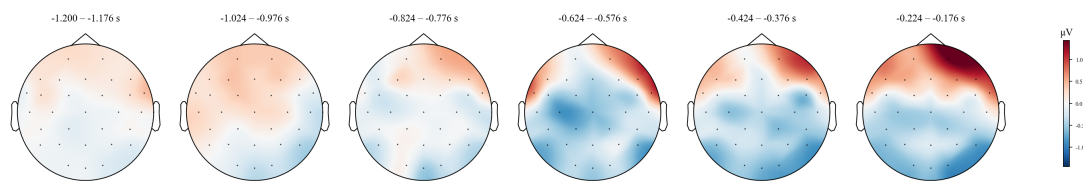
Figure 32 shows the *new – given* contrast:

Figure 32 – Baseline pipeline — cluster-permutation contrast *new – given*.

(a) Image plot



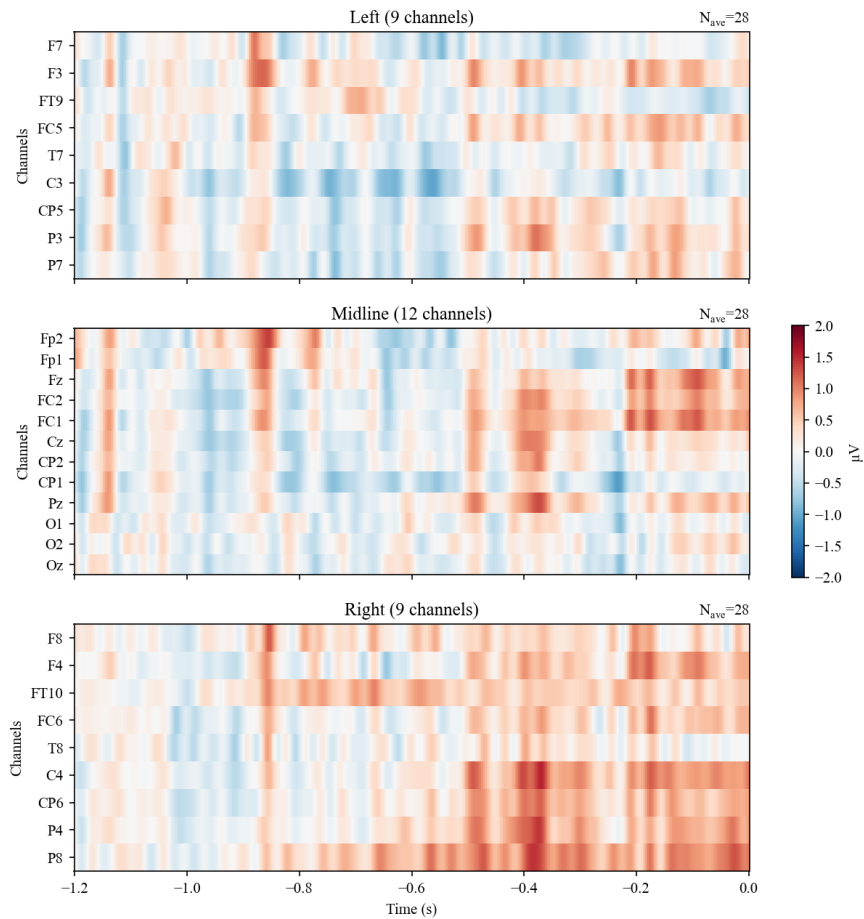
(b) Topography evolution



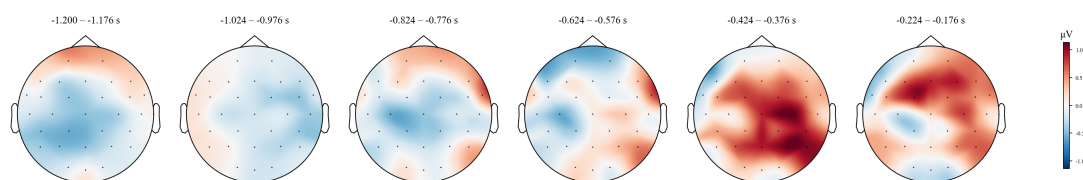
Finally, Figure 33 shows the *new – bridged* contrast:

Figure 33 – Baseline pipeline — cluster-permutation contrast *new* – *bridged*.

(a) Image plot



(b) Topography evolution



C.2.2 Categorical analysis without items 2, 61 and 62 (Alternative Pipeline)

An alternative pipeline was run without items 2, 61 and 62. The observed ERPs are very similar, as Figure 34 presents.

Figure 34 – Grand-average ERPs by condition: standard vs. alternative pipeline

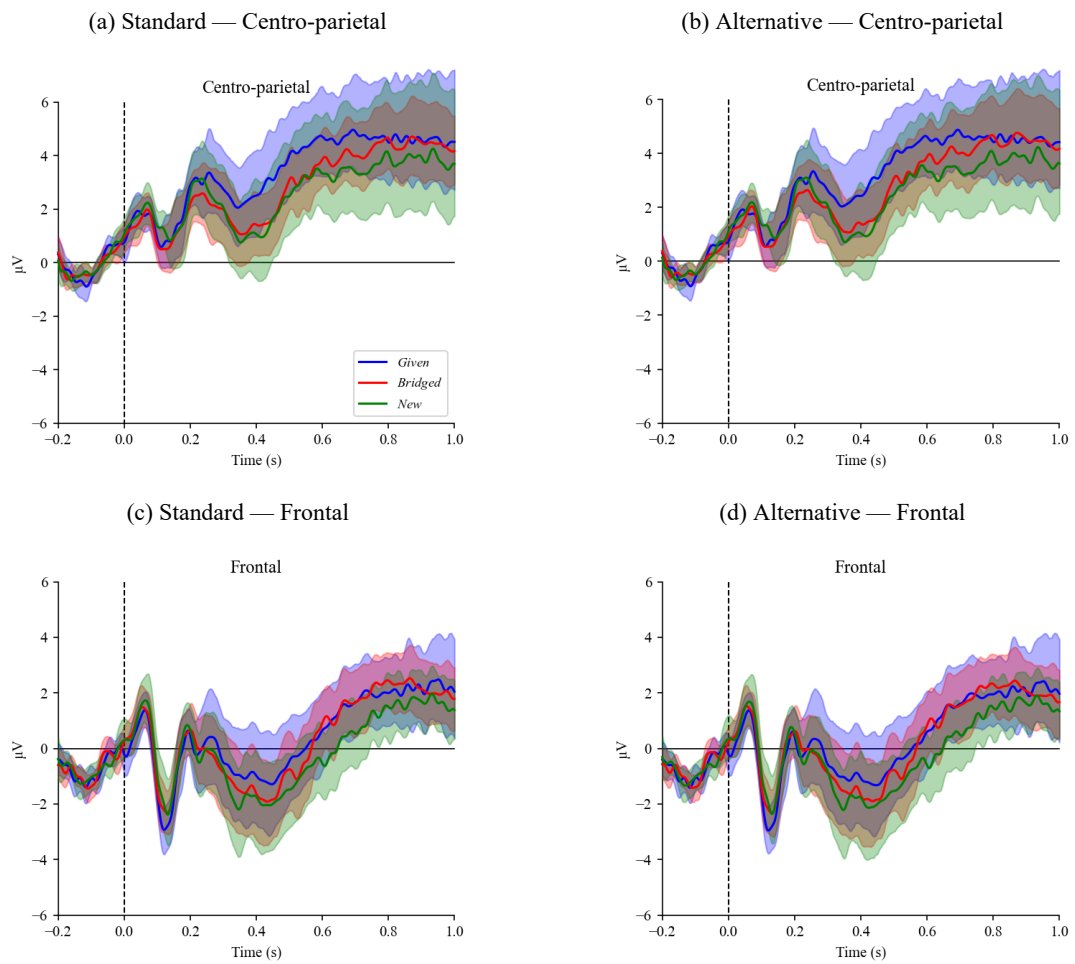
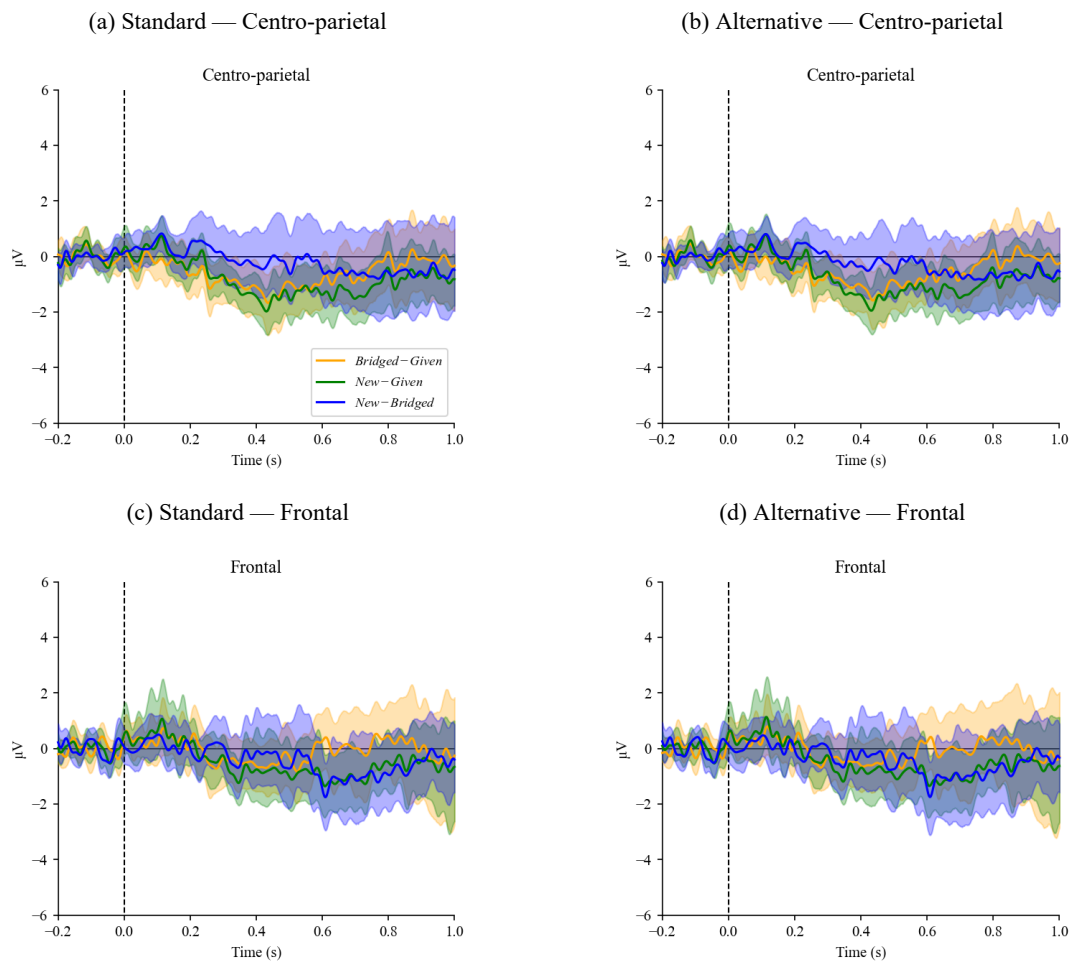
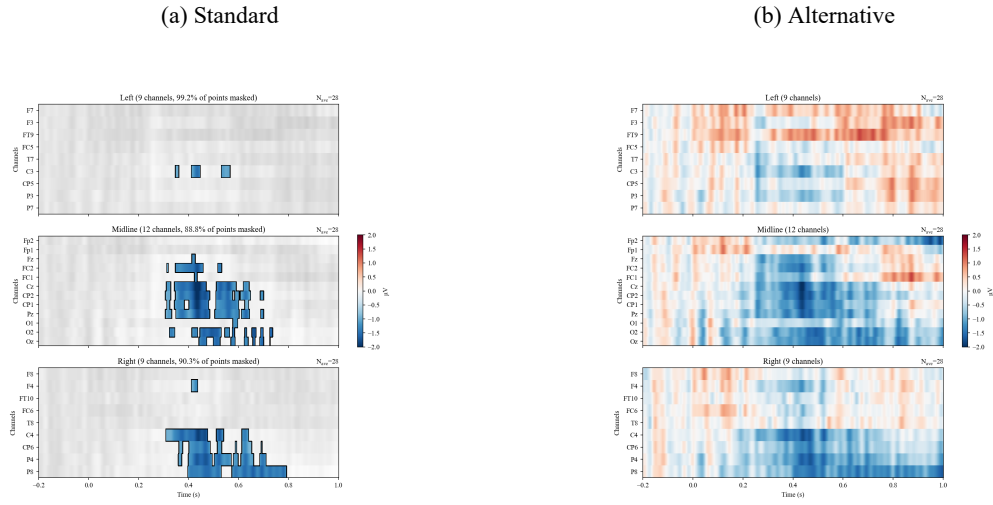


Figure 35 compares the two difference waves, also showing similar behavior.

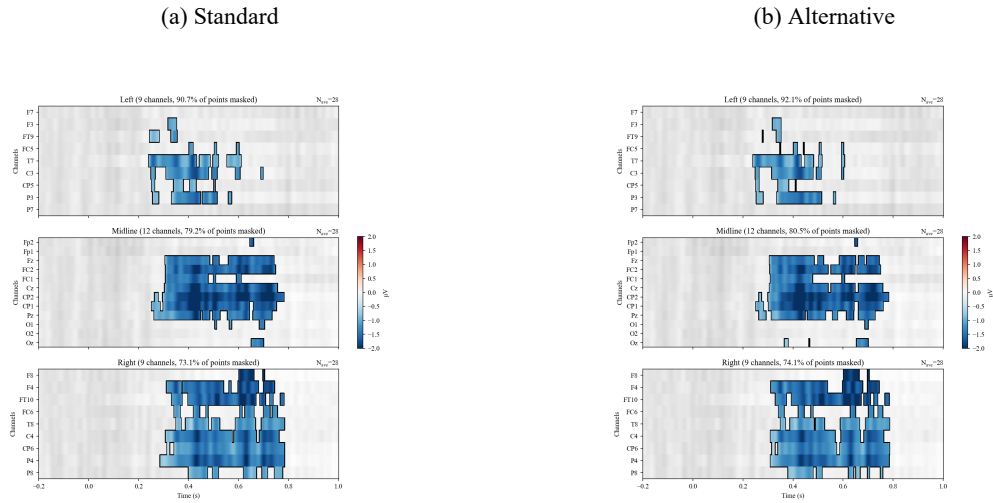
Figure 35 – Difference waves with 95% CI: standard vs. alternative pipeline



We also ran a Spatio-Temporal Cluster-Based Permutation test on the alternative pipeline's data. As Figure 36 shows, *bridged – given* did not reach significance, falling almost exactly at the established threshold $p = 0.053$.

Figure 36 – Cluster-permutation *bridged – given*: standard vs. alternative pipeline.

New – given, on the other hand, exhibits similar behavior to the standard analysis, as Figure 37 illustrates.

Figure 37 – Cluster-permutation contrast *new – given*: standard vs. alternative pipeline.

Finally, like in the standard analysis, there were no significant clusters for *new – bridged* with items 2, 61 and 62 removed.

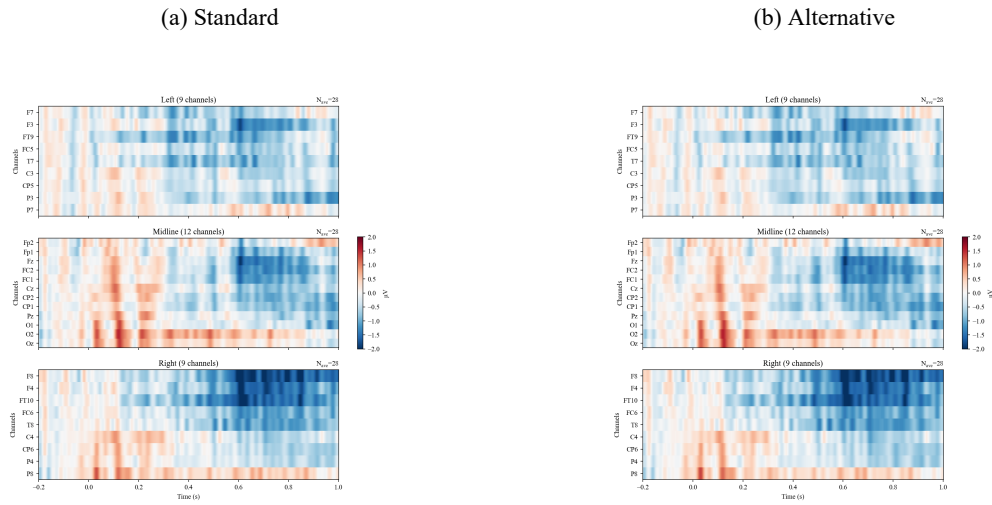
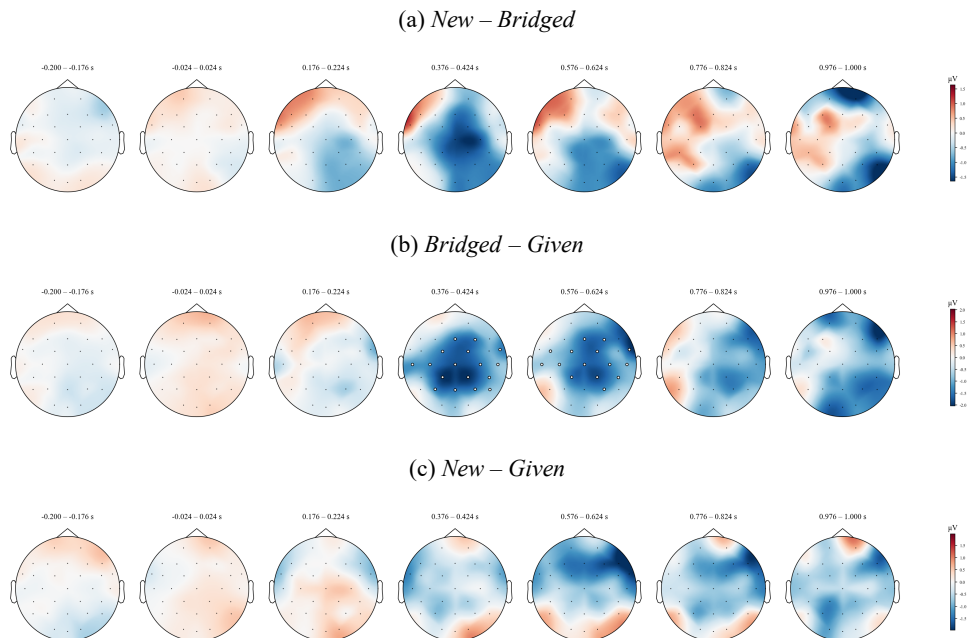
Figure 38 – Cluster-permutation contrast *new – bridged*: standard vs. alternative pipeline.

Figure 39 contrasts the topoplots.

Figure 39 – Alternative pipeline — topography evolution across contrasts.

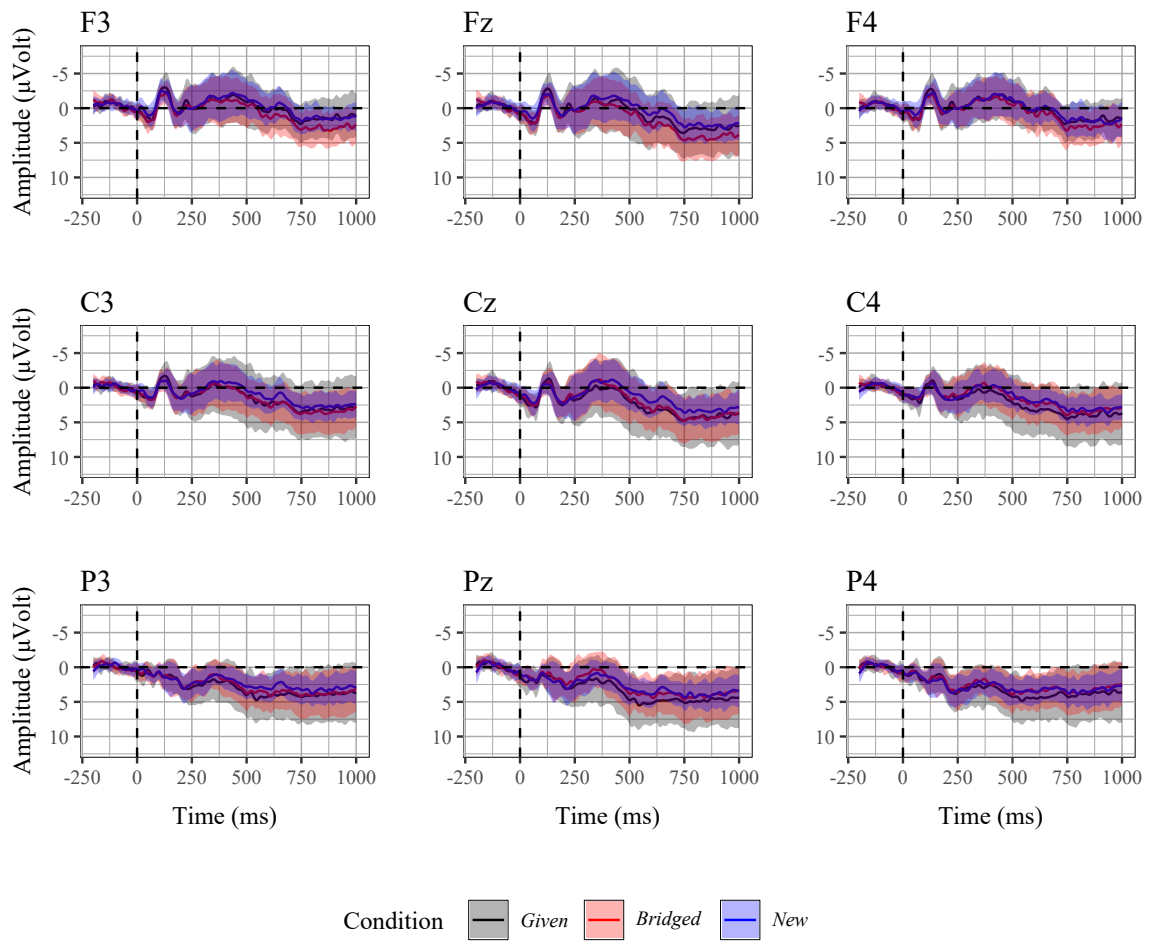


C.2.3 Observed ERPs grid (3x3)

Figure 40 gives an exploratory view of ERPs outside our predefined Regions of Interest.

Figure 40 – Observed ERPs grid (3x3)

Observed ERPs (3x3 Grid)



ANNEXES

Annexes

ANNEX A – THIRD-PARTY MATERIALS

The only third-party materials we used were Newman’s procedure for the Spatio-Temporal Cluster-Based Permutation analysis (<https://neuraldatascience.io/>) and Aurnhammer’s code for the rERP analysis (<https://github.com/caurnhammer/psyp23rerps>). Both adapted versions are made public through the repository: https://github.com/ghapoliveira/multimodal_inferences.